BANACH & C^*-ALGEBRAS

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Abstract. This document comprises notes on Banach algebras and C^*-algebras. Parts I and II of the notes were taken during my participation in the Groundwork for Operator Algebras Lecture Series (GOALS) workshop at IPAM, UCLA. Part III of the notes was written sporadically after the summer school. If you come across any typos, please send corrections to junaid.aftab1994@gmail.com. We work over $\mathbb{K} = \mathbb{C}$, the field of complex numbers.

Contents

Part 1. Banach Algebras

Banach algebras constitute a broad class of operator algebras and provide a natural setting for the study of several classical topics in functional analysis, particularly spectral theory. Fundamental notions such as the spectrum, resolvent set, and spectral radius can be developed in this setting and used to study the behavior of operators on general normed spaces. The class of Banach algebras also contains C^*-algebras as an important subclass. Consequently, Banach algebra theory provides a natural foundation for the broader study of operator algebras.

1. Definitions and Examples

In this section, we introduce the definition of a Banach algebra and discuss several examples.

1.1. **Definitions.** Let X be a Banach space. The Banach space $\mathscr{B}(X)$ of bounded linear operators on X carries additional algebraic structure given by composition. In particular, if $T, S \in \mathscr{B}(X)$, then $T \circ S \in \mathscr{B}(X)$ and $\|T \circ S\| \leq \|T\|\|S\|$. This observation motivates the following sequence of definitions.

Definition 1.1. Let V be a complex vector space. An **algebra** structure on V consists of a multiplication operation

$$\cdot \colon V \times V \longrightarrow V, \tag{1.1}$$

$$(x, y) \longmapsto x \cdot y, \tag{1.2}$$

that is bilinear and associative. That is,

$$(\alpha x + \beta y) \cdot z = \alpha(x \cdot z) + \beta(y \cdot z), \tag{1.3}$$

$$x \cdot (\alpha y + \beta z) = \alpha(x \cdot y) + \beta(x \cdot z), \tag{1.4}$$

$$(x \cdot y) \cdot z = x \cdot (y \cdot z). \tag{1.5}$$

for all $x, y, z \in V$. The algebra V is **unital** if there exists an element $e \in V$ such that

$$e \cdot x = x \cdot e = x. \tag{1.6}$$

The algebra V is **commutative** if

$$x \cdot y = y \cdot x \tag{1.7}$$

for all $x, y \in V$.

The element e in Definition 1.1 is called the identity of V. It is straightforward to verify that the identity of an algebra is unique. Definition 1.2 defines normed algebras by requiring the multiplication to be compatible with the norm.

Definition 1.2. An algebra V is a **normed algebra** if it is equipped with a norm $\|\cdot\| \colon V \to [0, \infty)$ that is submultiplicative. That is,

$$\|x \cdot y\| \leq \|x\|\|y\| \tag{1.8}$$

for all $x, y \in V$.

In what follows, we write the product $x \cdot y$ simply as xy for elements x, y of an algebra. The norm on a normed algebra V induces a metric on V and hence a topology, called the norm topology. The algebraic operations are continuous with respect to this topology, as established in Lemma 1.3.

Lemma 1.3. *Let V be a normed algebra. Addition, scalar multiplication, and multiplication are continuous with respect to the norm topology on V.*

Proof. The continuity of addition and scalar multiplication follows from the corresponding facts for normed spaces. We verify the continuity of multiplication. Suppose that $x_n \to x$ and $y_n \to y$ in V. Since every convergent sequence is bounded, the sequence $(y_n)_{n\geq 1}$ is bounded. By the submultiplicativity of the norm,

$$\begin{aligned} \|xy - x_n y_n\| &\leq \|xy - xy_n\| + \|xy_n - x_n y_n\| && (1.9)\\ &\leq \|x\|\|y - y_n\| + \|x - x_n\|\|y_n\|. && (1.10)\end{aligned}$$

The right-hand side converges to zero. Hence, $x_n y_n \to xy$, which proves that multiplication is continuous. □

We obtain Banach algebras by imposing completeness on normed algebras. Banach algebras and their morphisms are defined in Definition 1.4.

Definition 1.4. A **Banach algebra** is a normed algebra A that is complete with respect to the metric induced by its norm. A **morphism of Banach algebras** is a continuous complex-linear multiplicative map $\phi\colon A \to B$. The morphism ϕ is an **isometric isomorphism** if it is bijective and satisfies

$$\|\phi(a)\|_B = \|a\|_A \tag{1.11}$$

for every $a \in A$. If A and B are unital, then ϕ is called a **unital morphism** if $\phi(e_A) = e_B$.

Thus, a Banach algebra is a Banach space equipped with an associative multiplication for which the norm is submultiplicative. A unital Banach algebra is a Banach algebra whose underlying algebra is unital. If A be a unital Banach algebra with identity $e \in A$, then

$$\|e\| = \|e^2\| \leq \|e\|^2. \tag{1.12}$$

Since $e \neq 0$, it follows that $\|e\| \geq 1$. For a C^*-algebra, we will see that the identity automatically satisfies $\|e\| = 1$. We next introduce subalgebras, which provide a natural way to identify smaller algebras within a given algebra.

Definition 1.5. Let A be an algebra over $\mathbb{C}$. A subset $B \subseteq A$ is called a **subalgebra** of A if B is a vector subspace of A that is closed under multiplication.

If A is a normed algebra, then every subalgebra $B \subseteq A$, equipped with the restriction of the norm on A, is also a normed algebra. Moreover, the closure $\overline{B}$ of B in A is again a subalgebra. In particular, if A is a Banach algebra, then $\overline{B}$ is a Banach algebra. We now record several standard examples. We conclude this section by discussing ideals in Banach algebras.

Definition 1.6. Let A be a Banach algebra, and let $I \subseteq A$ be a vector subspace.

(1) The subspace I is a **left ideal** if $ba \in I$ for every $a \in I$ and $b \in A$.
(2) The subspace I is a **right ideal** if $ab \in I$ for every $a \in I$ and $b \in A$.
(3) The subspace I is a **two-sided ideal** if it is both a left ideal and a right ideal.

Closed two-sided ideals give rise to quotient Banach algebras.

Lemma 1.7. *Let A be a Banach algebra, and let $I \subseteq A$ be a closed two-sided ideal. Then the quotient space A/I is a Banach algebra with respect to the quotient norm.*

Proof. Since I is a two-sided ideal, multiplication on A/I is well-defined by

$$(a + I)(b + I) := ab + I. \tag{1.13}$$

It is a standard result from Banach space theory that A/I is a Banach space with respect to the quotient norm

$$\|a + I\|_{A/I} := \inf_{m\in I} \|a + m\|_A. \tag{1.14}$$

It remains to verify that this norm is submultiplicative. Let $a, b \in A$ and $\varepsilon > 0$. By the definition of the quotient norm, there exist $m, n \in I$ such that

$$\|a+m\|_A \leq \|a+I\|_{A/I} + \varepsilon, \tag{1.15}$$
$$\|b+n\|_A \leq \|b+I\|_{A/I} + \varepsilon. \tag{1.16}$$

Since $(a+m)(b+n) \in ab + I$, it follows that

$$\begin{aligned} \|ab+I\|_{A/I} &\leq \|(a+m)(b+n)\|_A && (1.17)\\ &\leq \|a+m\|_A \|b+n\|_A && (1.18)\\ &\leq \|a+I\|_{A/I}\|b+I\|_{A/I} && (1.19)\\ &\quad + \varepsilon\left(\|a+I\|_{A/I} + \|b+I\|_{A/I} + \varepsilon\right). && (1.20)\end{aligned}$$

Since this inequality holds for every $\varepsilon > 0$, we obtain

$$\|ab+I\|_{A/I} \leq \|a+I\|_{A/I}\|b+I\|_{A/I}. \tag{1.21}$$

Thus, the quotient norm is submultiplicative, and A/I is a Banach algebra. □

If I is only a closed left or right ideal, then A/I is a Banach space, but multiplication of cosets need not be well-defined. The quotient A/I is a Banach algebra when I is a closed two-sided ideal. If A is commutative, then every left or right ideal is automatically two-sided.

Example 1.8. Let A be a Banach algebra.

(1) The subspaces $\{0\}$ and A are two-sided ideals, called the trivial ideals.
(2) If A is unital, then every left, right, or two-sided ideal containing the identity e is equal to A.

Definition 1.9 introduces maximal and regular ideals in a Banach algebra.

Definition 1.9. Let A be a Banach algebra.

(1) A proper left ideal $I \subsetneq A$ is called a **maximal left ideal** if there is no proper left ideal $J \subsetneq A$ such that $I \subsetneq J$. Maximal right ideals and maximal two-sided ideals are defined similarly.
(2) Suppose that A is nonunital. A left ideal $I \subseteq A$ is called **regular** if there exists $u \in A$ such that $a - au \in I$ for every $a \in A$. A right ideal $I \subseteq A$ is called regular if there exists $u \in A$ such that $a - ua \in I$ for every $a \in A$.

Example 1.10. Let A be a unital Banach algebra, and let $I \subseteq A$ be a left ideal. Let $a \in \overline{I}$, and choose a sequence $(a_n)_{n\geq 1}$ in I such that $a_n \to a$. For every $b \in A$, the continuity of multiplication gives $ba_n \to ba$. Since $ba_n \in I$ for every $n \in \mathbb{N}$, it follows that $ba \in \overline{I}$. Hence, $\overline{I}$ is a left ideal. The right ideal case is similar.

By Zorn's lemma, every proper left or right ideal of a unital Banach algebra is contained in a maximal left or right ideal, respectively. This is proved in Lemma 1.11.

Lemma 1.11. *Let A be a unital Banach algebra. Every proper left ideal, I, is contained in a maximal left ideal, and every proper right/two-sided ideal is contained in a maximal right/two-sided ideal.*

Proof. Let

$$\mathscr{F} := \{J \subseteq A : J \text{ is a proper left ideal and } I \subseteq J\}. \tag{1.22}$$

We partially order $\mathscr{F}$ by inclusion. Since $I \in \mathscr{F}$, the collection $\mathscr{F} \neq \emptyset$. Let $\{J_\alpha\}_{\alpha\in\Lambda}$ be a chain in $\mathscr{F}$, and define

$$J := \bigcup_{\alpha\in\Lambda} J_\alpha. \tag{1.23}$$

Since the family is totally ordered by inclusion, J is a vector subspace of A. Moreover, if $a \in A$ and $x \in J$, then $x \in J_\alpha$ for some $\alpha \in \Lambda$. Hence, $ax \in J_\alpha \subseteq J$. Thus, J is a left ideal containing I_0. It remains to show that J is proper. Suppose that $J = A$. Then $e \in J$, so $e \in J_\alpha$ for some $\alpha \in \Lambda$. Since J_α is a left ideal, $a = ae \in J_\alpha$ for every $a \in A$. Hence, $J_\alpha = A$, contradicting the fact that J_α is proper. Therefore, $J \in \mathscr{F}$. Thus, every chain in $\mathscr{F}$ has an upper bound. By Zorn's lemma, $\mathscr{F}$ has a maximal element M. If

$$M \subseteq K \subsetneq A \tag{1.24}$$

where K is a left ideal, then $K \in \mathscr{F}$. The maximality of M in $\mathscr{F}$ gives $K = M$. Therefore, M is a maximal left ideal. The proof for right/two-sided ideal case is similar. □

Remark 1.12. The notion of regular ideals in Definition 1.9 is useful for extending results about maximal ideals to nonunital Banach algebras. Let A be a nonunital Banach algebra, and let $I \subsetneq A$ be a regular left ideal. If $u \in A$ satisfies $a - au \in I$ for every $a \in A$, then

$$J := I + \mathbb{C}(e' - u) \tag{1.25}$$

is a proper left ideal of the unitization A', and $J \cap A = I$. If I is a maximal left ideal of A, then J is a maximal left ideal of A'. Since maximal left ideals in a unital Banach algebra are closed, J is closed. Therefore, $I = J \cap A$ is closed. Thus, every maximal regular left ideal of a nonunital Banach algebra is closed. The analogous statement holds for maximal regular right ideals.

1.2. **Examples.** We now discuss several standard examples of Banach algebras. These examples include algebras of functions and algebras of bounded operators, which will recur throughout the study of C^*-algebras.

Example 1.13. Let S be a set, and let $\ell^\infty(S)$ denote the vector space of bounded complex-valued functions on S. Define addition, multiplication, and scalar multiplication pointwise by

$$(f + g)(s) := f(s) + g(s), \tag{1.26}$$

$$(fg)(s) := f(s)g(s), \tag{1.27}$$

$$(\lambda f)(s) := \lambda f(s) \tag{1.28}$$

for all $s \in S$. Equipped with the norm $\|f\|_\infty := \sup_{s\in S} |f(s)|$, the space $\ell^\infty(S)$ is a commutative unital Banach algebra. Its identity is the constant function $s \mapsto 1$.

Several important Banach algebras of continuous functions arise as closed subalgebras of $\ell^\infty(S)$.

Example 1.14. Let X be a topological space. The space of bounded continuous complex-valued functions on X is

$$C_b(X) := \{f : X \to \mathbb{C} : f \text{ is continuous and bounded}\}. \tag{1.29}$$

The space $C_b(X)$ is a closed subalgebra of $\ell^\infty(X)$. Hence, equipped with the supremum norm, $C_b(X)$ is a commutative unital Banach algebra. Now suppose that X is locally compact and Hausdorff. The space of continuous complex-valued functions on X that vanish at infinity is

$$C_0(X) := \left\{ f \in C(X) \;\middle|\; \begin{array}{c} \text{for every } \varepsilon > 0 \text{ there exists a compact set } K \subseteq X \\ \text{such that } |f(x)| < \varepsilon \text{ for every } x \notin K \end{array} \right\}. \tag{1.30}$$

The space $C_0(X)$ is a closed subalgebra of $C_b(X)$. Hence, equipped with the supremum norm, $C_0(X)$ is a commutative Banach algebra. It is unital if and only if X is compact.

If X is a compact Hausdorff space, then every continuous function on X is bounded and vanishes at infinity. Consequently, $C_0(X) = C_b(X)$. If X is a noncompact locally compact Hausdorff space, then $C_0(X) \subsetneq C_b(X)$ in general.

Remark 1.15. The Banach algebra $C_b(\mathbb{R})$ is not separable. To see this, consider bounded continuous functions whose values at the integers belong to $\{0, 1\}$. There exists an uncountable subset $S \subseteq C_b(\mathbb{R})$ such that

$$\|f - g\|_\infty \geq 1 \tag{1.31}$$

whenever $f, g \in S$ and $f \neq g$. If $B \subseteq C_b(\mathbb{R})$ is countable, then there exists $s \in S$ whose distance from every element of B is at least $\frac{1}{2}$. Therefore, B is not dense in $C_b(\mathbb{R})$.

Example 1.16. Let X be a Hilbert space. Then $\mathscr{B}(X)$ is a noncommutative unital Banach algebra. Its multiplication is given by composition, its identity is the identity operator, and its norm is the operator norm. In particular, if $X = \mathbb{C}^n$, then $M_n(\mathbb{C})$ is a Banach algebra under matrix multiplication and the operator norm.

Commutative Banach algebras play a central role in the general theory. We conclude this section with two fundamental commutative Banach algebras.

Example 1.17. Consider $L^1(\mathbb{R})$, the Banach space of Lebesgue integrable complex-valued functions on $\mathbb{R}$. For $f, g \in L^1(\mathbb{R})$, define their convolution by

$$(f * g)(x) := \int_{\mathbb{R}} f(x - y)g(y)\, dy. \tag{1.32}$$

Then $L^1(\mathbb{R})$ is a commutative Banach algebra under convolution. The operations of addition and convolution satisfy

$$(f * g) * h = f * (g * h), \tag{1.33}$$

$$f * (g + h) = f * g + f * h, \tag{1.34}$$

$$(f + g) * h = f * h + g * h, \tag{1.35}$$

$$f * g = g * f. \tag{1.36}$$

Moreover, the norm is submultiplicative. Indeed, Tonelli's theorem and the translation invariance of Lebesgue measure give

$$\|f * g\|_1 \leq \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x - y)||g(y)|\, dy\, dx \tag{1.37}$$

$$= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} |f(x - y)|\, dx \right] |g(y)|\, dy \tag{1.38}$$

$$= \|f\|_1 \int_{\mathbb{R}} |g(y)|\, dy = \|f\|_1 \|g\|_1. \tag{1.39}$$

The Banach algebra $L^1(\mathbb{R})$ is nonunital. Suppose, to the contrary, that $f \in L^1(\mathbb{R})$ is an identity for convolution. For every $g \in L^1(\mathbb{R})$, the convolution theorem gives

$$\widehat{g} = \widehat{f * g} = \widehat{f}\,\widehat{g}. \tag{1.40}$$

If g is a Gaussian, then $\widehat{g}$ does not vanish. It follows that $\widehat{f} \equiv 1$. On the other hand, the Riemann–Lebesgue lemma implies that $\widehat{f} \in C_0(\mathbb{R})$, which is a contradiction. Therefore, $L^1(\mathbb{R})$ is nonunital.

Remark 1.18. Convolution may first be defined on the dense subspace $C_c(\mathbb{R})$. The preceding norm estimate shows that it extends uniquely to a continuous bilinear map

$$L^1(\mathbb{R}) \times L^1(\mathbb{R}) \longrightarrow L^1(\mathbb{R}). \tag{1.41}$$

Example 1.19. Let

$$\mathbb{D} := \{z \in \mathbb{C} : |z| \leq 1\} \tag{1.42}$$

be the closed unit disk. Let $\mathscr{O}(\mathbb{D}) \subseteq C(\mathbb{D})$ denote the subalgebra of functions that are holomorphic on $\mathbb{D}^\circ$. By Morera's theorem, a uniform limit of functions in $\mathscr{O}(\mathbb{D})$ remains holomorphic on $\mathbb{D}^\circ$. Hence, $\mathscr{O}(\mathbb{D})$ is a closed subalgebra of $C(\mathbb{D})$. Let $\mathscr{A}(\mathbb{D})$ be the image of $\mathscr{O}(\mathbb{D})$ in $C(\mathbb{S}^1)$ under the restriction map. By the maximum modulus principle, the map $g \mapsto g|_{\mathbb{S}^1}$ is isometric. Hence, $\mathscr{A}(\mathbb{D})$ is complete and therefore closed in $C(\mathbb{S}^1)$. It follows that $\mathscr{A}(\mathbb{D})$ is a Banach algebra. This algebra is called the disk algebra. It consists of the functions in $C(\mathbb{S}^1)$ that admit continuous extensions to $\mathbb{D}$ that are holomorphic on $\mathbb{D}^\circ$.

2. Unitization

A priori, a Banach algebra need not contain an identity element. In this section, we show that every non-unital Banach algebra can be embedded isometrically as a closed ideal in a unital Banach algebra.

Definition 2.1. Let A be a non-unital Banach algebra. The **unitization** of A, denoted by A', is the vector space $A' := A \oplus \mathbb{C}$ equipped with multiplication

$$(a, \lambda)(b, \mu) := (ab + \lambda b + \mu a, \lambda\mu) \tag{2.1}$$

for $a, b \in A$ and $\lambda, \mu \in \mathbb{C}$, and with norm

$$\|(a, \lambda)\|_{A'} := \|a\|_A + |\lambda|. \tag{2.2}$$

We verify that the construction in Definition 2.1 indeed produces a unital Banach algebra.

Lemma 2.2. *Let A be a non-unital Banach algebra. Then its unitization A' is a unital Banach algebra with identity $e' := (0, 1)$. Moreover, the map*

$$\iota \colon A \longrightarrow A', \tag{2.3}$$
$$a \longmapsto (a, 0) \tag{2.4}$$

is an isometric morphism whose image is a closed two-sided ideal of A'.

Proof. It is straightforward to verify that the multiplication in Equation (2.1) is bilinear and associative. Moreover, for every $(a, \lambda) \in A'$,

$$(0, 1)(a, \lambda) = (a, \lambda) = (a, \lambda)(0, 1). \tag{2.5}$$

Thus, $e' = (0, 1)$ is the identity of A'. In particular, $\|e'\|_{A'} = 1$. We verify that the norm is submultiplicative. For $(a, \lambda), (b, \mu) \in A'$, we have

$$\begin{aligned}
\|(a, \lambda)(b, \mu)\|_{A'} &= \|ab + \lambda b + \mu a\|_A + |\lambda\mu| && (2.6)\\
&\leq \|a\|_A\|b\|_A + |\lambda|\|b\|_A + |\mu|\|a\|_A + |\lambda||\mu| && (2.7)\\
&= (\|a\|_A + |\lambda|)(\|b\|_A + |\mu|) && (2.8)\\
&= \|(a, \lambda)\|_{A'}\|(b, \mu)\|_{A'}. && (2.9)
\end{aligned}$$

To prove completeness, let $((a_n, \lambda_n))_{n \geq 1}$ be a Cauchy sequence in A'. By Equation (2.2),

$$\|(a_n, \lambda_n) - (a_m, \lambda_m)\|_{A'} = \|a_n - a_m\|_A + |\lambda_n - \lambda_m|. \tag{2.10}$$

It follows that $(a_n)_{n\geq 1}$ is Cauchy in A and $(\lambda_n)_{n\geq 1}$ is Cauchy in $\mathbb{C}$. Since A and $\mathbb{C}$ are complete, there exist $a \in A$ and $\lambda \in \mathbb{C}$ such that $a_n \to a$ and $\lambda_n \to \lambda$. Consequently,

$$\|(a_n, \lambda_n) - (a, \lambda)\|_{A'} = \|a_n - a\|_A + |\lambda_n - \lambda| \longrightarrow 0. \tag{2.11}$$

Hence, A' is complete. Finally,

$$\|\iota(a)\|_{A'} = \|(a, 0)\|_{A'} = \|a\|_A, \tag{2.12}$$

so ι is isometric. Its image is $A \oplus \{0\}$. For $a, b \in A$ and $\lambda \in \mathbb{C}$,

$$(a, 0)(b, \lambda) = (ab + \lambda a, 0) \in A \oplus \{0\}, \tag{2.13}$$

$$(b, \lambda)(a, 0) = (ba + \lambda a, 0) \in A \oplus \{0\}, \tag{2.14}$$

so $A \oplus \{0\}$ is a two-sided ideal of A'. It is closed because it is the kernel of the continuous projection

$$\pi \colon A' \longrightarrow \mathbb{C}, \tag{2.15}$$

$$(a, \lambda) \longmapsto \lambda. \tag{2.16}$$

This completes the proof. □

Remark 2.3. If A already has an identity e, then the construction in Definition 2.1 still produces an enlarged unital Banach algebra. In this case, A' is topologically isomorphic to the direct sum $A \oplus \mathbb{C}$ equipped with componentwise multiplication. The isomorphism is given by

$$\Psi \colon A' \longrightarrow A \oplus \mathbb{C}, \tag{2.17}$$

$$(a, \lambda) \longmapsto (a + \lambda e, \lambda). \tag{2.18}$$

Its inverse is

$$\Psi^{-1} \colon A \oplus \mathbb{C} \longrightarrow A', \tag{2.19}$$

$$(b, \lambda) \longmapsto (b - \lambda e, \lambda). \tag{2.20}$$

In what follows, we identify A with its image under ι and regard it as a closed two-sided ideal of A'. The unitization construction then gives the canonical short exact sequence

$$0 \longrightarrow A \xrightarrow{\iota} A' \xrightarrow{\pi} \mathbb{C} \longrightarrow 0. \tag{2.21}$$

Indeed, π is a surjective morphism of Banach algebras and $\ker(\pi) = A \oplus \{0\} = \iota(A)$. In particular, A is a maximal closed two-sided ideal of A' because

$$A'/A \cong \mathbb{C}. \tag{2.22}$$

Since $\mathbb{C}$ has no nontrivial proper ideals, the correspondence theorem for ideals implies that the only two-sided ideals of A' containing A are A and A'. Hence, A is a maximal proper two-sided ideal of A'. Proposition 2.4 formalizes the sense in which A' is the canonical unital Banach algebra obtained from A through a universal property.

Proposition 2.4. *Let A be a non-unital Banach algebra, let B be a unital Banach algebra, and let $\phi \colon A \to B$ be a morphism of Banach algebras. Then there exists a unique unital morphism $\phi' \colon A' \to B$ such that $\phi' \circ \iota = \phi$. It is given by*

$$\phi'(a, \lambda) = \phi(a) + \lambda e_B. \tag{2.23}$$

quivalently, there exists a unique unital morphism ϕ' for which the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "\iota"] \arrow[dr, "\phi"'] & A' \arrow[d, dashed, "\phi'"] \\ & B. \end{tikzcd} \tag{2.24}$$

Proof. The formula in Equation (2.23) defines a complex-linear map extending ϕ. It is unital because $\phi'(e') = \phi'(0,1) = e_B$. For $(a,\lambda),(b,\mu) \in A'$, we have

$$\phi'((a,\lambda)(b,\mu)) = \phi(ab + \lambda b + \mu a) + \lambda\mu e_B \tag{2.25}$$
$$= \phi(a)\phi(b) + \lambda\phi(b) + \mu\phi(a) + \lambda\mu e_B \tag{2.26}$$
$$= (\phi(a) + \lambda e_B)(\phi(b) + \mu e_B) \tag{2.27}$$
$$= \phi'(a,\lambda)\phi'(b,\mu). \tag{2.28}$$

Thus, ϕ' is multiplicative. It is also continuous because

$$\|\phi'(a,\lambda)\|_B \leq \|\phi(a)\|_B + |\lambda|\|e_B\|_B \tag{2.29}$$
$$\leq \|\phi\|\|a\|_A + |\lambda|\|e_B\|_B \tag{2.30}$$
$$\leq \max\{\|\phi\|, \|e_B\|_B\}\|(a,\lambda)\|_{A'}. \tag{2.31}$$

To prove uniqueness, let $\psi\colon A' \to B$ be another unital morphism satisfying $\psi \circ \iota = \phi$. Since $(a,\lambda) = (a,0) + \lambda(0,1)$, linearity and unitality give

$$\psi(a,\lambda) = \psi(a,0) + \lambda\psi(0,1) = \phi(a) + \lambda e_B. \tag{2.32}$$

Hence, $\psi = \phi'$. □

We conclude by computing the unitization of $C_0(X)$. Let X be a locally compact noncompact Hausdorff space. Recall that the one-point compactification of X is $X_\infty := X \cup \{\infty\}$, where ∞ is a new point. A subset $U \subseteq X_\infty$ is open if either U is open in X, or $\infty \in U$ and $X_\infty \setminus U$ is a compact subset of X. With this topology, X_∞ is a compact Hausdorff space. Every function $f \in C_0(X)$ extends uniquely to a function $\widetilde{f} \in C(X_\infty)$ satisfying $\widetilde{f}(\infty) = 0$. Thus, $C_0(X)$ may be identified with the closed ideal

$$\{g \in C(X_\infty) : g(\infty) = 0\} \tag{2.33}$$

of $C(X_\infty)$.

Proposition 2.5. *Let X be a locally compact noncompact Hausdorff space. Then there is a topological, generally non-isometric, isomorphism of Banach algebras*

$$C_0(X)' \cong C(X_\infty). \tag{2.34}$$

Proof. For $f \in C_0(X)$, let $\widetilde{f} \in C(X_\infty)$ denote its extension satisfying $\widetilde{f}(\infty) = 0$. Define

$$\psi\colon C_0(X)' \longrightarrow C(X_\infty), \tag{2.35}$$
$$(f,\lambda) \longmapsto \widetilde{f} + \lambda 1_{X_\infty}. \tag{2.36}$$

The map ψ is linear, multiplicative, and unital. We construct its inverse. Define

$$\phi\colon C(X_\infty) \longrightarrow C_0(X)', \tag{2.37}$$
$$g \longmapsto (g|_X - g(\infty)1_X, g(\infty)). \tag{2.38}$$

The function $g|_X - g(\infty)1_X$ belongs to $C_0(X)$ by the continuity of g at ∞. Direct computation shows that $\phi \circ \psi = \mathrm{Id}_{C_0(X)'}$ and $\psi \circ \phi = \mathrm{Id}_{C(X_\infty)}$. Hence, ψ is an algebra isomorphism. For $(f,\lambda) \in C_0(X)'$, we have

$$\|\psi(f,\lambda)\|_{C(X_\infty)} = \sup_{x \in X_\infty} |\widetilde{f}(x) + \lambda| \leq \|f\|_\infty + |\lambda| = \|(f,\lambda)\|_{C_0(X)'}. \tag{2.39}$$

Thus, ψ is continuous. Conversely, for $g \in C(X_\infty)$,

$$\|\phi(g)\|_{C_0(X)'} = \|g|_X - g(\infty)1_X\|_\infty + |g(\infty)| \leq 3\|g\|_{C(X_\infty)}. \tag{2.40}$$

Therefore, ϕ is continuous, and ψ is a topological isomorphism. □

3. Spectrum

The spectrum of an element of a Banach algebra generalizes the set of eigenvalues of a complex matrix. In this section, we introduce the spectrum and resolvent, establish their fundamental algebraic and topological properties, and prove that the spectrum of every element is a nonempty compact subset of the complex plane. We also discuss several examples and applications to operator equations.

3.1. Definitions.

We begin with the notion of invertibility.

Definition 3.1. Let A be a unital Banach algebra with identity e. An element $a \in A$ is **invertible** if there exists $b \in A$ such that $ab = ba = e$. The element b is called the **inverse** of a and is denoted by a^{-1}.

It is straightforward to verify that the inverse of an invertible element is unique. The set of invertible elements of A is denoted by

$$\mathrm{GL}(A) := \{a \in A : a \text{ is invertible}\}. \tag{3.1}$$

Under multiplication, $\mathrm{GL}(A)$ is a group with identity e, and $(ab)^{-1} = b^{-1}a^{-1}$ for all $a, b \in \mathrm{GL}(A)$. Definition 3.2 defines the resolvent set and spectrum of an element in a unital Banach algebra.

Definition 3.2. Let A be a unital Banach algebra with identity e, and let $a \in A$. The **resolvent set** of a is

$$\rho_A(a) := \{\lambda \in \mathbb{C} : \lambda e - a \text{ is invertible}\}. \tag{3.2}$$

The **spectrum** of a is

$$\sigma_A(a) := \{\lambda \in \mathbb{C} : \lambda e - a \text{ is not invertible}\} = \mathbb{C} \setminus \rho_A(a). \tag{3.3}$$

For $\lambda \in \rho_A(a)$, the element $R(\lambda, a) := (\lambda e - a)^{-1}$ is called the **resolvent** of a at λ.

If A is non-unital, then its spectrum and resolvent are defined using the unitization A'. More precisely, for $a \in A$, we define

$$\sigma_A(a) := \sigma_{A'}(a), \qquad \rho_A(a) := \rho_{A'}(a). \tag{3.4}$$

In this case, $0 \in \sigma_A(a)$ for every $a \in A$. Indeed, if a were invertible in A', then $e' = a^{-1}a \in A$ because A is a two-sided ideal of A'. This contradicts the fact that the identity e' of A' does not belong to A. Lemma 3.3 records several algebraic properties of the spectrum.

Lemma 3.3. *Let A and B be unital Banach algebras.*

(1) *For every $a, b \in A$, $\sigma_A(ab) \setminus \{0\} = \sigma_A(ba) \setminus \{0\}$.*
(2) *If $a, u \in A$ and u is invertible, then $\sigma_A(uau^{-1}) = \sigma_A(a)$.*
(3) *If $\phi\colon A \to B$ is a unital morphism of Banach algebras, then $\sigma_B(\phi(a)) \subseteq \sigma_A(a)$ for every $a \in A$.*

Proof. The proof is given below.

(1) Let $\lambda \neq 0$ and suppose that $\lambda \in \rho_A(ab)$. Set $v := (ab - \lambda e)^{-1}$. Then

$$(ab - \lambda e)v = v(ab - \lambda e) = e, \tag{3.5}$$

and hence $abv = vab = e + \lambda v$. A direct computation gives

$$(ba - \lambda e)(bva - e) = \lambda e, \tag{3.6}$$

$$(bva - e)(ba - \lambda e) = \lambda e. \tag{3.7}$$

Therefore, $ba - \lambda e$ is invertible, with

$$(ba - \lambda e)^{-1} = \lambda^{-1}(bva - e). \tag{3.8}$$

Thus, $\lambda \in \rho_A(ba)$. Interchanging a and b proves the reverse implication.

(2) For every $\lambda \in \mathbb{C}$,

$$\lambda e - uau^{-1} = u(\lambda e - a)u^{-1}. \tag{3.9}$$

The element on the left is invertible if and only if $\lambda e - a$ is invertible. Hence, $\lambda \in \rho_A(uau^{-1})$ if and only if $\lambda \in \rho_A(a)$.

(3) Suppose that $\lambda \notin \sigma_A(a)$. Then $\lambda e_A - a$ is invertible in A. Since ϕ is unital and multiplicative,

$$\phi(\lambda e_A - a) = \lambda e_B - \phi(a), \tag{3.10}$$

and $\phi((\lambda e_A - a)^{-1})$ is its inverse. Thus, $\lambda \notin \sigma_B(\phi(a))$, proving the desired inclusion.

This completes the proof. □

Remark 3.4. The exclusion of 0 in Lemma 3.3(1) is necessary. Let $S \in \mathscr{B}(\ell^2(\mathbb{N}))$ be the unilateral shift, and let S^* be the backward shift. Then

$$S^*S = I, \tag{3.11}$$

$$SS^* = I - P_{\mathrm{span}\{e_1\}}. \tag{3.12}$$

where $P_{\mathrm{span}\{e_1\}}$ is the orthogonal projection onto the first coordinate. Thus, S^*S is invertible, while SS^* is not invertible because it annihilates e_1. Taking $a = S^*$ and $b = S$, we obtain $0 \notin \sigma(ab)$ and $0 \in \sigma(ba)$.

Invertibility can depend on the ambient Banach algebra. If $A \subseteq B$ are unital Banach algebras with the same identity and $a \in A$, then an inverse of a in B need not belong to A. Consequently, the spectrum of an element may depend on the Banach algebra in which it is computed. Consequently, the inclusion $\sigma_B(a) \subseteq \sigma_A(a)$ may be strict, as shown in Example 3.5.

Example 3.5. Consider the disk algebra $\mathscr{A}(\mathbb{D})$ as a unital Banach subalgebra of $C(\mathbb{S}^1)$. The function $f(z) := z$ is invertible in $C(\mathbb{S}^1)$, with inverse $z \mapsto z^{-1}$. However, it is not invertible in $\mathscr{A}(\mathbb{D})$. Indeed, suppose that $g \in \mathscr{A}(\mathbb{D})$ were an inverse of f. Let G denote the continuous extension of g to $\mathbb{D}$ that is holomorphic on $\mathbb{D}^\circ$. Then $zG(z) = 1$ for every $z \in \mathbb{S}^1$. The function

$$h(z) := zG(z) - 1 \tag{3.13}$$

is continuous on $\mathbb{D}$, holomorphic on $\mathbb{D}^\circ$, and vanishes on $\mathbb{S}^1$. By the maximum modulus principle, h vanishes throughout $\mathbb{D}$. Evaluating at $z = 0$ gives $-1 = 0$, a contradiction. Equivalently, the candidate inverse $z \mapsto z^{-1}$ has a pole at the origin and therefore does not admit a holomorphic extension to $\mathbb{D}^\circ$. Thus, $0 \notin \sigma_{C(\mathbb{S}^1)}(f)$ and $0 \in \sigma_{\mathscr{A}(\mathbb{D})}(f)$.

We will show in Lemma 9.1 that the phenomenon exhibited in Example 3.5 does not occur for unital C^*-subalgebras that contain the same identity as the ambient C^*-algebra.

3.2. **Properties of Resolvent and Spectrum.** The geometric series admits a natural analogue in the setting of Banach algebras. It provides a useful criterion for the invertibility of elements sufficiently close to the identity. The result is formalized in Lemma 3.6.

Lemma 3.6 (Neumann Series)**.** *Let A be a unital Banach algebra with identity e. If $a \in A$ satisfies $\|a\| < 1$, then $e - a$ is invertible and*

$$(e - a)^{-1} = \sum_{n=0}^{\infty} a^n. \tag{3.14}$$

PROOF. Let $s_N := \sum_{n=0}^{N} a^n$. For $N > M$,

$$\|s_N - s_M\| \leq \sum_{n=M+1}^{N} \|a\|^n. \tag{3.15}$$

Since $\|a\| < 1$, the expression on the right converges to zero as $M, N \to \infty$. Hence, $(s_N)_{N \geq 0}$ is Cauchy. Since A is complete, there exists $s \in A$ such that $s_N \to s$. Moreover,

$$(e - a)s_N = s_N(e - a) = e - a^{N+1}. \tag{3.16}$$

Since $\|a^{N+1}\| \leq \|a\|^{N+1} \to 0$, the continuity of multiplication gives

$$(e - a)s = s(e - a) = e. \tag{3.17}$$

Therefore, $s = (e - a)^{-1}$. □

Lemma 3.6 has several immediate consequences. In particular, it gives useful criteria for invertibility and shows that the group of invertible elements is open. It also yields a convenient expansion for the inverse of a small perturbation of an invertible element and shows that the inversion map is continuous. We prove these properties in Proposition 3.7.

Proposition 3.7. *Let A be a unital Banach algebra with identity e.*

(1) *If $a \in A$ and $|\lambda| > \|a\|$, then $\lambda \in \rho_A(a)$ and*

$$R(\lambda, a) = \frac{1}{\lambda} \sum_{n=0}^{\infty} \left(\frac{a}{\lambda}\right)^n. \tag{3.18}$$

(2) *The group $\mathrm{GL}(A)$ is open in A.*

(3) *If $a \in \mathrm{GL}(A)$ and $h \in A$ such that $\|a^{-1}\|\|h\| < 1$, then $a + h$ is invertible and*

$$(a + h)^{-1} = \sum_{n=0}^{\infty} (-a^{-1}h)^n a^{-1}. \tag{3.19}$$

Moreover,

$$\|(a + h)^{-1} - a^{-1}\| \leq \frac{\|a^{-1}\|^2 \|h\|}{1 - \|a^{-1}\|\|h\|}. \tag{3.20}$$

In particular, the inversion map

$$\mathrm{GL}(A) \longrightarrow \mathrm{GL}(A), \tag{3.21}$$

$$a \longmapsto a^{-1} \tag{3.22}$$

is continuous and hence is a homeomorphism.

(4) *If $a \in \mathrm{GL}(A)$, then*

$$\sigma_A(a^{-1}) = \{\lambda^{-1} : \lambda \in \sigma_A(a)\}. \tag{3.23}$$

PROOF. The proof is given below.

(1) If $|\lambda| > \|a\|$, then $\left\|\frac{a}{\lambda}\right\| < 1$. Since $\lambda e - a = \lambda(e - \frac{a}{\lambda})$, Lemma 3.6 gives

$$(\lambda e - a)^{-1} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \left(\frac{a}{\lambda}\right)^n. \tag{3.24}$$

(2) Let $a \in \mathrm{GL}(A)$, and suppose that $b \in A$ satisfies $\|b - a\| < \|a^{-1}\|^{-1}$. Then

$$b = a(e + a^{-1}(b - a)) \tag{3.25}$$

and $\|a^{-1}(b - a)\| < 1$. By Lemma 3.6, $e + a^{-1}(b - a)$ is invertible. Hence, b is invertible. Therefore, $B(a, \|a^{-1}\|^{-1}) \subseteq \mathrm{GL}(A)$, so $\mathrm{GL}(A)$ is open.

(3) Since $a+h = a(e+a^{-1}h)$ and $\|a^{-1}h\| < 1$, the Neumann series gives

$$(a+h)^{-1} = \sum_{n=0}^{\infty}(-a^{-1}h)^n a^{-1}. \tag{3.26}$$

The term corresponding to $n=0$ is a^{-1}. Therefore,

$$(a+h)^{-1} - a^{-1} = \sum_{n=1}^{\infty}(-a^{-1}h)^n a^{-1}. \tag{3.27}$$

Using submultiplicativity and the geometric series formula, we obtain

$$\|(a+h)^{-1} - a^{-1}\| \le \sum_{n=1}^{\infty} \|a^{-1}h\|^n \|a^{-1}\| \tag{3.28}$$

$$\le \|a^{-1}\| \sum_{n=1}^{\infty}(\|a^{-1}\|\|h\|)^n \tag{3.29}$$

$$= \|a^{-1}\| \frac{\|a^{-1}\|\|h\|}{1-\|a^{-1}\|\|h\|} = \frac{\|a^{-1}\|^2\|h\|}{1-\|a^{-1}\|\|h\|}. \tag{3.30}$$

Therefore, $\|(a+h)^{-1} - a^{-1}\| \to 0$ whenever $\|h\| \to 0$. This proves that the inversion map is continuous at a. Since the inversion map is its own inverse, it is a homeomorphism.

(4) Let $\mu \neq 0$. Then

$$a^{-1} - \mu e = -\mu a^{-1}(a - \mu^{-1}e). \tag{3.31}$$

Since a and μe are invertible,

$$a^{-1} - \mu e \text{ is invertible} \iff a - \mu^{-1}e \text{ is invertible.} \tag{3.32}$$

The result follows.

This completes the proof. □

We record in Proposition 3.8 some basic properties of the resolvent $R(\lambda, a)$ of an element $a \in A$ at $\lambda \in \rho_A(a)$, including the resolvent identity and its consequences.

Proposition 3.8. *Let A be a unital Banach algebra, and let $a \in A$.*

(1) *For every $\lambda, \mu \in \rho_A(a)$,*

$$R(\lambda, a) - R(\mu, a) = (\mu - \lambda) R(\lambda, a) R(\mu, a). \tag{3.33}$$

(2) *The resolvent set $\rho_A(a)$ is open, and the map*

$$R(\,\cdot\,, a) \colon \rho_A(a) \longrightarrow A, \tag{3.34}$$

$$\lambda \longmapsto R(\lambda, a) \tag{3.35}$$

is holomorphic, with

$$\frac{\partial R}{\partial \lambda}(\lambda, a) = -R(\lambda, a)^2. \tag{3.36}$$

(3) *If $|\lambda| > \|a\|$, then*

$$\|R(\lambda, a)\| \le \frac{1}{|\lambda|}\left(\|e\| + \frac{\|a\|}{|\lambda| - \|a\|}\right). \tag{3.37}$$

In particular, if $\|e\| = 1$, then

$$\|R(\lambda, a)\| \le \frac{1}{|\lambda| - \|a\|}. \tag{3.38}$$

Proof. The proof is given below.

(1) Using the identity $x^{-1}-y^{-1} = x^{-1}(y-x)y^{-1}$ for invertible elements x and y, we obtain

$$R(\lambda, a) - R(\mu, a) = (\lambda e - a)^{-1}((\mu e - a) - (\lambda e - a))(\mu e - a)^{-1} \tag{3.39}$$
$$= (\mu - \lambda)R(\lambda, a)R(\mu, a). \tag{3.40}$$

(2) The map

$$F_a \colon \mathbb{C} \longrightarrow A, \tag{3.41}$$
$$\lambda \longmapsto \lambda e - a \tag{3.42}$$

is continuous. Since $\mathrm{GL}(A)$ is open,

$$\rho_A(a) = F_a^{-1}(\mathrm{GL}(A)) \tag{3.43}$$

is open. The continuity of inversion implies that the resolvent map is continuous. Dividing Equation (3.33) by $\lambda - \mu$ gives

$$\frac{R(\lambda, a) - R(\mu, a)}{\lambda - \mu} = -R(\lambda, a)R(\mu, a). \tag{3.44}$$

Letting $\mu \to \lambda$ proves Equation (3.36).

(3) By Equation (3.18),

$$\|R(\lambda, a)\| \leq \frac{1}{|\lambda|}\sum_{n=0}^{\infty}\left\|\frac{a}{\lambda}\right\|^n \leq \frac{1}{|\lambda|}\left(\|e\| + \sum_{n=1}^{\infty}\frac{\|a\|^n}{|\lambda|^n}\right) = \frac{1}{|\lambda|}\left(\|e\| + \frac{\|a\|}{|\lambda| - \|a\|}\right). \tag{3.45}$$

If $\|e\| = 1$, this expression reduces to Equation (3.38).

This completes the proof. $\square$

We now establish that for every element of a unital Banach algebra, the spectrum is nonempty and compact. These properties form the basis for much of spectral theory and will be used repeatedly in subsequent sections.

Proposition 3.9. *Let A be a unital Banach algebra, and let $a \in A$. Then $\sigma_A(a)$ is a compact subset of the closed disk*

$$\{z \in \mathbb{C} : |z| \leq \|a\|\}. \tag{3.46}$$

Proof. By Proposition 3.7(1), $\lambda e - a$ is invertible whenever $|\lambda| > \|a\|$. Hence,

$$\sigma_A(a) \subseteq \{z \in \mathbb{C} : |z| \leq \|a\|\}. \tag{3.47}$$

By Proposition 3.8(2), $\rho_A(a)$ is open. Therefore, $\sigma_A(a) = \mathbb{C} \setminus \rho_A(a)$ is closed. Thus, $\sigma_A(a)$ is closed and bounded in $\mathbb{C}$, and hence compact. $\square$

It is a little harder to prove that the spectrum is nonempty. To do this, we use Banach space-valued holomorphic functions together with the results from Proposition 3.8. We first recall the relevant notion of holomorphicity for Banach space-valued maps.

Definition 3.10. Let A be a complex Banach space, and let $U \subseteq \mathbb{C}$ be open. A function $f \colon U \to A$ is **Banach space valued holomorphic** if the limit

$$f'(z_0) := \lim_{z \to z_0}\frac{f(z) - f(z_0)}{z - z_0} \tag{3.48}$$

exists in A for every $z_0 \in U$.

Remark 3.11. The standard results of complex analysis, including Liouville's theorem, extend to Banach space valued holomorphic functions. These results may be proved by composing with bounded linear functionals and applying their scalar-valued counterparts.

Proposition 3.12. *Let A be a complex unital Banach algebra, and let $a \in A$. Then $\sigma_A(a) \neq \emptyset$.*

Proof. If $a = 0$, then $\sigma_A(a) = \{0\}$. Suppose that $a \neq 0$ and $\sigma_A(a) = \emptyset$. Then $\rho_A(a) = \mathbb{C}$, so the resolvent map

$$R(\,\cdot\,, a)\colon \mathbb{C} \longrightarrow A \tag{3.49}$$

is entire by Proposition 3.8(2). By Equation (3.37), $\|R(\lambda, a)\| \to 0$ as $|\lambda| \to \infty$. Choose $r > \|a\|$. Since $R(\,\cdot\,, a)$ is continuous, it is bounded on the compact disk

$$\{\lambda \in \mathbb{C} : |\lambda| \leq r\}. \tag{3.50}$$

Moreover, Equation (3.37) shows that

$$\|R(\lambda, a)\| \leq \frac{1}{|\lambda| - \|a\|} \leq \frac{1}{r - \|a\|} \tag{3.51}$$

whenever $|\lambda| \geq r$, assuming $\|e\| = 1$. Thus, $R(\,\cdot\,, a)$ is bounded both on the disk $|\lambda| \leq r$ and on its complement. Consequently, $R(\,\cdot\,, a)$ is a bounded entire Banach space valued holomorphic function. Liouville's theorem implies that it is constant. Since it converges to zero at infinity, it follows that $R(\lambda, a) = 0$ for every $\lambda \in \mathbb{C}$. This is impossible because

$$R(\lambda, a)(\lambda e - a) = e. \tag{3.52}$$

Therefore, $\sigma_A(a)$ is nonempty. ◻

Corollary 3.13. *Let A be a complex Banach algebra, not necessarily unital, and let $a \in A$. Then $\sigma_A(a)$ is a nonempty compact subset of $\mathbb{C}$.*

Proof. If A is non-unital, then $\sigma_A(a) = \sigma_{A'}(a)$ by definition. The result therefore follows by applying Propositions 3.9 and 3.12 to the unitization A'. ◻

3.3. **Examples and Applications.** We now compute the spectrum in several basic Banach algebras.

Example 3.14. Let $A = M_n(\mathbb{C})$. For $a \in A$, the element $\lambda I - a$ is not invertible precisely when $\det(\lambda I - a) = 0$. Consequently, $\sigma_A(a)$ is exactly the set of eigenvalues of a.

Example 3.15. Let X be a compact Hausdorff space, and let $A = C(X)$. For $f \in C(X)$, the function $\lambda 1_X - f$ is invertible if and only if it does not vanish on X. Hence,

$$\begin{aligned} \sigma_A(f) &= \{\lambda \in \mathbb{C} : \lambda - f(x) = 0 \text{ for some } x \in X\} && (3.53)\\ &= f(X). && (3.54) \end{aligned}$$

Example 3.16. Let X be a locally compact noncompact Hausdorff space, and let $f \in C_0(X)$. Under the identification $C_0(X)' \cong C(X_\infty)$, the function f corresponds to its extension $\widetilde{f} \in C(X_\infty)$ satisfying $\widetilde{f}(\infty) = 0$. By Example 3.15, we have[1]

$$\sigma_{C_0(X)}(f) = \widetilde{f}(X_\infty) = f(X) \cup \{0\} = \overline{f(X)}. \tag{3.55}$$

[1]Since f vanishes at infinity and X is noncompact, for every $\varepsilon > 0$ there exists $x \in X$ such that $|f(x)| < \varepsilon$. Hence, $0 \in \overline{f(X)}$. Moreover, $\widetilde{f}(X_\infty) = f(X) \cup \{0\}$ is compact, and therefore closed, because X_∞ is compact and $\widetilde{f}$ is continuous. It follows that $f(X) \cup \{0\} = \overline{f(X)}$.

Example 3.17. Let X be a compact Hausdorff space, and let $f \in C(X)$. Since $C(X)$ is a Banach space with respect to the supremum norm, we may define the bounded linear operator

$$M_f : C(X) \longrightarrow C(X), \tag{3.56}$$

$$g \longmapsto fg. \tag{3.57}$$

Thus, M_f is an element of the unital Banach algebra $\mathscr{B}(C(X))$. Moreover, $\|M_f\| = \|f\|_\infty$. For every $\lambda \in \mathbb{C}$,

$$\lambda I_{C(X)} - M_f = M_{\lambda 1_X - f}. \tag{3.58}$$

If $\lambda 1_X - f$ does not vanish on X, then

$$\frac{1}{\lambda 1_X - f} \in C(X), \tag{3.59}$$

and hence

$$(\lambda I_{C(X)} - M_f)^{-1} = M_{(\lambda 1_X - f)^{-1}}. \tag{3.60}$$

Conversely, if $\lambda I_{C(X)} - M_f$ is invertible, then it is surjective. Therefore, there exists $g \in C(X)$ such that

$$(\lambda 1_X - f)g = 1_X. \tag{3.61}$$

It follows that $\lambda - f(x) \neq 0$ for every $x \in X$. Consequently,

$$\sigma_{\mathscr{B}(C(X))}(M_f) = \{\lambda \in \mathbb{C} : \lambda - f(x) = 0 \text{ for some } x \in X\} = f(X). \tag{3.62}$$

In particular, $\sigma_{\mathscr{B}(C(X))}(M_f) = \sigma_{C(X)}(f)$.

In finite dimensions, the spectrum consists entirely of eigenvalues, whereas in infinite dimensions the spectrum may be much larger than the point spectrum. The unilateral shift gives a basic example of this phenomenon in Example 3.18.

Example 3.18. Let $S \in \mathscr{B}(\ell^2(\mathbb{N}))$ be the unilateral shift defined by

$$S(x_1, x_2, x_3, \ldots) = (0, x_1, x_2, \ldots). \tag{3.63}$$

The operator S is an isometry, so $\|S\| = 1$. It follows from Proposition 3.9 that

$$\sigma(S) \subseteq \{z \in \mathbb{C} : |z| \leq 1\}. \tag{3.64}$$

We claim that S has no eigenvalues. Suppose that $Sx = \lambda x$ for some $x = (x_1, x_2, \ldots) \in \ell^2(\mathbb{N})$. Comparing coordinates gives

$$0 = \lambda x_1, \qquad x_n = \lambda x_{n+1} \tag{3.65}$$

for every $n \geq 1$. If $\lambda \neq 0$, then $x_1 = 0$, and the remaining equations imply that $x_n = 0$ for every n. If $\lambda = 0$, then $Sx = 0$, which again implies that $x = 0$. Hence, the point spectrum of S is empty. Nevertheless, the spectrum of S is the entire closed unit disk. To see this, let $|\lambda| < 1$ and define

$$v_\lambda := (1, \overline{\lambda}, \overline{\lambda}^2, \ldots). \tag{3.66}$$

Since $|\lambda| < 1$, we have $v_\lambda \in \ell^2(\mathbb{N})$. The adjoint of S is the backward shift, and

$$S^* v_\lambda = \overline{\lambda} v_\lambda. \tag{3.67}$$

Thus, $S^* - \overline{\lambda} I$ is not invertible. If $S - \lambda I$ were invertible, then its adjoint $S^* - \overline{\lambda} I$ would also be invertible, which is a contradiction. Therefore,

$$\{z \in \mathbb{C} : |z| < 1\} \subseteq \sigma(S). \tag{3.68}$$

Since the spectrum is closed,

$$\sigma(S) = \{z \in \mathbb{C} : |z| \leq 1\}. \tag{3.69}$$

Thus, an infinite dimensional operator may have a large spectrum even when it has no eigenvalues.

We conclude this section by discussing some elementary applications of the spectrum. Proposition 3.19 shows how algebraic relations impose strong restrictions on the spectrum.

Proposition 3.19. *Let A be a complex unital Banach algebra with identity e.*

(1) *If $a \in A$ is nilpotent, then $\sigma_A(a) = \{0\}$.*
(2) *If $p \in A$ is idempotent, then $\sigma_A(p) \subseteq \{0, 1\}$.*
(3) *If $u \in A$ satisfies $u^2 = e$, then $\sigma_A(u) \subseteq \{-1, 1\}$.*

Proof. The proof is given below.

(1) Suppose that $a^N = 0$ for some $N \geq 1$. If $\lambda \neq 0$, then

$$(\lambda e - a)^{-1} = \sum_{n=0}^{N-1} \frac{a^n}{\lambda^{n+1}}. \tag{3.70}$$

Thus, $\sigma_A(a) \subseteq \{0\}$. Since the spectrum is nonempty, it follows that $\sigma_A(a) = \{0\}$.

(2) Let $|\lambda| > \|p\|$. Since every nonzero idempotent satisfies $\|p\| \geq 1$, the condition $|\lambda| > \|p\|$ ensures that $|1/\lambda| < 1$. The Neumann series and the identity $p^n = p$ for every $n \geq 1$ give

$$(\lambda e - p)^{-1} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{p^n}{\lambda^n} \tag{3.71}$$

$$= \frac{1}{\lambda} e + \frac{1}{\lambda} \sum_{n=1}^{\infty} \frac{p}{\lambda^n} \tag{3.72}$$

$$= \frac{1}{\lambda} e + \frac{1}{\lambda} \left(\sum_{n=1}^{\infty} \frac{1}{\lambda^n} \right) p = \frac{1}{\lambda} e + \frac{1}{\lambda(\lambda - 1)} p. \tag{3.73}$$

Although the Neumann series initially yields the formula only for $|\lambda| > \|p\|$, direct multiplication shows that it remains valid whenever $\lambda \notin \{0, 1\}$. Thus,

$$(\lambda e - p)^{-1} = \frac{1}{\lambda} e + \frac{1}{\lambda(\lambda - 1)} p \tag{3.74}$$

for every $\lambda \notin \{0, 1\}$. Hence, $\lambda \notin \sigma_A(u)$, and consequently $\sigma_A(u) \subseteq \{0, 1\}$.

(3) Let $|\lambda| > \|u\|$. Since $u^2 = e$ and $\|e\| = 1$, submultiplicativity gives $\|u\| \geq 1$. Thus, the condition $|\lambda| > \|u\|$ ensures that $|1/\lambda| < 1$. The Neumann series and the identities $u^{2n} = e$ and $u^{2n+1} = u$ for every $n \geq 0$ give

$$(\lambda e - u)^{-1} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{u^n}{\lambda^n} \tag{3.75}$$

$$= \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{e}{\lambda^{2n}} + \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{u}{\lambda^{2n+1}} \tag{3.76}$$

$$= \frac{1}{\lambda} \left(\sum_{n=0}^{\infty} \frac{1}{\lambda^{2n}} \right) e + \frac{1}{\lambda^2} \left(\sum_{n=0}^{\infty} \frac{1}{\lambda^{2n}} \right) u = \frac{\lambda}{\lambda^2 - 1} e + \frac{1}{\lambda^2 - 1} u = \frac{\lambda e + u}{\lambda^2 - 1}. \tag{3.77}$$

Although the Neumann series initially yields the formula only for $|\lambda| > \|u\|$, direct multiplication shows that it remains valid whenever $\lambda \notin \{-1, 1\}$. Thus,

$$(\lambda e - u)^{-1} = \frac{\lambda e + u}{\lambda^2 - 1} \tag{3.78}$$

for every $\lambda \notin \{-1, 1\}$. Hence, $\lambda \notin \sigma_A(u)$, and consequently $\sigma_A(u) \subseteq \{-1, 1\}$.

This completes the proof. □

We conclude with an operator theoretic interpretation of the spectrum that is fundamental in the study of linear equations. A scalar λ lies in the resolvent set of an operator precisely when the equation $(\lambda I - T)x = y$ is uniquely solvable for every $y \in X$ and its solution depends continuously on the data y. Thus, the spectrum records exactly those values of λ for which existence, uniqueness, or stability of solutions fails.

Proposition 3.20. *Let X be a complex Banach space, let $T \in \mathscr{B}(X)$, and let $\lambda \in \mathbb{C}$. Then the following statements are equivalent:*

(1) *$\lambda \notin \sigma(T)$.*
(2) *For every $y \in X$, the equation*

$$(\lambda I - T)x = y \tag{3.79}$$

has a unique solution $x \in X$, and the solution map $y \mapsto x$ is continuous.

When these conditions hold, the solution is given by $x = R(\lambda, T)y$.

Proof. The condition $\lambda \notin \sigma(T)$ means precisely that $\lambda I - T$ is invertible in $\mathscr{B}(X)$. Therefore, $\lambda I - T$ has a bounded inverse, and for every $y \in X$, the equation $(\lambda I - T)x = y$ has the unique solution

$$x = (\lambda I - T)^{-1} y = R(\lambda, T)y. \tag{3.80}$$

Since $R(\lambda, T)$ is bounded, the solution depends continuously on y. Conversely, suppose that the equation has a unique solution for every $y \in X$ and that the solution depends continuously on y. Define

$$S \colon X \longrightarrow X, \tag{3.81}$$
$$y \longmapsto x, \tag{3.82}$$

where x is the unique solution of $(\lambda I - T)x = y$. The linearity of $\lambda I - T$ and the uniqueness of solutions imply that S is linear. By assumption, S is continuous and hence bounded. Moreover,

$$S(\lambda I - T) = I \qquad \text{and} \qquad (\lambda I - T)S = I. \tag{3.83}$$

Thus, S is a bounded inverse of $\lambda I - T$. Therefore, $\lambda I - T$ is invertible in $\mathscr{B}(X)$, so $\lambda \notin \sigma(T)$. □

4. Polynomial Functional Calculus

Let A be a unital Banach algebra, and let $a \in A$. Given a polynomial $p \in \mathbb{C}[z]$, we may substitute a for the variable z to obtain a new element $p(a) \in A$. This construction is called the polynomial functional calculus. It provides a natural way to construct elements of A from a and relates their spectra through the polynomial spectral mapping theorem.

Definition 4.1. Let A be a unital Banach algebra with identity e, and let $a \in A$. For a polynomial $p(z) \in \mathbb{C}[z]$, define $p(a) := \sum_{k=0}^{n} c_k a^k$, where $a^0 := e$. The map

$$\Phi_a \colon \mathbb{C}[z] \longrightarrow A, \tag{4.1}$$
$$p \longmapsto p(a), \tag{4.2}$$

is called the **polynomial functional calculus** for a.

The map Φ_a is the unique unital algebra homomorphism satisfying $\Phi_a(z) = a$. In particular, for every $p, q \in \mathbb{C}[z]$ and $\alpha \in \mathbb{C}$,

$$\Phi_a(p+q) = \Phi_a(p) + \Phi_a(q), \tag{4.3}$$
$$\Phi_a(pq) = \Phi_a(p)\Phi_a(q), \tag{4.4}$$
$$\Phi_a(\alpha p) = \alpha\Phi_a(p). \tag{4.5}$$

Remark 4.2. Every unital complex Banach algebra also admits a holomorphic functional calculus, which defines $f(a)$ whenever f is holomorphic on a neighborhood of $\sigma_A(a)$. We do not develop the holomorphic functional calculus in these notes. The continuous functional calculus for normal elements of a C^*-algebra will be discussed later in Section 9.

The fundamental spectral property of the polynomial functional calculus is that applying a polynomial to an element applies the same polynomial to its spectrum. This is proven in Proposition 4.3.

Proposition 4.3 (Polynomial spectral mapping theorem)**.** *Let A be a unital Banach algebra, let $a \in A$, and let $p \in \mathbb{C}[z]$. Then*

$$\sigma_A(p(a)) = p(\sigma_A(a)), \tag{4.6}$$

where $p(\sigma_A(a)) := \{p(\lambda) : \lambda \in \sigma_A(a)\}$.

Proof. Suppose first that p is constant, with $p(z) = c$. Then $p(a) = ce$, and therefore

$$\sigma_A(p(a)) = \sigma_A(ce) = \{c\} = p(\sigma_A(a)). \tag{4.7}$$

Now suppose that p is nonconstant, and fix $\mu \in \mathbb{C}$. Since $\mathbb{C}$ is algebraically closed, there exist $c \in \mathbb{C} \setminus \{0\}$ and $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ such that

$$p(z) - \mu = c(z - \lambda_1)\cdots(z - \lambda_n). \tag{4.8}$$

Applying the polynomial functional calculus gives

$$p(a) - \mu e = c(a - \lambda_1 e)\cdots(a - \lambda_n e). \tag{4.9}$$

The factors $a - \lambda_j e$ commute because they are polynomials in a. A product of finitely many commuting elements is invertible if and only if each factor is invertible. Indeed, if commuting elements $x_1, \dots, x_n$ have an invertible product, then an inverse of x_j is obtained by multiplying the remaining factors by the inverse of $x_1 \cdots x_n$. Consequently,

$$\begin{aligned}
\mu \in \sigma_A(p(a)) &\iff p(a) - \mu e \text{ is not invertible} && (4.10)\\
&\iff a - \lambda_j e \text{ is not invertible for some } j && (4.11)\\
&\iff \lambda_j \in \sigma_A(a) \text{ for some } j && (4.12)\\
&\iff p(\lambda) = \mu \text{ for some } \lambda \in \sigma_A(a) && (4.13)\\
&\iff \mu \in p(\sigma_A(a)). && (4.14)
\end{aligned}$$

Therefore, $\sigma_A(p(a)) = p(\sigma_A(a))$. □

The polynomial functional calculus extends to a nonunital Banach algebra by passing to its unitization. Let A be a nonunital Banach algebra, let A' be its unitization, and let $a \in A$. For every $p \in \mathbb{C}[z]$, the polynomial $p(a)$ is evaluated as an element of A', and the Proposition 4.3 gives

$$\sigma_{A'}(p(a)) = p(\sigma_{A'}(a)). \tag{4.15}$$

If $p(0) = 0$, then $p(a) \in A$. With the convention $\sigma_A(a) := \sigma_{A'}(a)$, the preceding identity becomes

$$\sigma_A(p(a)) = p(\sigma_A(a)). \tag{4.16}$$

If $p(0) \neq 0$, then $p(a)$ need not belong to A, so its spectrum must be taken in A'. Applying Proposition 4.3 to the monomials $p(z) = z^n$ relates the spectrum of a to the growth of the powers of a. This leads to the spectral radius formula in Proposition 4.4.

Proposition 4.4 (Spectral radius formula)**.** *Let A be a unital Banach algebra, and let $a \in A$. Let*

$$r(a) := \max\{|\lambda| : \lambda \in \sigma_A(a)\}. \tag{4.17}$$

be the spectral radius of a. Then

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}. \tag{4.18}$$

Proof. By Proposition 4.3, for every $n \geq 1$ we have

$$\sigma_A(a^n) = \{\lambda^n : \lambda \in \sigma_A(a)\}. \tag{4.19}$$

It follows that $r(a^n) = r(a)^n$. Since the spectral radius of an element is bounded above by its norm, $r(a)^n = r(a^n) \leq \|a^n\|$. Therefore,

$$r(a) \leq \liminf_{n\to\infty} \|a^n\|^{1/n}. \tag{4.20}$$

We now prove the reverse inequality. We first derive a contour formula for the powers of a. Let $s > \|a\|$, and let

$$\Gamma_s := \{\lambda \in \mathbb{C} : |\lambda| = s\}, \tag{4.21}$$

oriented counterclockwise. For every $\lambda \in \Gamma_s$, we have

$$\left\|\frac{a}{\lambda}\right\| \leq \frac{\|a\|}{s} < 1. \tag{4.22}$$

By Lemma 3.6, we have

$$R(\lambda, a) = (\lambda e - a)^{-1} = \frac{1}{\lambda}\left(e - \frac{a}{\lambda}\right)^{-1} = \sum_{k=0}^{\infty} \frac{a^k}{\lambda^{k+1}}. \tag{4.23}$$

This series converges uniformly on Γ_s, since $\sum_{k=0}^{\infty} \frac{\|a\|^k}{s^{k+1}} < \infty$. We may therefore integrate the series term by term. For every $n \geq 0$,

$$\frac{1}{2\pi i}\int_{\Gamma_s} \lambda^n R(\lambda, a)\, d\lambda = \sum_{k=0}^{\infty} a^k \frac{1}{2\pi i}\int_{\Gamma_s} \lambda^{n-k-1}\, d\lambda = a^n, \tag{4.24}$$

because

$$\frac{1}{2\pi i}\int_{\Gamma_s} \lambda^{n-k-1}\, d\lambda = \begin{cases} 1, & k = n, \\ 0, & k \neq n. \end{cases} \tag{4.25}$$

Thus,

$$a^n = \frac{1}{2\pi i}\int_{\Gamma_s} \lambda^n R(\lambda, a)\, d\lambda \tag{4.26}$$

whenever $s > \|a\|$. We next extend Equation (4.26) to every radius greater than $r(a)$. Fix $r > r(a)$, and choose $s > \max\{r, \|a\|\}$. Since

$$\sigma_A(a) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq r(a)\}, \tag{4.27}$$

the function $\lambda \mapsto \lambda^n R(\lambda, a)$ is holomorphic on an open set containing the closed annulus

$$\{\lambda \in \mathbb{C} : r \leq |\lambda| \leq s\}. \tag{4.28}$$

Cauchy's theorem therefore gives

$$\int_{\Gamma_r} \lambda^n R(\lambda, a)\, d\lambda = \int_{\Gamma_s} \lambda^n R(\lambda, a)\, d\lambda. \tag{4.29}$$

The Banach space valued form of Cauchy's theorem follows by composing the integrand with each bounded linear functional on A, applying the scalar-valued Cauchy theorem, and then using the Hahn–Banach theorem. Consequently,

$$a^n = \frac{1}{2\pi i} \int_{\Gamma_r} \lambda^n R(\lambda, a)\, d\lambda \tag{4.30}$$

for every $r > r(a)$. Since $R(\,\cdot\,, a)$ is continuous on the compact circle Γ_r, the quantity

$$M_r := \max_{|\lambda|=r} \|R(\lambda, a)\| \tag{4.31}$$

is finite. Using the contour formula, we obtain

$$\|a^n\| \le \frac{1}{2\pi} \int_{\Gamma_r} |\lambda|^n \|R(\lambda, a)\|\, |d\lambda| \le \frac{1}{2\pi} r^n M_r (2\pi r) = r^{n+1} M_r. \tag{4.32}$$

Consequently,

$$\limsup_{n\to\infty} \|a^n\|^{1/n} \le \lim_{n\to\infty} r^{(n+1)/n} M_r^{1/n} = r. \tag{4.33}$$

Since this inequality holds for every $r > r(a)$, letting $r \downarrow r(a)$ gives

$$\limsup_{n\to\infty} \|a^n\|^{1/n} \le r(a). \tag{4.34}$$

Combining the lower and upper bounds yields $r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}$. □

We conclude with the Gelfand–Mazur theorem in Corollary 4.5, which shows that the complex numbers form the only complex unital Banach algebra in which every nonzero element is invertible. The proof is expressed as an application of Proposition 4.3 to a linear polynomial.

Corollary 4.5 (Gelfand–Mazur theorem)**.** *Let A be a unital Banach algebra in which every nonzero element is invertible. Then A is isomorphic to $\mathbb{C}$ as a unital Banach algebra. If $\|e\| = 1$, then the isomorphism is isometric.*

Proof. Let $a \in A$. Since $\sigma_A(a)$ is nonempty, there exists $\lambda_a \in \sigma_A(a)$. Consider the polynomial $q(z) = z - \lambda_a$. By Proposition 4.3, we have $\sigma_A(q(a)) = q(\sigma_A(a))$. Since $\lambda_a \in \sigma_A(a)$, we have

$$0 = q(\lambda_a) \in \sigma_A(q(a)). \tag{4.35}$$

Therefore, $q(a) = a - \lambda_a e$ is not invertible. By assumption, every nonzero element of A is invertible, so $a - \lambda_a e = 0$. Thus, every element of A is a scalar multiple of e. Define

$$\phi :: \mathbb{C} \longrightarrow A, \tag{4.36}$$

$$\lambda \longmapsto \lambda e. \tag{4.37}$$

The map ϕ is a unital algebra homomorphism. It is surjective because every element of A is a scalar multiple of e, and it is injective because $e \neq 0$. Moreover,

$$\|\phi(\lambda)\| = |\lambda| \|e\|. \tag{4.38}$$

Thus, both ϕ and ϕ^{-1} are continuous. Hence, ϕ is an isomorphism. □

5. Charcters and Gelfand Transform

The Gelfand transform is a fundamental tool for studying commutative Banach algebras. It associates to an abstract commutative Banach algebra an algebra of continuous functions on a topological space constructed from its one-dimensional representations. This representation need not be injective or surjective for a general commutative Banach algebra, but its kernel and spectral properties contain important structural information. For commutative C^*-algebras, the Gelfand transform will later become an isometric $*$-isomorphism.

5.1. **Characters and Maximal Ideals.** We begin by introducing the one-dimensional representations of a Banach algebra, usually called characters. These play a central role in the study of commutative Banach algebras and will later be used to define the Gelfand transform.

Definition 5.1. Let A be a complex Banach algebra. A **character** on A is a nonzero complex-linear map $\omega\colon A \to \mathbb{C}$ that is multiplicative. That is,

$$\omega(ab) = \omega(a)\omega(b) \tag{5.1}$$

for all $a, b \in A$. The set of all characters on A is denoted by $\widehat{A}$ and is called the **character space** of A.

Characters on a unital Banach algebra are automatically unital and continuous. Their values are also constrained by the spectra of the corresponding elements. Proposition 5.2 makes these statements precise.

Proposition 5.2. *Let A be a unital Banach algebra with identity e, and let $\omega \in \widehat{A}$. Then the following statements hold:*

(1) $\omega(e) = 1$.
(2) *If $a \in \mathrm{GL}(A)$, then $\omega(a) \neq 0$.*
(3) *For every $a \in A$, $\omega(a) \in \sigma_A(a)$.*
(4) *For every $a \in A$,*
$$|\omega(a)| \leq r(a) \leq \|a\|. \tag{5.2}$$
In particular, ω is continuous and $\|\omega\| \leq 1$. If $\|e\| = 1$, then $\|\omega\| = 1$.

Proof. The proof is given below:

(1) Since ω is nonzero, there exists $a \in A$ such that $\omega(a) \neq 0$. The multiplicativity of ω gives
$$\omega(a) = \omega(ae) = \omega(a)\omega(e). \tag{5.3}$$
Therefore, $\omega(e) = 1$.
(2) If $a \in \mathrm{GL}(A)$, then
$$\omega(a^{-1})\omega(a) = \omega(a^{-1}a) = \omega(e) = 1. \tag{5.4}$$
Hence, $\omega(a) \neq 0$.
(3) Now let $a \in A$. If $a - \omega(a)e$ were invertible, then the (2) would imply that $\omega(a - \omega(a)e) \neq 0$. On the other hand,
$$\omega(a - \omega(a)e) = \omega(a) - \omega(a)\omega(e) = 0, \tag{5.5}$$
which is a contradiction. Therefore, $a - \omega(a)e$ is not invertible, and hence $\omega(a) \in \sigma_A(a)$.
(4) By (3), we have
$$|\omega(a)| \leq r(a) \leq \|a\|. \tag{5.6}$$
Thus, ω is bounded and $\|\omega\| \leq 1$. If $\|e\| = 1$, then
$$1 = |\omega(e)| \leq \|\omega\|\|e\| = \|\omega\|, \tag{5.7}$$

so $\|\omega\| = 1$.

This completes the proof. □

In the commutative setting, characters are closely tied to maximal ideals and also recover the spectrum of each element. These relationships are made precise in Proposition 5.3.

Proposition 5.3. *Let A be a unital commutative Banach algebra. Then the following statements hold:*

(1) *The map*

$$\widehat{A} \longrightarrow \{M \subseteq A : M \text{ is a maximal ideal}\}, \tag{5.8}$$
$$\omega \longmapsto \ker \omega, \tag{5.9}$$

is a bijection.

(2) *For every $a \in A$, $\sigma_A(a) = \{\omega(a) : \omega \in \widehat{A}\}$.*

Proof. The proof is given below:

(1) Let $\omega \in \widehat{A}$. By Proposition 5.2, the map ω is continuous, so $\ker \omega$ is closed. Moreover, ω is surjective because $\omega(\lambda e) = \lambda$ for every $\lambda \in \mathbb{C}$. The first isomorphism theorem gives $A/\ker \omega \cong \mathbb{C}$. If I is an ideal satisfying

$$\ker \omega \subseteq I \subseteq A, \tag{5.10}$$

then $I/\ker \omega$ is a linear subspace of the one-dimensional vector space $A/\ker \omega$. Therefore, either

$$I/\ker \omega = \{0\}, \qquad \text{or} \qquad I/\ker \omega = A/\ker \omega. \tag{5.11}$$

Thus, either $I = \ker \omega$ or $I = A$, which proves that $\ker \omega$ is a maximal ideal. Conversely, let M be a maximal ideal of A. Since $\overline{M}$ is also an ideal and

$$M \subseteq \overline{M} \subseteq A, \tag{5.12}$$

maximality implies that either $\overline{M} = M$ or $\overline{M} = A$. Suppose that $\overline{M} = A$. Then $e \in \overline{M}$, so there exists $m \in M$ such that

$$\|e - m\| < 1. \tag{5.13}$$

By Lemma 3.6, $m = e - (e - m)$ is invertible. Since M is an ideal and $m \in M$, it follows that

$$e = m^{-1}m \in M. \tag{5.14}$$

This contradicts the assumption that M is proper. Therefore, $\overline{M} = M$, and hence M is closed. Consequently, A/M is a unital commutative Banach algebra. Since M is maximal, every nonzero element of A/M is invertible. Corollary 4.5 therefore gives an isomorphism

$$\psi \colon A/M \longrightarrow \mathbb{C}. \tag{5.15}$$

Let $q \colon A \to A/M$ denote the quotient map, and define $\omega := \psi \circ q$. Then ω is a character on A and $\ker \omega = M$. It remains to verify uniqueness. Suppose that $\omega_1, \omega_2 \in \widehat{A}$ satisfy

$$\ker \omega_1 = \ker \omega_2 = M. \tag{5.16}$$

For every $a \in A$,

$$a - \omega_1(a)e \in M. \tag{5.17}$$

Applying ω_2 gives $\omega_2(a) - \omega_1(a) = 0$. Thus, $\omega_1 = \omega_2$.

(2) By Proposition 5.2,

$$\{\omega(a) : \omega \in \widehat{A}\} \subseteq \sigma_A(a). \tag{5.18}$$

Conversely, let $\lambda \in \sigma_A(a)$. Then $a - \lambda e$ is not invertible. Since A is commutative, the ideal generated by $a - \lambda e$ is proper. It is therefore contained in a maximal ideal M. By (1), there exists $\omega \in \widehat{A}$ such that $M = \ker \omega$. Since $a - \lambda e \in M$, we obtain

$$0 = \omega(a - \lambda e) = \omega(a) - \lambda. \tag{5.19}$$

Thus, $\omega(a) = \lambda$, and hence

$$\sigma_A(a) \subseteq \{\omega(a) : \omega \in \widehat{A}\}. \tag{5.20}$$

This proves the desired equality.

This completes the proof. □

Remark 5.4. If A is a nonzero unital commutative Banach algebra, then $\widehat{A}$ is nonempty. Indeed, Proposition 5.3 shows that every spectral value of any element of A is obtained by evaluating that element at a character. Commutativity is essential for this argument. For example, $\widehat{M_2(\mathbb{C})} = \varnothing$. Indeed, the kernel of a character on $M_2(\mathbb{C})$ would be a proper two-sided ideal. Since $M_2(\mathbb{C})$ is simple, such a character would have to be injective, which is impossible for a complex-linear map from the four-dimensional space $M_2(\mathbb{C})$ into $\mathbb{C}$.

By Proposition 5.2, every character on a unital Banach algebra belongs to the dual space A^*. We equip $\widehat{A}$ with the relative weak-$*$ topology inherited from A^*. Thus, a net (ω_α) converges to ω in $\widehat{A}$ if and only if $\omega_\alpha(a) \to \omega(a)$ for every $a \in A$. Proposition 5.5 shows that $\widehat{A}$ is a compact Hausdorff space with respect to the relative weak-$*$ topology inherited from A^*.

Proposition 5.5. *Let A be a unital commutative Banach algebra. Then $\widehat{A}$ is a compact Hausdorff space with respect to the relative weak-$*$ topology.*

Proof. Let

$$B_{A^*} := \{\phi \in A^* : \|\phi\| \leq 1\} \tag{5.21}$$

denote the closed unit ball of A^*. By Proposition 5.2,

$$\widehat{A} = \{\phi \in B_{A^*} : \phi(e) = 1 \text{ and } \phi(ab) = \phi(a)\phi(b) \text{ for all } a, b \in A\}. \tag{5.22}$$

For each fixed $x \in A$, the evaluation map

$$\mathrm{ev}_x \colon A^* \longrightarrow \mathbb{C}, \tag{5.23}$$
$$\phi \longmapsto \phi(x), \tag{5.24}$$

is weak-$*$ continuous by the definition of the weak-$*$ topology. In particular, the maps

$$\phi \longmapsto \phi(e), \qquad \phi \longmapsto \phi(a), \qquad \phi \longmapsto \phi(b), \qquad \phi \longmapsto \phi(ab) \tag{5.25}$$

are weak-$*$ continuous for every fixed $a, b \in A$. Since addition and multiplication are continuous on $\mathbb{C}$, the map

$$\mathbb{C}^3 \longrightarrow \mathbb{C}, \tag{5.26}$$
$$(z_1, z_2, z_3) \longmapsto z_1 - z_2 z_3, \tag{5.27}$$

is continuous. Therefore, the map

$$\phi \longmapsto \phi(ab) - \phi(a)\phi(b) \tag{5.28}$$

is weak-$*$ continuous as a composition of continuous maps. It follows that $\widehat{A}$ is weak-$*$ closed in B_{A^*}. Indeed,

$$\widehat{A} = \{\phi \in B_{A^*} : \phi(e) = 1\} \cap \bigcap_{a,b \in A} \{\phi \in B_{A^*} : \phi(ab) - \phi(a)\phi(b) = 0\}. \tag{5.29}$$

The first set is the inverse image of the closed set $\{1\} \subseteq \mathbb{C}$ under the weak-$*$ continuous map $\phi \mapsto \phi(e)$. For every fixed $a, b \in A$, the corresponding set in the intersection is the inverse image of the closed set $\{0\} \subseteq \mathbb{C}$ under the weak-$*$ continuous map $\phi \mapsto \phi(ab) - \phi(a)\phi(b)$. Thus, every set appearing in the intersection is weak-$*$ closed. By the Banach–Alaoglu theorem, B_{A^*} is weak-$*$ compact. Therefore, $\widehat{A}$ is weak-$*$ compact. The weak-$*$ topology on A^* is Hausdorff, so the relative topology on $\widehat{A}$ is also Hausdorff. □

If A is separable, then the weak-$*$ topology on B_{A^*} is metrizable. In this case, the weak-$*$ closedness of $\widehat{A}$ may equivalently be verified using sequences. Without separability, one must use nets or the closed-condition argument in the proof of Proposition 5.5.

5.2. **The Gelfand Transform.** The character space provides a way to represent elements of a commutative Banach algebra as continuous functions. This leads to the Gelfand transform, which associates to each $a \in A$ a function on $\widehat{A}$ obtained by evaluating characters at a.

Definition 5.6. Let A be a unital commutative Banach algebra. For each $a \in A$, define

$$\Gamma(a) \colon \widehat{A} \longrightarrow \mathbb{C}, \tag{5.30}$$

$$\omega \longmapsto \omega(a). \tag{5.31}$$

The map

$$\Gamma \colon A \longrightarrow C(\widehat{A}), \tag{5.32}$$

$$a \longmapsto \Gamma(a), \tag{5.33}$$

is called the **Gelfand transform**.

For each $a \in A$, the function $\Gamma(a)$ is continuous by the definition of the relative weak-$*$ topology. The kernel of the Gelfand transform is described by the Jacobson radical.

Definition 5.7. Let A be a unital commutative Banach algebra. The **Jacobson radical** of A is

$$\operatorname{rad}(A) := \bigcap_{\omega \in \widehat{A}} \ker \omega. \tag{5.34}$$

Equivalently, $\operatorname{rad}(A)$ is the intersection of all maximal ideals of A. The algebra A is called **semisimple** if $\operatorname{rad}(A) = \{0\}$.

Proposition 5.8 establishes some basic properties of the Gelfand transform.

Proposition 5.8. *Let A be a unital commutative Banach algebra, and let $\Gamma \colon A \to C(\widehat{A})$ denote its Gelfand transform. Then the following statements hold:*

1. *The Gelfand transform is a unital algebra homomorphism.*
2. *The image $\Gamma(A)$ separates points of $\widehat{A}$.*
3. *For every $a \in A$,*
$$\sigma_A(a) = \{\Gamma(a)(\omega) : \omega \in \widehat{A}\}. \tag{5.35}$$
4. *For every $a \in A$,*
$$\|\Gamma(a)\|_\infty = r(a) \le \|a\|. \tag{5.36}$$
In particular, Γ is contractive.

(5) *An element* $a \in A$ *is invertible if and only if* $\Gamma(a)$ *does not vanish on* $\widehat{A}$.
(6) $\ker \Gamma = \mathrm{rad}(A)$. *Consequently,* Γ *is injective if and only if* A *is semisimple.*

Proof. The proof is given below:

(1) Let $a, b \in A$, $\lambda \in \mathbb{C}$, and $\omega \in \widehat{A}$. By the linearity and multiplicativity of ω,
$$\Gamma(a+b)(\omega) = \Gamma(a)(\omega) + \Gamma(b)(\omega), \tag{5.37}$$
$$\Gamma(\lambda a)(\omega) = \lambda\Gamma(a)(\omega), \tag{5.38}$$
$$\Gamma(ab)(\omega) = \Gamma(a)(\omega)\Gamma(b)(\omega), \tag{5.39}$$
$$\Gamma(e)(\omega) = 1. \tag{5.40}$$
Thus, Γ is a unital algebra homomorphism.
(2) Suppose that $\omega_1, \omega_2 \in \widehat{A}$ and $\omega_1 \neq \omega_2$. Since they are distinct linear functionals, there exists $a \in A$ such that $\omega_1(a) \neq \omega_2(a)$. Therefore,
$$\Gamma(a)(\omega_1) \neq \Gamma(a)(\omega_2), \tag{5.41}$$
so $\Gamma(A)$ separates points of $\widehat{A}$.
(3) By Proposition 5.3, we have
$$\sigma_A(a) = \{\omega(a) : \omega \in \widehat{A}\} = \{\Gamma(a)(\omega) : \omega \in \widehat{A}\}. \tag{5.42}$$
(4) By (3), we have
$$\|\Gamma(a)\|_\infty = \sup_{\omega \in \widehat{A}} |\omega(a)| = \max_{\lambda \in \sigma_A(a)} |\lambda| = r(a) \leq \|a\|. \tag{5.43}$$
(5) We have,
$$a \in \mathrm{GL}(A) \iff 0 \notin \sigma_A(a) \tag{5.44}$$
$$\iff \omega(a) \neq 0 \text{ for every } \omega \in \widehat{A} \tag{5.45}$$
$$\iff \Gamma(a) \text{ does not vanish on } \widehat{A}. \tag{5.46}$$
(6) We have,
$$\ker \Gamma = \{a \in A : \omega(a) = 0 \text{ for every } \omega \in \widehat{A}\} = \bigcap_{\omega \in \widehat{A}} \ker \omega = \mathrm{rad}(A). \tag{5.47}$$

This completes the proof. □

Example 5.9 shows that the Gelfand transform of a commutative Banach algebra need not be injective.

Example 5.9. Consider the algebra $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$ equipped with the norm $\|\lambda + \mu\varepsilon\| = |\lambda| + |\mu|$. This is a two-dimensional unital commutative Banach algebra. If $\omega \in \widehat{A}$, then
$$\omega(\varepsilon)^2 = \omega(\varepsilon^2) = 0, \tag{5.48}$$
and hence $\omega(\varepsilon) = 0$. Since $\omega(e) = 1$, the only character on A is $\omega(\lambda + \mu\varepsilon) = \lambda$. Thus, $\widehat{A}$ consists of one point and
$$\ker \Gamma = \mathbb{C}\varepsilon = \mathrm{rad}(A). \tag{5.49}$$
In particular, the nonzero element ε is not detected by the Gelfand transform.

The fundamental example of Gelfand theory is the algebra of continuous functions on a compact Hausdorff space.

Example 5.10. Let X be a compact Hausdorff space. For each $x \in X$, define the evaluation character

$$\mathrm{ev}_x \colon C(X) \longrightarrow \mathbb{C}, \tag{5.50}$$
$$f \longmapsto f(x). \tag{5.51}$$

Then the map

$$\varphi \colon X \longrightarrow \widehat{C(X)}, \tag{5.52}$$
$$x \longmapsto \mathrm{ev}_x, \tag{5.53}$$

is a homeomorphism. To prove surjectivity, let

$$I_x := \{f \in C(X) : f(x) = 0\} = \ker \mathrm{ev}_x\,. \tag{5.54}$$

Since ev_x is surjective, I_x is a maximal ideal. Let M be an arbitrary maximal ideal of $C(X)$. We claim that there exists $x \in X$ such that

$$M \subseteq I_x. \tag{5.55}$$

Suppose otherwise. Then, for every $x \in X$, there exists $f_x \in M$ such that $f_x(x) \neq 0$. By continuity, there exists an open neighborhood U_x of x on which f_x does not vanish. Compactness gives points $x_1, \dots, x_n \in X$ such that

$$X = U_{x_1} \cup \cdots \cup U_{x_n}. \tag{5.56}$$

Define $g := \sum_{j=1}^n \overline{f_{x_j}} f_{x_j}$. Since M is an ideal, $g \in M$. The construction of the sets U_{x_j} implies that $g(x) > 0$ for every $x \in X$. Since X is compact, g is bounded away from zero and is therefore invertible in $C(X)$. This contradicts the properness of M. Hence, $M \subseteq I_x$ for some $x \in X$. Since both M and I_x are maximal, $M = I_x$. By Proposition 5.3, every character ω has the form $\ker \omega = I_x$ for some $x \in X$. For each $f \in C(X)$,

$$f - f(x)1_X \in I_x = \ker \omega, \tag{5.57}$$

and hence $\omega(f) = f(x)$. Thus, $\omega = \mathrm{ev}_x$, proving that φ is surjective. If $x, y \in X$ and $x \neq y$, then there exists $f \in C(X)$ such that $f(x) \neq f(y)$. Therefore, $\mathrm{ev}_x \neq \mathrm{ev}_y$, so φ is injective. For each $f \in C(X)$, the map

$$x \longmapsto \mathrm{ev}_x(f) = f(x) \tag{5.58}$$

is continuous. By the definition of the weak-$*$ topology, φ is continuous. Since X is compact and $\widehat{C(X)}$ is Hausdorff, the continuous bijection φ is a homeomorphism. Therefore,

$$\widehat{C(X)} \cong X. \tag{5.59}$$

Under this identification, the Gelfand transform of $f \in C(X)$ is the function f itself because $\Gamma(f)(\mathrm{ev}_x) = f(x)$.

We conclude this section by considering the nonunital case. The character space of a nonunital commutative Banach algebra is most naturally studied by passing to its unitization.

Proposition 5.11. *Let A be a nonunital commutative Banach algebra, and let A' be its unitization. Then the following statements hold:*

(1) *Every character on A is continuous and has norm at most one and every $\omega \in \widehat{A}$ has a unique extension*

$$\widetilde{\omega} \colon A' \longrightarrow \mathbb{C}, \tag{5.60}$$
$$(a, \lambda) \longmapsto \omega(a) + \lambda. \tag{5.61}$$

Moreover,

$$\widehat{A'} = \{\widetilde{\omega} : \omega \in \widehat{A}\} \cup \{\phi_\infty\}, \tag{5.62}$$

where $\phi_\infty(a, \lambda) = \lambda$.

(2) *The character space* $\widehat{A}$ *is locally compact and Hausdorff and* $\widehat{A'}$ *is the one-point compactification of* $\widehat{A}$.

(3) *The Gelfand transform is a contractive algebra homomorphism* $\Gamma\colon A \longrightarrow C_0(\widehat{A})$.

(4) *For every* $a \in A$,

$$\sigma_A(a) = \{\omega(a) : \omega \in \widehat{A}\} \cup \{0\}, \tag{5.63}$$

where the spectrum of a *is computed in the unitization* A'.

Proof. The proof is given below:

(1) Let $\omega \in \widehat{A}$, and define

$$\widetilde{\omega}(a, \lambda) = \omega(a) + \lambda. \tag{5.64}$$

Using the multiplication on A', we obtain

$$\widetilde{\omega}((a, \lambda)(b, \mu)) = \widetilde{\omega}(ab + \lambda b + \mu a, \lambda\mu) \tag{5.65}$$
$$= \omega(a)\omega(b) + \lambda\omega(b) + \mu\omega(a) + \lambda\mu \tag{5.66}$$
$$= (\omega(a) + \lambda)(\omega(b) + \mu) \tag{5.67}$$
$$= \widetilde{\omega}(a, \lambda)\widetilde{\omega}(b, \mu). \tag{5.68}$$

Thus, $\widetilde{\omega}$ is a character on the unital Banach algebra A'. By Proposition 5.2, it is continuous and has norm at most one. Its restriction to A is ω, so ω is also continuous and has norm at most one. The extension is unique because every unital extension must satisfy

$$\widetilde{\omega}(a, \lambda) = \widetilde{\omega}(a, 0) + \lambda\widetilde{\omega}(0, 1) = \omega(a) + \lambda. \tag{5.69}$$

Now let $\phi \in \widehat{A'}$. If the restriction of ϕ to A is nonzero, then it is a character $\omega \in \widehat{A}$, and the uniqueness of the extension gives $\phi = \widetilde{\omega}$. If the restriction of ϕ to A is zero, then $\phi(a, \lambda) = \lambda$, so $\phi = \phi_\infty$. Therefore,

$$\widehat{A'} = \{\widetilde{\omega} : \omega \in \widehat{A}\} \cup \{\phi_\infty\}. \tag{5.70}$$

(2) The map

$$\widehat{A} \longrightarrow \widehat{A'} \setminus \{\phi_\infty\}, \tag{5.71}$$
$$\omega \longmapsto \widetilde{\omega}, \tag{5.72}$$

is a homeomorphism with respect to the relative weak-$*$ topologies. Since $\widehat{A'}$ is compact Hausdorff, its open subspace $\widehat{A'} \setminus \{\phi_\infty\}$ is locally compact and Hausdorff. Under this identification, $\widehat{A'}$ is the one-point compactification of $\widehat{A}$.

(3) For $a \in A$, the Gelfand transform of $(a, 0) \in A'$ is continuous on $\widehat{A'}$ and satisfies

$$\Gamma_{A'}(a, 0)(\phi_\infty) = 0. \tag{5.73}$$

Its restriction to $\widehat{A}$ therefore vanishes at infinity. Hence, $\Gamma(a) \in C_0(\widehat{A})$. The linearity and multiplicativity of Γ follow from those of the characters, and $\|\Gamma(a)\|_\infty \le \|a\|$.

(4) Applying Proposition 5.3 to the unital commutative Banach algebra A' gives

$$\sigma_A(a) := \sigma_{A'}(a, 0) \tag{5.74}$$
$$= \{\phi(a, 0) : \phi \in \widehat{A'}\} \tag{5.75}$$
$$= \{\omega(a) : \omega \in \widehat{A}\} \cup \{0\}. \tag{5.76}$$

This completes the proof. □

Unlike the unital case, the character space of a nonunital commutative Banach algebra may be empty. For example, let A be a nonzero Banach space equipped with the zero multiplication. If ω were a character and $\omega(a) \neq 0$, then $0 = \omega(a^2) = \omega(a)^2$, which is impossible. Thus, $\widehat{A} = \varnothing$.

Example 5.12. Let X be a locally compact Hausdorff space. For every $x \in X$, define

$$\mathrm{ev}_x \colon C_0(X) \longrightarrow \mathbb{C}, \tag{5.77}$$
$$f \longmapsto f(x). \tag{5.78}$$

Then the map

$$\varphi \colon X \longrightarrow \widehat{C_0(X)}, \tag{5.79}$$
$$x \longmapsto \mathrm{ev}_x, \tag{5.80}$$

is a homeomorphism. If X is compact, then $C_0(X) = C(X)$, and the result follows from Example 5.10. Suppose that X is noncompact. By Proposition 2.5, The unitization of $C_0(X)$ is isomorphic as a unital Banach algebra to $C(X_\infty)$, where X_∞ is the one-point compactification of X. By Example 5.10,

$$\widehat{C(X_\infty)} \cong X_\infty. \tag{5.81}$$

Under this identification, evaluation at ∞ restricts to the zero functional on $C_0(X)$. Every other evaluation character restricts to ev_x for some $x \in X$. It follows from Proposition 5.11 that every character on $C_0(X)$ is of this form and that

$$\widehat{C_0(X)} \cong X \tag{5.82}$$

as locally compact Hausdorff spaces. Under this identification, the Gelfand transform of $f \in C_0(X)$ is the function f itself because $\Gamma(f)(\mathrm{ev}_x) = f(x)$.

Example 5.10 and Example 5.12 anticipate the commutative Gelfand–Naimark theorem, which states that if A is a unital commutative C^*-algebra, then the Gelfand transform is an isometric $*$-isomorphism

$$\Gamma \colon A \longrightarrow C(\widehat{A}). \tag{5.83}$$

For a nonunital commutative C^*-algebra, it is an isometric $*$-isomorphism

$$\Gamma \colon A \longrightarrow C_0(\widehat{A}). \tag{5.84}$$

These results form the basis of Gelfand duality for commutative C^*-algebras.

Part 2. C^*-Algebras I

Linear algebra studies linear operators on finite dimensional vector spaces, such as maps $T: \mathbb{C}^n \to \mathbb{C}^n$. If $n \geq 2$ and $T, S: \mathbb{C}^n \to \mathbb{C}^n$ are two such operators, then $T \circ S \neq S \circ T$ in general. This is often one of the first familiar examples of noncommutativity in mathematics. More generally, functional analysis studies infinite dimensional vector spaces equipped with additional analytic structures, together with the linear operators acting on them. A fundamental example is a Hilbert space $\mathscr{H}$. If $\mathscr{H}$ is an infinite dimensional Hilbert space, we consider the Banach space of bounded linear operators on $\mathscr{H}$.

$$\mathscr{B}(\mathscr{H}) := \{T : H \to H \mid T \text{ is linear and bounded}\}. \tag{5.85}$$

The phenomenon of noncommutativity appears throughout mathematics and physics, including quantum mechanics, linear algebra, and the representation theory of groups. The theory of operator algebras, including the theory of C^*-algebras, provides an analytic framework for studying this phenomenon. In their first paper on operator algebras in 1936, Francis Murray and John von Neumann wrote[2]

"various aspects of the quantum mechanical formalism suggest strongly the elucidation of this subject."

Hence, the study of operator algebras can be considered an essential part of *non-commutative mathematics*, which is also called *quantum mathematics*[3] more colloquially and popularly.

Why should one study C^*-algebras in particular? One important reason is that C^*-algebras provide a noncommutative generalization of topology. This perspective is made precise by the commutative Gelfand–Naimark theorem in Theorem 8.1.

Theorem 5.13. *(Gelfand & Naimark) Let A be a C^*-algebra. Then A is commutative if and only if A is isometrically $*$-isomorphic to $C_0(X)$ for some locally compact Hausdorff topological space, X.*

Corollary 5.14. *Let A be a unital C^*-algebra. Then A is commutative if and only if A is isometrically $*$-isomorphic to $C(X)$ for some compact Hausdorff topological space, X.*

Thus, every locally compact Hausdorff space gives rise to a commutative C^*-algebra, and every commutative C^*-algebra arises in this way. In this sense, commutative C^*-algebras provide an algebraic description of ordinary topology, while noncommutative C^*-algebras may be viewed as algebras of functions on hypothetical noncommutative spaces. This perspective, commonly called *noncommutative topology*, forms part of a broader collection of analogies between classical, noncommutative, and quantum mathematics. Some of these analogies are summarized in Part 2.

Classical Mathematics	Quantum Mathematics
Topology	C^*-algebras
Differential Geometry	Non Commutative Geometry
Probability Theory	Free Probability
Groups	Quantum groups
Cohomology	Quantum Cohomology

[2]The paper concerned the algebras now known as von Neumann algebras.

[3]Thus, phrases such as *quantum groups*, *quantum cohomology*, etc., are used to describe objects studied in *quantum mathematics*.

6. Definitions and Examples

In this section, we introduce the definition of a C^*-algebra, establish some of its basic properties, and discuss several examples.

6.1. **$*$-Algebras.** We first introduce algebras equipped with an involution. The involution on a $*$-algebra abstracts the operation of taking adjoints of complex matrices and bounded operators on a Hilbert space.

Definition 6.1. Let A be a complex algebra. Let $a, b \in A$ and $\lambda \in \mathbb{C}$. An **involution** on A is a map

$$* \colon A \longrightarrow A, \tag{6.1}$$

$$a \longmapsto a^*, \tag{6.2}$$

satisfying the following properties:

(1) $(a+b)^* = a^* + b^*$.
(2) $(ab)^* = b^* a^*$.
(3) $(a^*)^* = a$.
(4) $(\lambda a)^* = \overline{\lambda} a^*$.

An algebra equipped with an involution is called a **$*$-algebra**.

Thus, an involution is conjugate-linear, anti-multiplicative, and involutive. Equipping a Banach algebra with a continuous involution leads naturally to the notion of a Banach $*$-algebra.

Definition 6.2. A **Banach $*$-algebra** is a Banach algebra equipped with a continuous involution. A **morphism of Banach $*$-algebras** is a continuous complex-linear multiplicative map $\phi\colon \to B$ that preserves the involution. That is,

$$\phi(a^*) = \phi(a)^* \tag{6.3}$$

for every $a \in A$. If A and B are unital, then ϕ is called unital if

$$\phi(e_A) = e_B. \tag{6.4}$$

The involution gives rise to several natural classes of elements that play an important role in the structure of a $*$-algebra. We record the basic definitions in Definition 6.3.

Definition 6.3. Let A be a $*$-algebra, and let $a \in A$.

(1) The element a is **normal** if $a^* a = a a^*$.
(2) The element a is **self-adjoint** if $a = a^*$.
(3) The element a is a **projection** if $a = a^* = a^2$.
(4) If A is unital, then a is **unitary** if $a^* a = a a^* = e$.
(5) If A is unital, then a is an **isometry** if $a^* a = e$.

Proposition 6.4 records several algebraic properties of the involution in the unital setting.

Proposition 6.4. *Let A be a unital $*$-algebra with identity e. Then the following statements hold:*

(1) *$e^* = e$.*
(2) *If $a \in \mathrm{GL}(A)$, then $a^* \in \mathrm{GL}(A)$ and $(a^*)^{-1} = (a^{-1})^*$.*
(3) *For every $a \in A$, $\sigma_A(a^*) = \overline{\sigma_A(a)}$.*

Proof. The proof is given below:

(1) For every $a \in A$,
$$a^* = (ea)^* = a^*e^*, \qquad a^* = (ae)^* = e^*a^*. \tag{6.5}$$
Since every element of A is of the form a^*[4], the element e^* is both a left and right identity for A. The uniqueness of the identity therefore gives $e^* = e$.

(2) Suppose that $a \in \mathrm{GL}(A)$. Then
$$a^*(a^{-1})^* = (a^{-1}a)^* = e^* = e, \tag{6.6}$$
$$(a^{-1})^*a^* = (aa^{-1})^* = e^* = e. \tag{6.7}$$
Thus, a^* is invertible and $(a^*)^{-1} = (a^{-1})^*$.

(3) For every $\lambda \in \mathbb{C}$,
$$a^* - \lambda e = (a - \overline{\lambda} e)^*. \tag{6.8}$$
By (2) $a^* - \lambda e$ is invertible if and only if $a - \overline{\lambda} e$ is invertible. Therefore,
$$\lambda \in \sigma_A(a^*) \iff \overline{\lambda} \in \sigma_A(a), \tag{6.9}$$
which proves that $\sigma_A(a^*) = \overline{\sigma_A(a)}$.

This completes the proof. □

6.2. **The C^*-Identity.** The fundamental example of a noncommutative Banach algebra is $\mathscr{B}(\mathscr{H})$, where $\mathscr{H}$ is a Hilbert space. This algebra has additional structure arising from the adjoint operation. For $T \in \mathscr{B}(\mathscr{H})$, the adjoint of T is the unique operator $T^* \in \mathscr{B}(\mathscr{H})$ satisfying
$$\langle \xi, T^*\eta \rangle = \langle T\xi, \eta \rangle \tag{6.10}$$
for all $\xi, \eta \in \mathscr{H}$, where the inner product is taken to be linear in the first variable. For all $T, S \in \mathscr{B}(\mathscr{H})$ and $\lambda \in \mathbb{C}$, the adjoint satisfies
$$(T+S)^* = T^* + S^*, \tag{6.11}$$
$$(TS)^* = S^*T^*, \tag{6.12}$$
$$T^{**} = T, \tag{6.13}$$
$$(\lambda T)^* = \overline{\lambda} T^*. \tag{6.14}$$
Thus, the adjoint operation is an involution on $\mathscr{B}(\mathscr{H})$. We now examine how the adjoint interacts with the operator norm. For every $\xi \in \mathscr{H}$, the Cauchy–Schwarz inequality gives
$$\|T\xi\|^2 = \langle T\xi, T\xi \rangle \tag{6.15}$$
$$= \langle \xi, T^*T\xi \rangle \tag{6.16}$$
$$\leq \|\xi\| \|T^*T\xi\| \tag{6.17}$$
$$\leq \|T^*T\| \|\xi\|^2. \tag{6.18}$$
Taking the supremum over all $\xi \in \mathscr{H}$ satisfying $\|\xi\| = 1$ gives
$$\|T\|^2 \leq \|T^*T\| \leq \|T^*\| \|T\|. \tag{6.19}$$
If $T \neq 0$, dividing by $\|T\|$ gives $\|T\| \leq \|T^*\|$. Applying the same inequality to T^* and using $T^{**} = T$ gives $\|T^*\| \leq \|T\|$. The equality is immediate when $T = 0$. Consequently, $\|T^*\| = \|T\|$ for every $T \in \mathscr{B}(\mathscr{H})$. Substituting this equality into Equation (6.19) yields
$$\|T^*T\| = \|T\|^2. \tag{6.20}$$

[4]The involution is surjective because it is involutive. Indeed, for every $x \in A$, taking $a = x^*$ gives $a^* = (x^*)^* = x$.

This fundamental relation is called the C^*-identity and motivates the definition of a C^*-algebra in Definition 6.5.

Definition 6.5. A **C^*-algebra** is a Banach $*$-algebra A satisfying the C^*-identity

$$\|a^*a\| = \|a\|^2 \tag{6.21}$$

for every $a \in A$. A **$*$-homomorphism** between C^*-algebras is a complex-linear multiplicative map $\phi\colon A \to B$ satisfying $\phi(a^*) = \phi(a)^*$ for every $a \in A$.

No continuity assumption is required in Definition 6.5. In fact, we will prove in Proposition 6.13 that every $*$-homomorphism between C^*-algebras is automatically contractive and hence continuous. Lemma 6.6 gives an equivalent formulation of the C^*-identity.

Lemma 6.6. *Let A be a Banach $*$-algebra. Then A is a C^*-algebra if and only if*

$$\|a\|^2 \leq \|a^*a\| \tag{6.22}$$

for every $a \in A$.

Proof. If A is a C^*-algebra, then the assertion follows immediately from the C^*-identity. Conversely, suppose that $\|a\|^2 \leq \|a^*a\|$ for every $a \in A$. By submultiplicativity,

$$\|a\|^2 \leq \|a^*a\| \leq \|a^*\|\|a\|. \tag{6.23}$$

If $a \neq 0$, it follows that $\|a\| \leq \|a^*\|$. Applying the same argument to a^* gives $\|a^*\| \leq \|a^{**}\| = \|a\|$. Thus, $\|a^*\| = \|a\|$. Consequently, we have

$$\|a\|^2 \leq \|a^*a\| \leq \|a^*\|\|a\| = \|a\|^2. \tag{6.24}$$

Therefore, $\|a^*a\| = \|a\|^2$, so A is a C^*-algebra. □

Closed $*$-subalgebras provide a natural way to pass to smaller C^*-algebras while retaining the inherited norm and involution. This leads to the notion of C^*-subalgebra of a C^*-algebra A, which is a norm-closed $*$-subalgebra of A. It is not required to contain the identity of A.

Definition 6.7. Let A be a C^*-algebra, and let $F \subseteq A$. The **C^*-subalgebra generated by** F, denoted by $C^*(F)$, is the smallest C^*-subalgebra of A containing F. Equivalently,

$$C^*(F) = \bigcap\{B \subseteq A : B \text{ is a } C^*\text{-subalgebra containing } F\}. \tag{6.25}$$

The generated C^*-subalgebra can be described in terms of words in the elements of F and their adjoints. We make this precise in Proposition 6.8.

Proposition 6.8. *Let A be a C^*-algebra, and let $F \subseteq A$. Define $F^* := \{x^* : x \in F\}$ and, for every $n \geq 1$, define*

$$W_n := \{x_1x_2\cdots x_n : x_j \in F \cup F^*\}. \tag{6.26}$$

If $W := \bigcup_{n=1}^{\infty} W_n$, then $C^*(F) = \overline{\operatorname{span} W}$. If A is unital, then the unital C^*-subalgebra generated by F is

$$C^*(F, e) = \overline{\operatorname{span}(\{e\} \cup W)}. \tag{6.27}$$

Proof. The set W is closed under multiplication and involution. Therefore, $\operatorname{span} W$ is a $*$-subalgebra of A containing F, and its norm closure is a C^*-subalgebra of A containing F. Conversely, every C^*-subalgebra of A containing F must contain F^*, every word in $F \cup F^*$, the linear span of those words, and its norm closure. Hence, every C^*-subalgebra containing F contains $\overline{\operatorname{span} W}$. The first identity follows. The proof of the unital statement is analogous. □

Remark 6.9. For elements $a_1, \dots, a_n \in A$, we write

$$C^*(a_1, \dots, a_n) := C^*(\{a_1, \dots, a_n\}). \tag{6.28}$$

Example 6.10 records several basic examples and constructions of C^*-algebras.

Example 6.10. The following are basic examples of C^*-algebras:

(1) The complex numbers form a unital commutative C^*-algebra with involution $\lambda^* = \overline{\lambda}$.
(2) The matrix algebra $M_n(\mathbb{C})$ is a unital C^*-algebra with respect to the operator norm and the involution $A \mapsto A^*$, where A^* denotes the conjugate transpose of A.
(3) For every Hilbert space $\mathscr{H}$, the algebra $\mathscr{B}(\mathscr{H})$ is a unital C^*-algebra with respect to the adjoint operation.
(4) Let X be a locally compact Hausdorff space. The algebra $C_0(X)$ is a commutative C^*-algebra with involution
$$f^*(x) = \overline{f(x)}. \tag{6.29}$$
Indeed,
$$\|f^*f\|_\infty = \sup_{x\in X} |f(x)|^2 = \|f\|_\infty^2. \tag{6.30}$$
The algebra $C_0(X)$ is unital if and only if X is compact.
(5) Let X be a topological space. The algebra $C_b(X)$ of bounded continuous complex-valued functions on X is a unital commutative C^*-algebra with pointwise operations, the supremum norm, and pointwise complex conjugation.
(6) For every Hilbert space $\mathscr{H}$, the space $\mathscr{K}(\mathscr{H})$ of compact operators on $\mathscr{H}$ is a norm-closed two-sided ideal of $\mathscr{B}(\mathscr{H})$ and hence a C^*-algebra. It is unital if and only if $\mathscr{H}$ is finite dimensional.
(7) If A and B are C^*-algebras, then their direct sum $A \oplus B$, equipped with the operations $(a, b)^* = (a^*, b^*)$ and the norm
$$\|(a, b)\| = \max\{\|a\|, \|b\|\}, \tag{6.31}$$
is a C^*-algebra.
(8) Every norm-closed $*$-subalgebra of a C^*-algebra is a C^*-algebra with respect to the inherited norm and involution.

We conclude this discussion in Example 6.11 with an example showing that an isometric Banach $*$-algebra need not satisfy the C^*-identity.

Example 6.11. Let $\mathscr{A}(\mathbb{D})$ denote the disk algebra consisting of the continuous functions on the closed unit disk $\mathbb{D}$ that are holomorphic on its interior. Define
$$f^*(z) := \overline{f(\overline{z})}. \tag{6.32}$$
This operation preserves $\mathscr{A}(\mathbb{D})$ and defines an isometric involution because
$$\|f^*\|_\infty = \sup_{z\in\mathbb{D}} |f^*(z)| = \sup_{z\in\mathbb{D}} |f(\overline{z})| = \|f\|_\infty. \tag{6.33}$$
Thus, $\mathscr{A}(\mathbb{D})$ is an isometric Banach $*$-algebra. Consider the function $f(z) = e^{iz}$. Its adjoint is $f^*(z) = \overline{e^{i\overline{z}}} = e^{-iz}$. Therefore, $f^*(z)f(z) = 1$ for every $z \in \mathbb{D}$, and hence $\|f^*f\|_\infty = 1$. On the other hand,
$$\|f\|_\infty = \sup_{z\in\mathbb{D}} |e^{iz}| = \sup_{z\in\mathbb{D}} e^{-\operatorname{Im} z} = e. \tag{6.34}$$
Consequently,
$$\|f^*f\|_\infty = 1 \neq e^2 = \|f\|_\infty^2. \tag{6.35}$$
Thus, $\mathscr{A}(\mathbb{D})$ does not satisfy the C^*-identity and is not a C^*-algebra.

6.3. **Properties.** We now discuss several basic properties of C^*-algebras, beginning with consequences of the C^*-identity in Proposition 6.12.

Proposition 6.12. *Let A be a nonzero C^*-algebra. Then the following statements hold:*

(1) *For every $a \in A$, $\|a^*\| = \|a\|$.*
(2) *If A is unital with identity e, then $\|e\| = 1$.*
(3) *Every $a \in A$ can be written uniquely as $a = \mathrm{Re}(a) + i\,\mathrm{Im}(a)$, where*

$$\mathrm{Re}(a) := \frac{a+a^*}{2} \qquad \text{and} \qquad \mathrm{Im}(a) := \frac{a-a^*}{2i} \tag{6.36}$$

are self-adjoint.
(4) *If A is unital and $u \in A$ is unitary, then $\sigma_A(u) \subseteq \mathbb{S}^1$.*
(5) *If $a \in A$ is self-adjoint, then $r(a) = \|a\|$.*
(6) *If $a \in A$ is normal, then $r(a) = \|a\|$.*
(7) *If A is unital, then its C^*-norm is uniquely determined by its algebraic operations and involution.*

Proof. The proof is given below:

(1) By the C^*-identity and submultiplicativity,

$$\|a\|^2 = \|a^*a\| \leq \|a^*\|\|a\|. \tag{6.37}$$

If $a \neq 0$, it follows that $\|a\| \leq \|a^*\|$. Applying the same argument to a^* gives $\|a^*\| \leq \|a^{**}\| = \|a\|$. The equality is immediate when $a = 0$. Therefore, $\|a^*\| = \|a\|$.
(2) Suppose that A is unital. Since $e^* = e$, the C^*-identity gives

$$\|e\|^2 = \|e^*e\| = \|e\|. \tag{6.38}$$

Since A is nonzero, $e \neq 0$ and hence $\|e\| \neq 0$. Therefore, $\|e\| = 1$.
(3) For every $a \in A$, define

$$\mathrm{Re}(a) = \frac{a+a^*}{2} \qquad \text{and} \qquad \mathrm{Im}(a) = \frac{a-a^*}{2i}. \tag{6.39}$$

A direct computation gives $\mathrm{Re}(a)^* = \mathrm{Re}(a)$ and and $\mathrm{Im}(a)^* = \mathrm{Im}(a)$. $a = \mathrm{Re}(a) + i\,\mathrm{Im}(a)$. The uniqueness follows by comparing an expression with its adjoint.
(4) Suppose that u is unitary. Then $u^{-1} = u^*$. Moreover,

$$\|u\|^2 = \|u^*u\| = \|e\| = 1, \tag{6.40}$$

Hence, $\|u\| = \|u^{-1}\| = 1$. Let $\lambda \in \sigma_A(u)$. Since u is invertible, $\lambda \neq 0$, and

$$|\lambda| \leq r(u) \leq \|u\| = 1. \tag{6.41}$$

By Proposition 3.7(4), we have $\lambda^{-1} \in \sigma_A(u^{-1})$. Therefore,

$$|\lambda|^{-1} \leq \|u^{-1}\| = 1. \tag{6.42}$$

Thus, $|\lambda| = 1$, proving that $\sigma_A(u) \subseteq \mathbb{S}^1$.
(5) Suppose that a is self-adjoint. Then $\|a^2\| = \|a^*a\| = \|a\|^2$. Repeated application of this identity gives

$$\|a^{2^n}\| = \|a\|^{2^n} \tag{6.43}$$

for every $n \geq 1$. By Proposition 4.4, we have

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n} = \lim_{n\to\infty} \|a^{2^n}\|^{1/2^n} = \|a\|. \tag{6.44}$$

(6) Suppose that a is normal. The element a^*a is self-adjoint, and the normality of a gives

$$\|a^2\|^2 = \|(a^2)^*a^2\| = \|(a^*)^2a^2\| = \|(a^*a)^2\| = \|a^*a\|^2 = \|a\|^4. \tag{6.45}$$

Thus, $\|a^2\| = \|a\|^2$. Every power of a normal element is normal, so repeated application of this equality gives $\|a^{2^n}\| = \|a\|^{2^n}$. Proposition 4.4 therefore yields $r(a) = \|a\|$.

(7) Suppose that $\|\cdot\|_1$ and $\|\cdot\|_2$ are two complete C^*-norms on the same unital $*$-algebra A. Invertibility is determined by the algebraic structure, so the spectrum of each element is the same with respect to both norms. Since a^*a is self-adjoint,

$$\|a\|_1^2 = \|a^*a\|_1 = r(a^*a), \tag{6.46}$$

$$\|a\|_2^2 = \|a^*a\|_2 = r(a^*a). \tag{6.47}$$

Therefore, $\|a\|_1 = \|a\|_2$ for every $a \in A$.

This completes the proof. □

The C^*-identity imposes strong restrictions on maps between C^*-algebras. In particular, preserving the algebraic operations and the involution automatically implies continuity. We prove this in Proposition 6.13.

Proposition 6.13. *Let A and B be C^*-algebras.*

(1) *A complex-linear map $\phi\colon A \to B$ is $*$-preserving if and only if it maps self-adjoint elements of A to self-adjoint elements of B.*

(2) *Every $*$-homomorphism $\phi\colon A \to B$ is contractive. That is,*

$$\|\pi(a)\| \le \|a\| \tag{6.48}$$

for every $a \in A$. In particular, every $*$-homomorphism is continuous.

Proof. The proof is given below:

(1) If ϕ is $*$-preserving and $a = a^*$, then $\phi(a)^* = \phi(a^*) = \phi(a)$, so $\phi(a)$ is self-adjoint. Conversely, suppose that ϕ maps self-adjoint elements to self-adjoint elements. By Proposition 6.12(3), write $a = \mathrm{Re}(a) + i\,\mathrm{Im}(a)$. Then $a^* = \mathrm{Re}(a) - i\,\mathrm{Im}(a)$. Since $\phi(\mathrm{Re}(a))$ and $\phi(\mathrm{Im}(a))$ are self-adjoint,

$$\phi(a)^* = \phi(\mathrm{Re}(a))^* - i\phi(\mathrm{Im}(a))^* = \phi(\mathrm{Re}(a)) - i\phi(\mathrm{Im}(a)) = \phi(a^*). \tag{6.49}$$

Thus, ϕ is $*$-preserving.

(2) Let $\pi\colon A \to B$ be a $*$-homomorphism. Passing to the unitizations gives a unital algebra homomorphism

$$\pi'\colon A' \longrightarrow B', \tag{6.50}$$

$$(a, \lambda) \longmapsto (\pi(a), \lambda). \tag{6.51}$$

A unital algebra homomorphism preserves invertibility and therefore gives

$$\sigma_B(\pi(a^*a)) \subseteq \sigma_A(a^*a). \tag{6.52}$$

Since a^*a and $\pi(a^*a)$ are self-adjoint, Proposition 6.12 gives

$$\|\pi(a)\|^2 = \|\pi(a)^*\pi(a)\| = \|\pi(a^*a)\| = r(\pi(a^*a)) \le r(a^*a) = \|a^*a\| = \|a\|^2. \tag{6.53}$$

Therefore,

$$\|\pi(a)\| \le \|a\|. \tag{6.54}$$

This completes the proof. □

Kernels and images of $*$-homomorphisms provide fundamental constructions in the theory of C^*-algebras. Proposition 6.14 gives the algebraic form of the first isomorphism theorem for C^*-algebras.

Proposition 6.14. *Let $\pi\colon A \to B$ be a $*$-homomorphism between C^*-algebras. Then $\ker\pi$ is a norm-closed self-adjoint two-sided ideal of A, and $\pi(A)$ is a $*$-subalgebra of B. Moreover, there is a unique injective algebraic $*$-homomorphism*

$$\widetilde{\pi}\colon A/\ker\pi \longrightarrow B, \tag{6.55}$$

$$a + \ker\pi \longmapsto \pi(a). \tag{6.56}$$

Its image is $\pi(A)$.

Proof. Since π is continuous, $\ker\pi = \pi^{-1}(\{0\})$ is closed. The linearity and multiplicativity of π imply that $\ker\pi$ is a two-sided ideal. If $a \in \ker\pi$, then

$$\pi(a^*) = \pi(a)^* = 0, \tag{6.57}$$

so $a^* \in \ker\pi$. Hence, $\ker\pi$ is self-adjoint. The image $\pi(A)$ is closed under addition, scalar multiplication, multiplication, and involution, so it is a $*$-subalgebra of B. The existence and uniqueness of $\widetilde{\pi}$ follow from the algebraic first isomorphism theorem. $\square$

Characters on a unital C^*-algebra automatically preserve the involution and are positive on elements of the form a^*a. Proposition 6.15 establishes several basic properties of such characters.

Proposition 6.15. *Let A be a unital C^*-algebra, and let $\phi \in \widehat{A}$. Then the following statements hold:*

(1) *If $a \in A$ is self-adjoint, then $\phi(a) \in \mathbb{R}$.*
(2) *ϕ is $*$-preserving and $\|\phi\| = 1$.*
(3) *For every $a \in A$, $\phi(a^*a) = |\phi(a)|^2 \geq 0$.*
(4) *If $u \in A$ is unitary, then $|\phi(u)| = 1$.*

Proof. The proof is given below:

(1) Let $a = a^*$, and write $\phi(a) = \alpha + i\beta$, where $\alpha, \beta \in \mathbb{R}$. For every $t \in \mathbb{R}$,

$$\alpha^2 + (\beta + t)^2 = |\phi(a + ite)|^2 \tag{6.58}$$

$$\leq \|a + ite\|^2 \tag{6.59}$$

$$= \|(a + ite)^*(a + ite)\| \tag{6.60}$$

$$= \|(a - ite)(a + ite)\| \tag{6.61}$$

$$= \|a^2 + t^2 e\| \leq \|a\|^2 + t^2. \tag{6.62}$$

Therefore,

$$\alpha^2 + \beta^2 + 2\beta t \leq \|a\|^2 \tag{6.63}$$

for every $t \in \mathbb{R}$. This is possible only if $\beta = 0$. Hence, $\phi(a) \in \mathbb{R}$.

(2) For an arbitrary $a \in A$, write $a = \mathrm{Re}(a) + i\,\mathrm{Im}(a)$. Since $\phi(\mathrm{Re}(a))$ and $\phi(\mathrm{Im}(a))$ are real,

$$\phi(a^*) = \phi(\mathrm{Re}(a)) - i\phi(\mathrm{Im}(a)) = \overline{\phi(a)}. \tag{6.64}$$

Thus, ϕ is $*$-preserving. We already know that $\|\phi\| \leq 1$. Since $\phi(e) = 1$ and $\|e\| = 1$,

$$1 = |\phi(e)| \leq \|\phi\|, \tag{6.65}$$

so $\|\phi\| = 1$.

(3) We have

$$\phi(a^*a) = \phi(a^*)\phi(a) = \overline{\phi(a)}\phi(a) = |\phi(a)|^2 \geq 0. \tag{6.66}$$

(4) If u is unitary, we have

$$|\phi(u)|^2 = \overline{\phi(u)}\phi(u) = \phi(u^*)\phi(u) = \phi(u^*u) = \phi(e) = 1. \tag{6.67}$$

Therefore, $|\phi(u)| = 1$.

This completes the proof. □

7. Unitization

Not every C^*-algebra has a multiplicative identity. A standard example is $C_0(\mathbb{R})$, the C^*-algebra of continuous complex-valued functions on $\mathbb{R}$ that vanish at infinity, equipped with pointwise operations and the supremum norm. As in the Banach algebra setting, it is often useful to adjoin an identity to a nonunital C^*-algebra. The algebraic construction is the same, but the unitization must now be equipped with a norm that satisfies the C^*-identity.

Definition 7.1. Let A be a nonunital C^*-algebra. Define $A' := A \oplus \mathbb{C}$ with multiplication and involution given by

$$(a, \alpha)(b, \beta) = (ab + \alpha b + \beta a, \alpha\beta), \qquad (a, \alpha)^* = (a^*, \overline{\alpha}). \tag{7.1}$$

For $(a, \alpha) \in A'$, define

$$\|(a, \alpha)\|_{A'} := \sup_{\substack{b \in A \\ \|b\| \leq 1}} \|ab + \alpha b\|. \tag{7.2}$$

The resulting unital C^*-algebra is called the **unitization** of A.

The norm in Equation (7.2) is the operator norm obtained from the action of A' on the Banach space A by left multiplication. We now verify that it is a norm and that it satisfies the C^*-identity.

Proposition 7.2. *Let A be a nonunital C^*-algebra. Then A', equipped with the operations and norm in Definition 7.1, is a unital C^*-algebra with identity $e_{A'} = (0, 1)$. Moreover, the canonical map*

$$\iota_A \colon A \longrightarrow A', \tag{7.3}$$
$$a \longmapsto (a, 0), \tag{7.4}$$

is an isometric $*$-homomorphism whose image is a closed two-sided ideal of A'.

Proof. For each $a \in A$, let $L_a \colon A \to A$ denote the left multiplication operator. Define

$$\Theta \colon A' \longrightarrow \mathscr{B}(A), \tag{7.5}$$
$$(a, \alpha) \longmapsto L_a + \alpha I_A. \tag{7.6}$$

For $(a, \alpha), (b, \beta) \in A'$, we have

$$\Theta(a, \alpha)\Theta(b, \beta) = (L_a + \alpha I_A)(L_b + \beta I_A) = L_{ab+\alpha b+\beta a} + \alpha\beta I_A = \Theta((a, \alpha)(b, \beta)). \tag{7.7}$$

Thus, Θ is an algebra homomorphism. Moreover, $\|(a, \alpha)\|_{A'} = \|L_a + \alpha I_A\|$. We show that Θ is injective. Suppose that $L_a + \alpha I_A = 0$. If $\alpha = 0$, then $L_a = 0$. In particular, $aa^* = L_a(a^*) = 0$. The C^*-identity gives

$$\|a\|^2 = \|aa^*\| = 0, \tag{7.8}$$

so $a = 0$. Now suppose that $\alpha \neq 0$, and set

$$u := -\alpha^{-1}a. \tag{7.9}$$

Then $L_u = I_A$, so $ub = b$ for every $b \in A$. Taking adjoints gives $b^*u^* = b^*$ for every $b \in A$. Since the involution is surjective, every $c \in A$ can be written as $c = b^*$ for some $b \in A$. Hence, $cu^* = c$ for every $c \in A$. In particular,

$$uu^* = u \qquad \text{and} \qquad uu^* = u^*. \tag{7.10}$$

Hence, $u = u^*$, and therefore u is both a left and right identity for A. This contradicts the assumption that A is nonunital. Thus, $\alpha = 0$ and $a = 0$, proving that Θ is injective. Consequently, Equation (7.2) defines a norm on A'. Since Θ is an algebra homomorphism and the norm on A' is induced by the operator norm,

$$\|xy\|_{A'} \leq \|x\|_{A'}\|y\|_{A'} \tag{7.11}$$

for all $x, y \in A'$. We next show that ι_A is isometric. For every $a, b \in A$, $\|ab\| \leq \|a\|\|b\|$, so $\|L_a\| \leq \|a\|$. If $a \neq 0$, set $b = \frac{a^*}{\|a\|}$. Since the involution on a C^*-algebra is isometric, $\|b\| = 1$, and

$$\|L_a(b)\| = \left\| a\frac{a^*}{\|a\|} \right\| = \frac{\|aa^*\|}{\|a\|} = \frac{\|a\|^2}{\|a\|} = \|a\|. \tag{7.12}$$

Therefore, $\|L_a\| = \|a\|$. The equality is immediate when $a = 0$. It follows that

$$\|(a, 0)\|_{A'} = \|a\|. \tag{7.13}$$

Thus, ι_A is isometric. The image

$$L(A) := \{L_a : a \in A\} \tag{7.14}$$

is closed in $\mathscr{B}(A)$ because $a \mapsto L_a$ is isometric and A is complete. Since the sum of a closed subspace and a finite dimensional subspace is closed,

$$\Theta(A') = L(A) + \mathbb{C}I_A \tag{7.15}$$

is closed in $\mathscr{B}(A)$. Hence, $\Theta(A')$ is complete, and therefore A' is a Banach algebra. It remains to verify the C^*-identity. Let $x \in A'$, and identify each $b \in A$ with $(b, 0) \in A'$. Since A is a two-sided ideal in A', both xb and x^*xb belong to A. For every $b \in A$ satisfying $\|b\| \leq 1$, the C^*-identity in A gives

$$\|xb\|^2 = \|(xb)^*(xb)\| = \|b^*x^*xb\| \leq \|b^*\|\|x^*xb\| \leq \|x^*x\|_{A'}. \tag{7.16}$$

Taking the supremum over all $b \in A$ with $\|b\| \leq 1$ gives $\|x\|_{A'}^2 \leq \|x^*x\|_{A'}$. By submultiplicativity, $\|x\|_{A'}^2 \leq \|x^*\|_{A'}\|x\|_{A'}$. If $x \neq 0$, it follows that

$$\|x\|_{A'} \leq \|x^*\|_{A'}. \tag{7.17}$$

Applying the same argument to x^* gives

$$\|x^*\|_{A'} \leq \|x^{**}\|_{A'} = \|x\|_{A'}. \tag{7.18}$$

Thus, $\|x^*\|_{A'} = \|x\|_{A'}$. Consequently,

$$\|x\|_{A'}^2 \leq \|x^*x\|_{A'} \leq \|x^*\|_{A'}\|x\|_{A'} = \|x\|_{A'}^2. \tag{7.19}$$

Therefore, $\|x^*x\|_{A'} = \|x\|_{A'}^2$. This proves that A' is a C^*-algebra. The element $(0, 1)$ is the identity of A'. The map ι_A is clearly linear, multiplicative, and $*$-preserving, and we have already shown that it is isometric. Its image is $\iota_A(A) = A \oplus \{0\}$, which is a closed two-sided ideal of A'. $\square$

Remark 7.3. Many results formulated for unital C^*-algebras can be extended to nonunital C^*-algebras by passing to their unitizations. In particular, spectra of elements of a nonunital C^*-algebra are computed in its unitization.

The norm in Equation (7.2) is the unique C^*-norm on the algebraic unitization for which the canonical embedding of A is isometric. In this sense, it is the canonical norm on the unitization. The unitization contains A as a closed two-sided ideal of codimension one. More precisely, define

$$q_A\colon A' \longrightarrow \mathbb{C}, \tag{7.20}$$
$$(a,\lambda) \longmapsto \lambda. \tag{7.21}$$

Then q_A is a surjective unital $*$-homomorphism and $\ker q_A = \iota_A(A)$. Consequently, the unitization gives the short exact sequence

$$0 \longrightarrow A \xrightarrow{\iota_A} A' \xrightarrow{q_A} \mathbb{C} \longrightarrow 0. \tag{7.22}$$

This sequence is split. Indeed, the map

$$s_A\colon \mathbb{C} \longrightarrow A', \tag{7.23}$$
$$\lambda \longmapsto (0,\lambda), \tag{7.24}$$

is a unital $*$-homomorphism satisfying $q_A \circ s_A = \mathrm{id}_{\mathbb{C}}$. In particular, $\iota_A(A)$ is a maximal closed two-sided ideal of A' because

$$A'/\iota_A(A) \cong \mathbb{C}. \tag{7.25}$$

The unitization is characterized by the fact that every $*$-homomorphism from A into a unital C^*-algebra extends uniquely to a unital $*$-homomorphism from A'. We make this precise in Proposition 7.4.

Proposition 7.4. *Let A be a nonunital C^*-algebra, let B be a unital C^*-algebra, and let $\pi\colon A \to B$ be a $*$-homomorphism. Then there exists a unique unital $*$-homomorphism*

$$\widetilde{\pi}\colon A' \longrightarrow B, \tag{7.26}$$
$$(a,\lambda) \longmapsto \pi(a) + \lambda e_B, \tag{7.27}$$

such that $\widetilde{\pi} \circ \iota_A = \pi$. Moreover, $\widetilde{\pi}$ is contractive.

Proof. The map $\widetilde{\pi}$ is clearly complex-linear and unital. For $(a,\lambda),(b,\mu) \in A'$, we have

$$\begin{aligned}
\widetilde{\pi}((a,\lambda)(b,\mu)) &= \widetilde{\pi}(ab + \lambda b + \mu a, \lambda\mu) && (7.28)\\
&= \pi(ab) + \lambda\pi(b) + \mu\pi(a) + \lambda\mu e_B && (7.29)\\
&= (\pi(a) + \lambda e_B)(\pi(b) + \mu e_B) && (7.30)\\
&= \widetilde{\pi}(a,\lambda)\widetilde{\pi}(b,\mu). && (7.31)
\end{aligned}$$

Furthermore,

$$\begin{aligned}
\widetilde{\pi}((a,\lambda)^*) &= \widetilde{\pi}(a^*, \overline{\lambda}) && (7.32)\\
&= \pi(a^*) + \overline{\lambda} e_B && (7.33)\\
&= \pi(a)^* + \overline{\lambda} e_B && (7.34)\\
&= \widetilde{\pi}(a,\lambda)^*. && (7.35)
\end{aligned}$$

Thus, $\widetilde{\pi}$ is a unital $*$-homomorphism. Every $*$-homomorphism between C^*-algebras is contractive, so $\widetilde{\pi}$ is contractive. Finally, suppose that $\psi\colon A' \to B$ is another unital $*$-homomorphism satisfying $\psi \circ \iota_A = \pi$. Then

$$\begin{aligned}\psi(a, \lambda) &= \psi((a, 0) + \lambda e_{A'}) &(7.36)\\ &= \psi(a, 0) + \lambda \psi(e_{A'}) &(7.37)\\ &= \pi(a) + \lambda e_B &(7.38)\\ &= \widetilde{\pi}(a, \lambda). &(7.39)\end{aligned}$$

Hence, $\psi = \widetilde{\pi}$. □

Proposition 7.4 implies that the unitization construction is compatible with $*$-homomorphisms. In particular, a $*$-homomorphism between nonunital C^*-algebras extends canonically to a unital $*$-homomorphism between their unitizations. We make this precise in Corollary 7.5.

Corollary 7.5. *Let A and B be nonunital C^*-algebras, and let $\pi\colon A \to B$ be a $*$-homomorphism. Then there exists a unique unital $*$-homomorphism*

$$\begin{aligned}\pi'\colon A' &\longrightarrow B', &(7.40)\\ (a, \lambda) &\longmapsto (\pi(a), \lambda), &(7.41)\end{aligned}$$

such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "\iota_A"] \arrow[d, "\pi"'] & A' \arrow[r, "q_A"] \arrow[d, "\pi'"] & \mathbb{C} \arrow[d, equal] \\ B \arrow[r, "\iota_B"'] & B' \arrow[r, "q_B"'] & \mathbb{C}. \end{tikzcd}$$

Proof. Apply Proposition 7.4 to the $*$-homomorphism $\iota_B \circ \pi\colon A \to B'$. The resulting unital extension is

$$\pi'(a, \lambda) = (\pi(a), \lambda). \tag{7.42}$$

The identities $\pi' \circ \iota_A = \iota_B \circ \pi$ and $q_B \circ \pi' = q_A$ follow directly from the definition. □

If A is already unital, its minimal unitization is simply A. If one nevertheless applies the formal construction above, then the resulting algebra is $*$-isomorphic to the direct sum of the C^*-algebras A and $\mathbb{C}$. We make this precise in Proposition 7.6.

Proposition 7.6. *Let A be a unital C^*-algebra, and let A' denote its formal unitization. Then $A' \cong A \oplus \mathbb{C}$ as a direct sum of C^*-algebras.*

Proof. Let e_A denote the identity of A, let $e_{A'}$ denote the identity of A', and define $p := e_{A'} - e_A$. Since both e_A and $e_{A'}$ are self-adjoint, we have $p^* = e_{A'} - e_A = p$. Moreover,

$$\begin{aligned}p^2 &= (e_{A'} - e_A)^2 &(7.43)\\ &= e_{A'}^2 - e_{A'}e_A - e_Ae_{A'} + e_A^2 &(7.44)\\ &= e_{A'} - e_A - e_A + e_A = e_{A'} - e_A = p. &(7.45)\end{aligned}$$

Thus, p is a projection. For every $a \in A$, both e_A and $e_{A'}$ act as the identity on a. Therefore,

$$\begin{aligned}pa &= (e_{A'} - e_A)a = a - a = 0, &(7.46)\\ ap &= a(e_{A'} - e_A) = a - a = 0. &(7.47)\end{aligned}$$

Every element of A' has the form $a + \lambda e_{A'}$ for some $a \in A$ and $\lambda \in \mathbb{C}$. Since $e_{A'} = e_A + p$, we have

$$a + \lambda e_{A'} = (a + \lambda e_A) + \lambda p. \tag{7.48}$$

Since $a + \lambda e_A \in A$, every element of A' can be written as $b + \lambda p$ for some $b \in A$ and $\lambda \in \mathbb{C}$. Conversely, every element of this form belongs to A'. Hence,

$$A' = \{a + \lambda p : a \in A,\ \lambda \in \mathbb{C}\}. \tag{7.49}$$

Equip $A \oplus \mathbb{C}$ with componentwise multiplication and involution, and define

$$\Phi\colon A \oplus \mathbb{C} \longrightarrow A', \tag{7.50}$$

$$(a, \lambda) \longmapsto a + \lambda p. \tag{7.51}$$

The map Φ is linear. Since $pa = ap = 0$ for every $a \in A$ and $p^2 = p$, we have

$$\Phi(a, \lambda)\Phi(b, \mu) = (a + \lambda p)(b + \mu p) \tag{7.52}$$

$$= ab + \lambda pb + \mu ap + \lambda\mu p^2 \tag{7.53}$$

$$= ab + \lambda\mu p \tag{7.54}$$

$$= \Phi(ab, \lambda\mu). \tag{7.55}$$

Thus, Φ is multiplicative. Moreover, since $p^* = p$,

$$\Phi(a, \lambda)^* = (a + \lambda p)^* = a^* + \overline{\lambda} p = \Phi(a^*, \overline{\lambda}), \tag{7.56}$$

so Φ preserves the involution. The preceding description of A' shows that Φ is surjective. To prove injectivity, suppose that $a + \lambda p = 0$. Multiplying this equality by e_A and using $pe_A = 0$ gives

$$a = ae_A + \lambda pe_A = 0. \tag{7.57}$$

It follows that $\lambda p = 0$. Since $p = e_{A'} - e_A \neq 0$, we obtain $\lambda = 0$. Therefore, Φ is injective and hence is a $*$-isomorphism. Consequently, $A' \cong A \oplus \mathbb{C}$ as a direct sum of C^*-algebras. ☐

8. Gelfand–Naimark Theorem

We now prove the commutative Gelfand–Naimark theorem. This result characterizes commutative C^*-algebras as algebras of continuous functions and provides the foundation for interpreting general C^*-algebras as noncommutative topological spaces.

Theorem 8.1 (Gelfand–Naimark)**.** *Let A be a C^*-algebra. Then A is commutative if and only if there exists a locally compact Hausdorff space X such that $A \cong C_0(X)$ isometrically as $*$-algebras. In particular, then one may take $X = \widehat{A}$, and the isometric $*$-isomorphism is the Gelfand transform*

$$\Gamma\colon A \longrightarrow C_0(\widehat{A}), \tag{8.1}$$

$$a \longmapsto \widehat{a}, \tag{8.2}$$

where $\widehat{a}(\omega) = \omega(a)$ for every $\omega \in \widehat{A}$.

Proof. Suppose first that A is commutative. We have already established in Section 5 that $\widehat{A}$ is a locally compact Hausdorff space and that the Gelfand transform

$$\Gamma\colon A \longrightarrow C_0(\widehat{A}) \tag{8.3}$$

is a contractive algebra homomorphism. We now prove that Γ is an isometric $*$-isomorphism. Since A is commutative, every element of A is normal. Proposition 6.12(6) gives

$$\|\Gamma(a)\|_\infty = \sup_{\omega \in \widehat{A}} |\omega(a)| = r(a) = \|a\| \tag{8.4}$$

for every $a \in A$. Thus, Γ is an isometry and, in particular, is injective. We next show that Γ preserves the involution. By Proposition 6.15(2), for every $a \in A$ and $\omega \in \widehat{A}$ we have

$$\Gamma(a^*)(\omega) = \omega(a^*) = \overline{\omega(a)} = \overline{\Gamma(a)(\omega)} = \Gamma(a)^*(\omega). \tag{8.5}$$

Therefore, $\Gamma(a^*) = \Gamma(a)^*$ for every $a \in A$, and hence $\Gamma(A)$ is a $*$-subalgebra of $C_0(\widehat{A})$. We now verify the hypotheses of the locally compact version of the Stone–Weierstrass theorem. First, $\Gamma(A)$ separates points of $\widehat{A}$. Indeed, if $\omega_1, \omega_2 \in \widehat{A}$ and $\omega_1 \neq \omega_2$, then there exists $a \in A$ such that $\omega_1(a) \neq \omega_2(a)$. Consequently, $\Gamma(a)(\omega_1) \neq \Gamma(a)(\omega_2)$. Second, $\Gamma(A)$ vanishes nowhere. If $\omega \in \widehat{A}$, then ω is nonzero, so there exists $a \in A$ such that $\omega(a) \neq 0$. Therefore, $\Gamma(a)(\omega) \neq 0$. Since $\Gamma(A)$ is self-adjoint, separates points, and vanishes nowhere, the locally compact Stone–Weierstrass theorem gives

$$\overline{\Gamma(A)} = C_0(\widehat{A}). \tag{8.6}$$

It remains to show that $\Gamma(A)$ is closed. Let $(\Gamma(a_n))_{n\geq 1}$ be a sequence in $\Gamma(A)$ converging to some $f \in C_0(\widehat{A})$. Since Γ is an isometry,

$$\|a_n - a_m\| = \|\Gamma(a_n) - \Gamma(a_m)\|_\infty \tag{8.7}$$

for every $m, n \in \mathbb{N}$. Hence, $(a_n)_{n\geq 1}$ is a Cauchy sequence in A. Since A is complete, there exists $a \in A$ such that $a_n \to a$. The continuity of Γ then gives $\Gamma(a_n) \longrightarrow \Gamma(a)$. By uniqueness of limits, $f = \Gamma(a)$. Thus, $\Gamma(A)$ is closed. Since it is also dense in $C_0(\widehat{A})$, we conclude that $\Gamma(A) = C_0(\widehat{A})$. Therefore, Γ is an isometric $*$-isomorphism. Conversely, suppose that there exists a locally compact Hausdorff space X such that A is $*$-isomorphic to $C_0(X)$. Since multiplication in $C_0(X)$ is pointwise, $C_0(X)$ is commutative. It follows that A is commutative. □

When A is unital, its character space is compact, so Theorem 8.1 takes the following form.

Corollary 8.2. *Let A be a unital commutative C^*-algebra. Then $\widehat{A}$ is a compact Hausdorff space, and the Gelfand transform defines an isometric unital $*$-isomorphism*

$$\Gamma\colon A \longrightarrow C(\widehat{A}). \tag{8.8}$$

Proof. This follows from Theorem 8.1. □

We conclude the section with an approximation consequence that will be used in Section 9. In particular, Lemma 8.3 shows that the inverse of an invertible self-adjoint element can be approximated in norm by polynomials in that element.

Lemma 8.3. *Let A be a unital commutative C^*-algebra, and let $a \in A$ be self-adjoint and invertible. Then there exists a sequence of polynomials $(p_n)_{n\geq 1}$ in $\mathbb{C}[z]$ such that*

$$a^{-1} = \lim_{n\to\infty} p_n(a). \tag{8.9}$$

Proof. By Corollary 8.2, we may identify A with $C(X)$ for some compact Hausdorff space X. Under this identification, the element a corresponds to a real-valued function $f \in C(X)$. Since a is invertible, the function f does not vanish on X. Compactness of X therefore implies that there exists $\varepsilon > 0$ such that $|f(x)| \geq \varepsilon$ for every $x \in X$. Set

$$K := [-\|f\|_\infty, -\varepsilon] \cup [\varepsilon, \|f\|_\infty]. \tag{8.10}$$

Then K is compact, $f(X) \subseteq K$, and the function

$$g\colon K \longrightarrow \mathbb{C}, \tag{8.11}$$

$$t \longmapsto \frac{1}{t}, \tag{8.12}$$

is continuous. By the Stone–Weierstrass theorem, there exists a sequence of polynomials $(p_n)_{n\geq 1}$ such that

$$\sup_{t\in K}\left|p_n(t)-\frac{1}{t}\right|\longrightarrow 0. \tag{8.13}$$

Since $f(X)\subseteq K$, it follows that

$$\|p_n(f)-f^{-1}\|_\infty=\sup_{x\in X}\left|p_n(f(x))-\frac{1}{f(x)}\right|\leq\sup_{t\in K}\left|p_n(t)-\frac{1}{t}\right|\longrightarrow 0. \tag{8.14}$$

Consequently, $p_n(a)\to a^{-1}$ in norm. □

9. Continuous Functional Calculus

Polynomial functional calculus from Section 4 allows us to apply a polynomial to an element of a unital Banach algebra. For a normal element of a C^*-algebra, the presence of the involution and the C^*-identity allows us to extend this construction from polynomials to arbitrary continuous functions on the spectrum. This extension is called the continuous functional calculus.

9.1. Motivation.

Let A be a unital C^*-algebra, and let $a\in A$ be normal. Consider polynomials in the two variables z and $\overline{z}$ of the form

$$p(z,\overline{z})=\sum_{k,l=0}^{N}c_{k,l}z^k\overline{z}^l,\qquad c_{k,l}\in\mathbb{C}. \tag{9.1}$$

Since a is normal, a and a^* commute. We may therefore define

$$p(a,a^*)=\sum_{k,l=0}^{N}c_{k,l}a^k(a^*)^l. \tag{9.2}$$

Normality is essential here because the polynomial algebra generated by z and $\overline{z}$ is commutative. The algebra of polynomials in z and $\overline{z}$, restricted to the compact set $\sigma_A(a)$, is self-adjoint, contains the constant functions, and separates points. The Stone–Weierstrass theorem therefore implies that it is dense in $C(\sigma_A(a))$. Consequently, every function $f\in C(\sigma_A(a))$ can be approximated uniformly by a sequence of such polynomials. To define $f(a)$ by approximating f with polynomials, one must show that uniform convergence on $\sigma_A(a)$ corresponds to norm convergence in A. The C^*-identity provides precisely the required norm control. The resulting construction is the continuous functional calculus, which is detailed in Section 9.2.

9.2. Construction.

Let A be a C^*-algebra, and let $a\in A$ be normal. We denote by $C^*(a)$ the smallest C^*-subalgebra of A containing a. Equivalently,

$$C^*(a)=\overline{\{p(a,a^*):p\in\mathbb{C}[z,\overline{z}],\ p(0,0)=0\}}. \tag{9.3}$$

If A is unital with identity e, we denote by $C^*(e,a)$ the smallest unital C^*-subalgebra of A containing a. Thus,

$$C^*(e,a)=\overline{\{p(a,a^*):p\in\mathbb{C}[z,\overline{z}]\}}. \tag{9.4}$$

Since a is normal, the C^*-algebras $C^*(a)$ and $C^*(e,a)$ are commutative. Before constructing the continuous functional calculus, we establish that the spectrum of an element of a unital C^*-subalgebra does not depend on whether it is computed in the subalgebra or in the ambient C^*-algebra. Note that Lemma 9.1 is specific to C^*-algebras and stands in contrast to the general Banach algebra setting, as illustrated by Example 3.5.

Lemma 9.1 (Spectral permanence)**.** *Let A be a unital C^*-algebra, and let $B \subseteq A$ be a C^*-subalgebra containing the identity of A. Then*

$$\sigma_B(b) = \sigma_A(b) \tag{9.5}$$

for every $b \in B$.

Proof. Since every inverse of an element of B in B is also an inverse in A, we immediately have $\sigma_A(b) \subseteq \sigma_B(b)$. To prove the reverse inclusion, it suffices to show that if $b \in B$ is invertible in A, then $b^{-1} \in B$. Suppose first that $b = b^*$. Define

$$E := C^*(e, b, b^{-1}) \subseteq A. \tag{9.6}$$

Since b is self-adjoint, its inverse is also self-adjoint. Moreover, b and b^{-1} commute. Hence, E is a unital commutative C^*-algebra. By Lemma 8.3, there exists a sequence of polynomials $(p_n)_{n\geq 1}$ such that

$$b^{-1} = \lim_{n\to\infty} p_n(b). \tag{9.7}$$

Since $b \in B$ and B contains e, we have

$$p_n(b) \in C^*(e, b) \subseteq B \tag{9.8}$$

for every $n \in \mathbb{N}$. Since B is closed, it follows that $b^{-1} \in B$. Now let $b \in B$ be arbitrary and suppose that b is invertible in A. Then b^*b is self-adjoint and invertible in A. By the preceding argument, $(b^*b)^{-1} \in B$. Consequently, $(b^*b)^{-1}b^* \in B$. Moreover, $(b^*b)^{-1}b^*b = e$. Since b is invertible in A, its left inverse is its inverse. Therefore, $b^{-1} = (b^*b)^{-1}b^* \in B$. Thus, b is invertible in B, which proves $\sigma_B(b) \subseteq \sigma_A(b)$. Combining the two inclusions proves the result. □

Having established spectral permanence in Lemma 9.1, we can now identify the commutative C^*-algebra generated by a normal element with an algebra of continuous functions on its spectrum. This provides the setting for the continuous functional calculus, which we construct in Proposition 9.2.

Proposition 9.2 (Continuous functional calculus)**.** *Let A be a unital C^*-algebra, and let $a \in A$ be normal. There exists a unique unital isometric $*$-isomorphism*

$$\Phi_a \colon C(\sigma_A(a)) \longrightarrow C^*(e, a) \tag{9.9}$$

such that

$$\Phi_a(1_{\sigma_A(a)}) = e, \qquad \Phi_a(\mathrm{id}_{\sigma_A(a)}) = a, \tag{9.10}$$

where $\mathrm{id}_{\sigma_A(a)}(z) = z$ for every $z \in \sigma_A(a)$. For $f \in C(\sigma_A(a))$, we write $f(a) := \Phi_a(f)$. The map Φ_a is called the continuous functional calculus of a.

Proof. Set $B := C^*(e, a)$. Since a is normal, B is a unital commutative C^*-algebra. By Theorem 8.1, the Gelfand transform defines an isometric $*$-isomorphism

$$\Psi \colon B \longrightarrow C(\widehat{B}). \tag{9.11}$$

We identify the character space $\widehat{B}$ with $\sigma_A(a)$. Define

$$\tau \colon \widehat{B} \longrightarrow \sigma_A(a), \tag{9.12}$$

$$\omega \longmapsto \omega(a). \tag{9.13}$$

By Lemma 9.1 and the characterization of the spectrum in a unital commutative Banach algebra, we have

$$\sigma_A(a) = \sigma_B(a) = \{\omega(a) : \omega \in \widehat{B}\}. \tag{9.14}$$

Thus, τ is surjective. To prove injectivity, suppose that $\omega_1, \omega_2 \in \widehat{B}$ satisfy $\omega_1(a) = \omega_2(a)$. Since characters on a C^*-algebra preserve the involution,

$$\omega_1(a^*) = \overline{\omega_1(a)} = \overline{\omega_2(a)} = \omega_2(a^*). \tag{9.15}$$

Both characters are unital, so they agree on every polynomial in a and a^*. Since these polynomials are dense in B and the characters are continuous, it follows that $\omega_1 = \omega_2$. Therefore, τ is injective. The map τ is continuous because the weak-$*$ topology on $\widehat{B}$ makes every evaluation map $\omega \mapsto \omega(a)$ continuous. Since $\widehat{B}$ is compact and $\sigma_A(a)$ is Hausdorff, the continuous bijection τ is a homeomorphism. Composition with τ therefore defines an isometric $*$-isomorphism

$$\Theta\colon C(\sigma_A(a)) \longrightarrow C(\widehat{B}), \tag{9.16}$$

$$f \longmapsto f \circ \tau. \tag{9.17}$$

Define $\Phi_a := \Psi^{-1} \circ \Theta$. Then Φ_a is a unital isometric $*$-isomorphism. Moreover, $\Theta(1_{\sigma_A(a)}) = 1_{\widehat{B}} = \Psi(e)$, and

$$\Theta(\mathrm{id}_{\sigma_A(a)})(\omega) = \mathrm{id}_{\sigma_A(a)}(\tau(\omega)) = \omega(a) = \Psi(a)(\omega). \tag{9.18}$$

Consequently, $\Phi_a(1_{\sigma_A(a)}) = e$ and $\Phi_a(\mathrm{id}_{\sigma_A(a)}) = a$. It remains to prove uniqueness. Let

$$\Phi\colon C(\sigma_A(a)) \longrightarrow C^*(e, a) \tag{9.19}$$

be another unital $*$-homomorphism satisfying $\Phi(1_{\sigma_A(a)}) = e$ and $\Phi(\mathrm{id}_{\sigma_A(a)}) = a$. Since Φ preserves the involution, $\Phi(\overline{\mathrm{id}_{\sigma_A(a)}}) = a^*$. It follows that Φ and Φ_a agree on every polynomial in z and $\overline{z}$. These polynomials form a dense $*$-subalgebra of $C(\sigma_A(a))$ by the Stone–Weierstrass theorem. Since $*$-homomorphisms between C^*-algebras are continuous, we conclude that $\Phi = \Phi_a$. □

Proposition 9.2 allows problems concerning a normal element of a possibly noncommutative C^*-algebra to be reduced to corresponding problems about continuous functions on its spectrum. We next record its fundamental properties in Corollary 9.3.

Corollary 9.3. *Let A be a unital C^*-algebra, let $a \in A$ be normal, and let $K := \sigma_A(a)$. Then the following statements hold:*

(1) *For every $f \in C(K)$, the element $f(a)$ is normal.*
(2) *For every $f, g \in C(K)$ and $\lambda \in \mathbb{C}$,*

$$(f + g)(a) = f(a) + g(a), \tag{9.20}$$

$$(fg)(a) = f(a)g(a), \tag{9.21}$$

$$(\lambda f)(a) = \lambda f(a), \tag{9.22}$$

$$f(a)^* = \overline{f}(a), \tag{9.23}$$

$$\|f(a)\| = \|f\|_\infty. \tag{9.24}$$

(3) *The continuous spectral mapping theorem holds. That is,*

$$\sigma_A(f(a)) = f(\sigma_A(a)). \tag{9.25}$$

(4) *If $g \in C(f(K))$, then $g(f(a)) = (g \circ f)(a)$.*
(5) *The element $f(a)$ is invertible if and only if f does not vanish on K. When these conditions hold, $f(a)^{-1} = (\frac{1}{f})(a)$.*

Proof. The proof is given below:

(1) We have

$$f(a)^*f(a) = \overline{f}(a)f(a) = (\overline{f}f)(a) = (f\overline{f})(a) = f(a)\overline{f}(a) = f(a)f(a)^*, \tag{9.26}$$

Hence, $f(a)$ is normal.

(2) The statements follow directly from the fact that Φ_a is an isometric $*$-homomorphism.

(3) Lemma 9.1 and Proposition 9.2 give

$$\begin{aligned} \sigma_A(f(a)) &= \sigma_{C^*(e,a)}(f(a)) && (9.27)\\ &= \sigma_{C(K)}(f) && (9.28)\\ &= f(K) && (9.29)\\ &= f(\sigma_A(a)). && (9.30) \end{aligned}$$

(4) Consider the maps

$$\begin{aligned} \Phi_1\colon C(f(K)) &\longrightarrow C^*(e, f(a)), & \Phi_2\colon C(f(K)) &\longrightarrow C^*(e, a), && (9.31)\\ g &\longmapsto g(f(a)), & g &\longmapsto (g\circ f)(a). && (9.32) \end{aligned}$$

Both are unital $*$-homomorphisms that send the identity function on $f(K)$ to $f(a)$. By the uniqueness of the continuous functional calculus, they agree. Hence, $g(f(a)) = (g\circ f)(a)$.

(5) (3) implies $0 \notin \sigma_A(f(a))$ if and only if $0 \notin f(K)$. Thus, $f(a)$ is invertible if and only if f does not vanish on K. Applying the functional calculus gives

$$f(a)\left(\frac{1}{f}\right)(a) = e = \left(\frac{1}{f}\right)(a)f(a), \tag{9.33}$$

which proves the inverse formula.

This completes the proof. □

We next use the continuous functional calculus to characterize several important classes of normal elements directly in terms of their spectra. These characterizations are collected in Corollary 9.4.

Corollary 9.4. *Let A be a unital C^*-algebra, and let $a \in A$ be normal. Then:*

(1) *The element a is self-adjoint if and only if $\sigma_A(a) \subseteq \mathbb{R}$.*
(2) *The element a is unitary if and only if $\sigma_A(a) \subseteq \mathbb{S}^1$.*
(3) *The element a is a projection if and only if $\sigma_A(a) \subseteq \{0, 1\}$.*

Proof. The proof is given below:

(1) Suppose first that a is self-adjoint. Then its spectrum is contained in $\mathbb{R}$ by Proposition 6.15(1). Conversely, suppose that $\sigma_A(a) \subseteq \mathbb{R}$. The identity function on $\sigma_A(a)$ is then real-valued, and hence

$$a^* = \mathrm{id}_{\sigma_A(a)}(a)^* = \overline{\mathrm{id}_{\sigma_A(a)}}(a) = \mathrm{id}_{\sigma_A(a)}(a) = a. \tag{9.34}$$

Thus, a is self-adjoint.

(2) Suppose that a is unitary. Then $\sigma_A(a) \subseteq \mathbb{S}^1$. by Proposition 6.12(4). Conversely, suppose that the spectrum of a is contained in $\mathbb{S}^1$. Since $\sigma_A(a) \subseteq \mathbb{S}^1$, we have

$$\overline{\mathrm{id}_{\sigma_A(a)}}\,\mathrm{id}_{\sigma_A(a)} = \mathrm{id}_{\sigma_A(a)}\,\overline{\mathrm{id}_{\sigma_A(a)}} = 1_{\sigma_A(a)}. \tag{9.35}$$

Applying the continuous functional calculus gives

$$\begin{aligned} a^*a &= \Phi_a(\mathrm{id}_{\sigma_A(a)})^*\Phi_a(\mathrm{id}_{\sigma_A(a)}) &(9.36)\\ &= \Phi_a(\overline{\mathrm{id}_{\sigma_A(a)}})\Phi_a(\mathrm{id}_{\sigma_A(a)}) &(9.37)\\ &= \Phi_a(\overline{\mathrm{id}_{\sigma_A(a)}}\,\mathrm{id}_{\sigma_A(a)}) &(9.38)\\ &= \Phi_a(1_{\sigma_A(a)}) = e, &(9.39)\end{aligned}$$

Similarly, $aa^* = e$. Therefore, a is unitary.

(3) Suppose first that a is a projection. Since $a^2 = a$, Proposition 4.3 gives

$$\{0\} = \sigma_A(a^2 - a) = \{\lambda^2 - \lambda : \lambda \in \sigma_A(a)\}. \tag{9.40}$$

Thus, every $\lambda \in \sigma_A(a)$ satisfies $\lambda^2 - \lambda = 0$. Therefore, $\sigma_A(a) \subseteq \{0, 1\}$. Conversely, suppose that $\sigma_A(a) \subseteq \{0, 1\}$. The identity function on $\sigma_A(a)$ is real-valued, so $\overline{\mathrm{id}_{\sigma_A(a)}} = \mathrm{id}_{\sigma_A(a)}$. Moreover, since every element of $\sigma_A(a)$ is either 0 or 1, we have $\mathrm{id}^2_{\sigma_A(a)} = \mathrm{id}_{\sigma_A(a)}$. Applying the continuous functional calculus gives

$$\begin{aligned} a^* &= \Phi_a(\mathrm{id}_{\sigma_A(a)})^* = \Phi_a(\overline{\mathrm{id}_{\sigma_A(a)}}) = \Phi_a(\mathrm{id}_{\sigma_A(a)}) = a, &(9.41)\\ a^2 &= \Phi_a(\mathrm{id}_{\sigma_A(a)})^2 = \Phi_a(\mathrm{id}^2_{\sigma_A(a)}) = \Phi_a(\mathrm{id}_{\sigma_A(a)}) = a. &(9.42)\end{aligned}$$

Hence, a is a projection.

This completes the proof. □

The continuous functional calculus also provides a way to construct logarithms of certain unitary elements. We make this statement precise in Corollary 9.5.

Corollary 9.5. *Let A be a unital C^*-algebra, and let $a \in A$ be unitary. If $\sigma_A(a) \subsetneq \mathbb{S}^1$, then there exists a self-adjoint element $b \in A$ such that $a = e^{ib}$.*

Proof. Since $\sigma_A(a)$ is a proper compact subset of $\mathbb{S}^1$, there exists $\zeta \in \mathbb{S}^1 \setminus \sigma_A(a)$. The punctured circle $\mathbb{S}^1 \setminus \{\zeta\}$ admits a continuous argument. Therefore, there exists a continuous real-valued function $h\colon \sigma_A(a) \to \mathbb{R}$ such that $e^{ih(z)} = z$ for every $z \in \sigma_A(a)$. Applying the continuous functional calculus of a, define

$$b := \Phi_a(h) = h(a) \in C^*(e, a) \subseteq A. \tag{9.43}$$

Since h is real-valued,

$$b^* = \overline{h}(a) = h(a) = b. \tag{9.44}$$

Thus, b is self-adjoint. By the compatibility of the continuous functional calculus with composition,

$$e^{ib} = e^{ih(a)} = (e^{ih})(a) = \mathrm{id}_{\sigma_A(a)}(a) = a. \tag{9.45}$$

This completes the proof. □

The continuous functional calculus is compatible with the algebraic structure of C^*-algebras. More precisely, applying a continuous function to a normal element and then applying a unital $*$-homomorphism gives the same result as first applying the $*$-homomorphism and then using the continuous functional calculus. This naturality property is formalized in Corollary 9.6.

Corollary 9.6. *Let A and B be unital C^*-algebras, let $\varphi\colon A \to B$ be a unital $*$-homomorphism, and let $a \in A$ be normal. Then*

$$\varphi(f(a)) = f(\varphi(a)) \tag{9.46}$$

for every $f \in C(\sigma_A(a))$, where the function on the right-hand side is restricted to $\sigma_B(\varphi(a))$.

Proof. The element $\varphi(a)$ is normal, and $\sigma_B(\varphi(a)) \subseteq \sigma_A(a)$. Define two unital $*$-homomorphisms from $C(\sigma_A(a))$ to B by

$$\Phi_1(f) := \varphi(f(a)), \tag{9.47}$$
$$\Phi_2(f) := f|_{\sigma_B(\varphi(a))}(\varphi(a)). \tag{9.48}$$

Both maps send the constant function $1_{\sigma_A(a)}$ to e_B and the identity function to $\varphi(a)$. They therefore agree on every polynomial in z and $\overline{z}$. Since these polynomials are dense in $C(\sigma_A(a))$ and both maps are continuous, we conclude that $\Phi_1 = \Phi_2$. Thus, $\varphi(f(a)) = f(\varphi(a))$. ☐

An important consequence of Corollary 9.6 is that injective $*$-homomorphisms between C^*-algebras are automatically isometric. We establish this in Corollary 9.7.

Corollary 9.7. *Every injective $*$-homomorphism $\varphi\colon A \to B$ between C^*-algebras is isometric.*

Proof. Suppose first that A and B are unital and that φ is unital. Let $a \in A$ be self-adjoint. Since every $*$-homomorphism is contractive, $\|\varphi(a)\| \leq \|a\|$. Suppose, toward a contradiction, that $\|\varphi(a)\| < \|a\|$. Since a is self-adjoint, it is normal, and therefore $\|a\| = r(a)$. Moreover, $\sigma_A(a)$ is compact, so the continuous function

$$\sigma_A(a) \longrightarrow [0, \infty), \tag{9.49}$$
$$\lambda \longmapsto |\lambda|, \tag{9.50}$$

attains its maximum. Hence, there exists $\lambda_0 \in \sigma_A(a)$ such that

$$|\lambda_0| = \max_{\lambda \in \sigma_A(a)} |\lambda| = r(a) = \|a\|. \tag{9.51}$$

Since φ preserves the involution and a is self-adjoint, we have $\varphi(a)^* = \varphi(a^*) = \varphi(a)$. Thus, $\varphi(a)$ is self-adjoint, and hence $\sigma_B(\varphi(a)) \subseteq \mathbb{R}$. Moreover, the spectral radius is bounded above by the norm, so every $\mu \in \sigma_B(\varphi(a))$ satisfies

$$|\mu| \leq r(\varphi(a)) \leq \|\varphi(a)\|. \tag{9.52}$$

Since μ is real, it follows that

$$\sigma_B(\varphi(a)) \subseteq [-\|\varphi(a)\|, \|\varphi(a)\|]. \tag{9.53}$$

Hence, $\lambda_0 \notin \sigma_B(\varphi(a))$. Since $\sigma_B(\varphi(a))$ is a closed subset of $\sigma_A(a)$, there exists a continuous function $f\colon \sigma_A(a) \to \mathbb{C}$ such that

$$f|_{\sigma_B(\varphi(a))} = 0, \tag{9.54}$$
$$f(\lambda_0) = 1. \tag{9.55}$$

By Corollary 9.6, $\varphi(f(a)) = f(\varphi(a)) = 0$. Since φ is injective, this implies that $f(a) = 0$. However, the isometry of the continuous functional calculus gives $\|f(a)\| = \|f\|_\infty \geq 1$, which is a contradiction. Therefore, $\|\varphi(a)\| = \|a\|$ for every self-adjoint $a \in A$. For an arbitrary $a \in A$, the element a^*a is self-adjoint. Hence,

$$\|\varphi(a)\|^2 = \|\varphi(a)^*\varphi(a)\| = \|\varphi(a^*a)\| = \|a^*a\| = \|a\|^2. \tag{9.56}$$

Thus, φ is isometric in the unital case. For general C^*-algebras, extend φ to the unital $*$-homomorphism

$$\varphi'\colon A' \longrightarrow B', \tag{9.57}$$
$$(a, \lambda) \longmapsto (\varphi(a), \lambda). \tag{9.58}$$

The injectivity of φ implies that φ' is injective. By the unital case, φ' is isometric. Since the canonical embeddings of A and B into their unitizations are isometric, we obtain $\|\varphi(a)\| = \|a\|$ for every $a \in A$. $\square$

We conclude this section by recording the nonunital form of the continuous functional calculus in Proposition 9.8.

Proposition 9.8 (Nonunital continuous functional calculus). *Let A be a nonunital C^*-algebra, let A' be its unitization, and let $a \in A$ be normal. Define $\sigma_A(a) := \sigma_{A'}(a)$. If $f \in C(\sigma_A(a))$ satisfies $f(0) = 0$, then*

$$f(a) \in C^*(a) \subseteq A. \tag{9.59}$$

Moreover, the map

$$\begin{aligned} \Phi_a^0 \colon \{f \in C(\sigma_A(a)) : f(0) = 0\} &\longrightarrow C^*(a), && (9.60)\\ f &\longmapsto f(a), && (9.61) \end{aligned}$$

is an isometric $*$-isomorphism.

Proof. Apply Proposition 9.2 to a as an element of A'. This gives an isometric $*$-isomorphism

$$\Phi_a \colon C(\sigma_A(a)) \longrightarrow C^*(e_{A'}, a). \tag{9.62}$$

Since A is a proper ideal of A', the element a is not invertible in A'. Therefore, $0 \in \sigma_A(a)$. Let

$$I_0 := \{f \in C(\sigma_A(a)) : f(0) = 0\}. \tag{9.63}$$

The $*$-algebra of polynomials in z and $\overline{z}$ with zero constant term is dense in I_0. Under Φ_a, these polynomials are mapped to polynomials in a and a^* with zero constant term. By the definition of $C^*(a)$, it follows that $\Phi_a(I_0) = C^*(a)$. Since Φ_a is an isometric $*$-isomorphism, its restriction to I_0 is an isometric $*$-isomorphism onto $C^*(a)$. $\square$

10. Positivity, States and Representations

Continuous functional calculus provides a powerful method for studying normal elements of a C^*-algebra. Although an arbitrary element of a noncommutative C^*-algebra need not be normal, every element a determines the self-adjoint elements a^*a and aa^*. This observation leads to the theory of positive elements. Positivity, in turn, allows us to define states. Later on, states will be used to construct representations of C^*-algebras through the Gelfand–Naimark–Segal construction and to prove the Gelfand–Naimark–Segal theorem in Section 11

10.1. Positive Elements.

We begin by defining positive elements in terms of their spectra. For a nonunital C^*-algebra, the spectrum is understood to be computed in its unitization.

Definition 10.1. Let A be a C^*-algebra. A self-adjoint element $a \in A$ is **positive** if $\sigma_A(a) \subseteq [0, \infty)$. The set of positive elements of A is denoted by A_+. If $a \in A_+$, we write $a \geq 0$.

If $a = a^*$, we write $a \leq 0$ when $-a \geq 0$. More generally, for self-adjoint elements $a, b \in A$, we write $a \leq b$ if $b - a \geq 0$.

Example 10.2. Let X be a locally compact Hausdorff space. Then

$$C_0(X)_+ = \{f \in C_0(X) : f(x) \geq 0 \text{ for every } x \in X\}. \tag{10.1}$$

Thus, the positive elements of $C_0(X)$ are precisely the nonnegative real-valued functions.

Continuous functional calculus allows a self-adjoint element to be decomposed into positive and negative parts and provides positive roots of positive elements. We formalize this statement in Proposition 10.3.

Proposition 10.3. *Let A be a unital C^*-algebra.*

(1) *Every self-adjoint element $a \in A$ admits a unique decomposition $a = a_+ - a_-$, where $a_+, a_- \in A_+$ and $a_+a_- = a_-a_+ = 0$. The elements a_+ and a_- are called the positive part and negative part of a, respectively.*
(2) *If $a \in A_+$ and $n \geq 1$, then there exists a unique element $b \in A_+$ such that $b^n = a$. The element b is denoted by $a^{1/n}$ and is called the positive n-th root of a.*

Proof. The proof is given below:

(1) Define continuous functions on $\sigma_A(a) \subseteq \mathbb{R}$ by $f_+(t) := \max\{t, 0\}$ and $f_-(t) := \max\{-t, 0\}$. These functions satisfy
$$f_+(t) - f_-(t) = t, \tag{10.2}$$
$$f_+(t)f_-(t) = 0 \tag{10.3}$$
for every $t \in \sigma_A(a)$. Define $a_\pm := f_\pm(a)$. By the continuous spectral mapping theorem, we have
$$\sigma_A(a_\pm) = f_\pm(\sigma_A(a)) \subseteq [0, \infty). \tag{10.4}$$
Thus, $a_\pm \in A_+$. Moreover, the continuous functional calculus gives
$$a_+ - a_- = (f_+ - f_-)(a) = a, \tag{10.5}$$
$$a_+a_- = (f_+f_-)(a) = 0. \tag{10.6}$$
Taking adjoints also gives $a_-a_+ = 0$. To prove uniqueness, define $|a| := (a^2)^{1/2}$. The scalar identities $f_\pm(t) = \frac{|t|\pm t}{2}$ give $a_\pm = \frac{|a|\pm a}{2}$, Hence, a_+ and a_- are uniquely determined by a.
(2) Let
$$f_n \colon \sigma_A(a) \longrightarrow [0, \infty), \tag{10.7}$$
$$t \longmapsto t^{1/n}. \tag{10.8}$$
The function f_n is continuous. Define $b := f_n(a)$. By the continuous spectral mapping theorem,
$$\sigma_A(b) = f_n(\sigma_A(a)) \subseteq [0, \infty), \tag{10.9}$$
so $b \in A_+$. Moreover,
$$b^n = f_n(a)^n = f_n^n(a) = a. \tag{10.10}$$
Suppose that $c \in A_+$ also satisfies $c^n = a$. Compatibility of the continuous functional calculus with composition gives
$$c = (c^n)^{1/n} = a^{1/n} = b. \tag{10.11}$$
Therefore, the positive n-th root is unique.

This completes the proof. □

Lemma 10.4 gives a norm characterization of positivity and provides a useful method for proving that a self-adjoint element is positive.

Lemma 10.4. *Let A be a unital C^*-algebra, and let $a \in A$ be self-adjoint. Then the following statements are equivalent:*

(1) $a \geq 0$.
(2) *For every $\alpha \geq \|a\|$, $\|\alpha e - a\| \leq \alpha$.*
(3) *There exists $\alpha \geq \|a\|$ such that $\|\alpha e - a\| \leq \alpha$.*

Proof. Suppose first that $a \geq 0$, and let $\alpha \geq \|a\|$. Since a is positive, $\sigma_A(a) \subseteq [0, \|a\|]$. For every $\lambda \in \sigma_A(a)$, we therefore have

$$0 \leq \lambda \leq \|a\| \leq \alpha. \tag{10.12}$$

It follows that $|\alpha - \lambda| = \alpha - \lambda \leq \alpha$. The element $\alpha e - a$ is self-adjoint. Hence, it is normal, and the continuous spectral mapping theorem gives

$$\sigma_A(\alpha e - a) = \{\alpha - \lambda : \lambda \in \sigma_A(a)\}. \tag{10.13}$$

Since the norm of a normal element equals its spectral radius, we obtain

$$\|\alpha e - a\| = \max_{\mu \in \sigma_A(\alpha e - a)} |\mu| = \max_{\lambda \in \sigma_A(a)} |\alpha - \lambda| \leq \alpha. \tag{10.14}$$

Thus, the first statement implies the second. The second statement immediately implies the third. Conversely, suppose that there exists $\alpha \geq \|a\|$ such that $\|\alpha e - a\| \leq \alpha$. Let $\lambda \in \sigma_A(a)$. Since a is self-adjoint, $\lambda \in \mathbb{R}$. By the continuous spectral mapping theorem, $\alpha - \lambda \in \sigma_A(\alpha e - a)$. Therefore,

$$|\alpha - \lambda| \leq r(\alpha e - a) = \|\alpha e - a\| \leq \alpha. \tag{10.15}$$

Since $\alpha - \lambda$ is real, the inequality $|\alpha - \lambda| \leq \alpha$ is equivalent to

$$-\alpha \leq \alpha - \lambda \leq \alpha \iff -2\alpha \leq -\lambda \leq 0 \iff 0 \leq \lambda \leq 2\alpha. \tag{10.16}$$

Thus, $\sigma_A(a) \subseteq [0, \infty)$, and hence $a \geq 0$. □

Proposition 10.5 establishes several fundamental properties of positive elements and of the positive cone A_+.

Proposition 10.5. *Let A be a unital C^*-algebra.*

(1) *If $a \in A_+$ and $t \geq 0$, then $ta \in A_+$.*
(2) *The set A_+ is norm closed.*
(3) *If $a, b \in A_+$, then $a + b \in A_+$.*
(4) *If $a, b \in A_+$ and $ab = ba$, then $ab \in A_+$.*
(5) *For $a \in A$, the following statements are equivalent:*
   (a) *$a \geq 0$.*
   (b) *There exists a self-adjoint element $b \in A$ such that $a = b^2$.*
   (c) *There exists $c \in A$ such that $a = c^*c$.*
(6) *The relation*

$$a \leq b \iff b - a \in A_+ \tag{10.17}$$

defines a partial order on $A_{\mathrm{sa}} := \{a \in A : a = a^*\}$.

Proof. The proof is given below:

(1) Suppose that $a \geq 0$ and $t \geq 0$. The polynomial spectral mapping theorem gives

$$\sigma_A(ta) = t\sigma_A(a) \subseteq [0, \infty). \tag{10.18}$$

Thus, $ta \geq 0$.

(2) Let $(a_n)_{n\geq 1}$ be a sequence in A_+ such that $a_n \to a$. Since the involution is continuous, we have $a_n^* \to a^*$. Since $a_n^* = a_n$ for every n, uniqueness of limits gives $a^* = a$. By Lemma 10.4, we have

$$\| \|a_n\| e - a_n \| \leq \|a_n\| \tag{10.19}$$

for every $n \in \mathbb{N}$. Since $\|a_n\| \to \|a\|$, passing to the limit gives

$$\| \|a\| e - a \| \leq \|a\|. \tag{10.20}$$

Another application of Lemma 10.4 gives $a \geq 0$. Therefore, A_+ is closed.

(3) Let $a, b \in A_+$ and set $\alpha := \|a\| + \|b\|$. Using Lemma 10.4, we obtain

$$\begin{aligned} \|\alpha e - (a+b)\| &= \|(\|a\|e - a) + (\|b\|e - b)\| && (10.21)\\ &\le \|\|a\|e - a\| + \|\|b\|e - b\| && (10.22)\\ &\le \|a\| + \|b\| && (10.23)\\ &= \alpha. && (10.24) \end{aligned}$$

Since $\alpha \ge \|a+b\|$, Lemma 10.4 implies that $a + b \ge 0$.

(4) Suppose that $a, b \in A_+$ and $ab = ba$. Since a and b are self-adjoint and commute, the unital C^*-algebra $C^*(e, a, b)$ is commutative. Under the Gelfand transform, a and b correspond to nonnegative continuous functions. Their product is therefore nonnegative. Hence, $ab \ge 0$.

(5) Suppose that $a \ge 0$. Let $b := a^{1/2}$ by Proposition 10.3(2). Then $b = b^*$ and $a = b^2$. Thus, (a) implies (b). Suppose that $a = b^2$ for some self-adjoint $b \in A$. Then $a = b^*b$. Thus, (b) implies (c). Suppose that $a = c^*c$ for some $c \in A$. We first establish the claim that if $x \in A$ satisfies $x^*x \le 0$, then $x = 0$. Write $x = h + ik$, where

$$h = \frac{x + x^*}{2}, \qquad k = \frac{x - x^*}{2i} \tag{10.25}$$

are self-adjoint. Since the nonzero spectra of x^*x and xx^* agree by Lemma 3.3(1), the assumption $x^*x \le 0$ implies $xx^* \le 0$. Consequently, $x^*x + xx^* \le 0$. On the other hand,

$$x^*x + xx^* = 2(h^2 + k^2). \tag{10.26}$$

The polynomial spectral mapping theorem shows that h^2 and k^2 are positive. By (3), we have $2(h^2 + k^2) \ge 0$. Thus, the self-adjoint element $x^*x + xx^*$ is both positive and negative. Its spectrum is therefore contained in $\{0\}$. Since it is normal, we have

$$\|2(h^2 + k^2)\| = \|x^*x + xx^*\| = r(x^*x + xx^*) = 0. \tag{10.27}$$

Hence, $h^2 + k^2 = 0$. Since $h^2, k^2 \ge 0$, we have $h^2 = -k^2$ and $k^2 = -h^2$ Thus, h^2 and k^2 are both positive and negative, and hence $h^2 = k^2 = 0$. The C^*-identity gives

$$\begin{aligned} \|h\|^2 &= \|h^2\| = 0, && (10.28)\\ \|k\|^2 &= \|k^2\| = 0. && (10.29) \end{aligned}$$

Therefore, $h = k = 0$, and hence $x = 0$. We now apply the claim to $a = c^*c$. The element a is self-adjoint. Let $a = a_+ - a_-$ be its positive-negative decomposition by Proposition 10.3(1), and define $y := ca_-^{1/2}$. Since a_+ and a_- are continuous functions of a, they commute with a and with each other. Moreover, $a_+a_- = 0$. The functions defining a_+ and $a_-^{1/2}$ have disjoint supports. Thus, continuous functional calculus gives $a_+a_-^{1/2} = 0$. Therefore,

$$\begin{aligned} y^*y &= a_-^{1/2}c^*ca_-^{1/2} && (10.30)\\ &= a_-^{1/2}aa_-^{1/2} && (10.31)\\ &= a_-^{1/2}(a_+ - a_-)a_-^{1/2} && (10.32)\\ &= a_-^{1/2}a_+a_-^{1/2} - a_-^{1/2}a_-a_-^{1/2} = -a_-^{1/2}a_-a_-^{1/2} = -a_-^2 \le 0. && (10.33) \end{aligned}$$

Hence, $y = 0$ and $a_-^2 = 0$. Since a_- is self-adjoint, the C^*-identity gives $\|a_-\|^2 = \|a_-^2\| = 0$. Thus, $a_- = 0$, and consequently $a = a_+ \ge 0$. Therefore, (c) implies (a).

(6) Reflexivity follows from $0 \in A_+$. Suppose that $a \le b$ and $b \le a$. Then $b - a \ge 0$ and $a - b \ge 0$. Thus,
$$\sigma_A(b-a) \subseteq [0,\infty) \cap (-\infty, 0] = \{0\}. \tag{10.34}$$
Since $b-a$ is self-adjoint, $\|b-a\| = r(b-a) = 0$. Hence, $a = b$, proving antisymmetry. Finally, if $a \le b$ and $b \le c$, then $c - a = (c-b) + (b-a)$. Both terms on the right are positive, so (3) gives $c - a \ge 0$. Therefore, $a \le c$, proving transitivity.

This completes the proof. □

Remark 10.6. The commutativity assumption in Proposition 10.5(4) is necessary. For example, consider
$$a = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{10.35}$$
as elements of $M_2(\mathbb{C})$. Both a and b are positive, but
$$ab = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \tag{10.36}$$
is not self-adjoint and is therefore not positive.

The preceding results remain valid for nonunital C^*-algebras. They may be proved by passing to the unitization and using the nonunital continuous functional calculus. In particular, the functions used to define positive parts and positive roots vanish at zero, so the resulting elements belong to the original nonunital algebra.

10.2. **Positive Linear Functionals and States.** Positivity of elements allows us to define positive linear functionals. These functionals will later provide the analytic data used to construct representations of a C^*-algebra in the Gelfand–Naimark–Segal construction in Section 11.

Definition 10.7. Let A be a C^*-algebra. A linear functional $\varphi\colon A \to \mathbb{C}$ is positive if $\varphi(a) \ge 0$ for every $a \in A_+$.

Equivalently, by Proposition 10.5(5), φ is positive if $\varphi(a^*a) \ge 0$ for every $a \in A$. Proposition 10.8 records the fundamental properties of positive linear functionals. Proposition 10.8 records some basic positive linear functionals.

Proposition 10.8. *Let A be a C^*-algebra, and let φ be a positive linear functional on A.*

(1) *The functional φ preserves the involution. That is, $\varphi(a^*) = \overline{\varphi(a)}$ for every $a \in A$.*
(2) *For every $a, b \in A$, the Cauchy–Schwarz inequality holds:*
$$|\varphi(b^*a)|^2 \le \varphi(a^*a)\varphi(b^*b). \tag{10.37}$$
(3) *If A is unital, then φ is bounded and $\|\varphi\| = \varphi(e)$.*

Proof. The proof is given below:

(1) By Proposition 6.13, it suffices to show that φ maps self-adjoint elements to self-adjoint elements. Let $a = a^*$. By Proposition 10.3, there exist $a_+, a_- \in A_+$ such that $a = a_+ - a_-$. Since φ is positive, $\varphi(a) = \varphi(a_+) - \varphi(a_-) \in \mathbb{R}$. Thus, φ maps self-adjoint elements to real numbers.
(2) Let $a, b \in A$ and $\lambda \in \mathbb{C}$. Since φ is positive,
$$0 \le \varphi((\lambda a + b)^*(\lambda a + b)) \tag{10.38}$$
$$= |\lambda|^2\varphi(a^*a) + \overline{\lambda}\varphi(a^*b) + \lambda\varphi(b^*a) + \varphi(b^*b). \tag{10.39}$$

By (1), $\varphi(a^*b) = \overline{\varphi(b^*a)}$. Hence,

$$|\lambda|^2\varphi(a^*a) + 2\operatorname{Re}(\lambda\varphi(b^*a)) + \varphi(b^*b) \geq 0. \tag{10.40}$$

Choose $\gamma \in \mathbb{S}^1$ such that $\gamma\varphi(b^*a) = |\varphi(b^*a)|$. For every $t \in \mathbb{R}$, set $\lambda = t\gamma$. Then

$$t^2\varphi(a^*a) + 2t|\varphi(b^*a)| + \varphi(b^*b) \geq 0 \tag{10.41}$$

for every $t \in \mathbb{R}$. The discriminant of this real quadratic polynomial is nonpositive. Therefore,

$$4|\varphi(b^*a)|^2 \leq 4\varphi(a^*a)\varphi(b^*b). \tag{10.42}$$

Thus,

$$|\varphi(b^*a)|^2 \leq \varphi(a^*a)\varphi(b^*b). \tag{10.43}$$

(3) By Proposition 10.5(5), the element a^*a is positive. Hence, $0 \leq a^*a$. Moreover, the C^*-identity gives $\|a^*a\| = \|a\|^2$. Since a^*a is positive,

$$\sigma_A(a^*a) \subseteq [0, \|a^*a\|] = [0, \|a\|^2]. \tag{10.44}$$

Thus, the function

$$\sigma_A(a^*a) \longrightarrow [0, \infty), \tag{10.45}$$
$$t \longmapsto \|a\|^2 - t \tag{10.46}$$

is nonnegative. Applying the continuous functional calculus gives $\|a\|^2 e - a^*a \geq 0$. Therefore,

$$0 \leq a^*a \leq \|a\|^2 e. \tag{10.47}$$

Positivity of φ gives $\varphi(a^*a) \leq \|a\|^2\varphi(e)$. Applying (2) with $b = e$ gives

$$|\varphi(a)|^2 = |\varphi(e^*a)|^2 \leq \varphi(a^*a)\varphi(e) \leq \|a\|^2\varphi(e)^2. \tag{10.48}$$

Consequently, $|\varphi(a)| \leq \varphi(e)\|a\|$. Thus, φ is bounded and $\|\varphi\| \leq \varphi(e)$. Since $\|e\| = 1$, we also have $\varphi(e) \leq \|\varphi\|$. Therefore, $\|\varphi\| = \varphi(e)$.

This completes the proof. $\square$

Positive linear functionals provide the natural class of functionals compatible with the order structure of a C^*-algebra. By imposing a normalization condition, we obtain the distinguished class of positive functionals known as states. Definition 10.9 introduces this notion.

Definition 10.9. Let A be a C^*-algebra. A **state** on A is a positive $\varphi\colon A \to \mathbb{C}$ such that $\|\varphi\| = 1$. The set of states on A is denoted by

$$S(A) := \{\varphi \in A^* : \varphi \geq 0,\ \|\varphi\| = 1\} \tag{10.49}$$

and is called the **state space** of A.

Corollary 10.10. *Let A be a unital C^*-algebra, and let φ be a positive linear functional on A. Then φ is a state if and only if $\varphi(e) = 1$.*

Proof. This follows immediately from Proposition 10.8(3). $\square$

10.3. **Representations of C^*-Algebras.** We now introduce representations of abstract C^*-algebras as concrete algebras of bounded operators on Hilbert spaces. Representations are essential for the Gelfand–Naimark–Segal construction, which will be developed in Section 11. In this subsection, we review the basic definitions and properties of representations of C^*-algebras.

Definition 10.11. Let A be a C^*-algebra. A **representation** of A is a pair $(\pi, \mathscr{H})$, where $\mathscr{H}$ is a Hilbert space and $\pi\colon A \to \mathscr{B}(\mathscr{H})$ is a $*$-homomorphism. Two representations $(\pi_1, \mathscr{H}_1)$ and $(\pi_2, \mathscr{H}_2)$ of A are **unitarily equivalent** if there exists a unitary operator $U\colon \mathscr{H}_1 \to \mathscr{H}_2$ such that

$$U\pi_1(a) = \pi_2(a)U \tag{10.50}$$

for every $a \in A$. In this case, we write $(\pi_1, \mathscr{H}_1) \sim_U (\pi_2, \mathscr{H}_2)$.

A representation $(\pi, \mathscr{H})$ of a C^*-algebra A is nondegenerate if

$$\overline{\operatorname{span}\{\pi(a)x : a \in A,\ x \in \mathscr{H}\}} = \mathscr{H}. \tag{10.51}$$

We abbreviate the linear span appearing above by $\pi(A)\mathscr{H}$. Lemma 10.12 gives two useful characterizations of nondegeneracy.

Lemma 10.12. *Let A be a C^*-algebra, and let $(\pi, \mathscr{H})$ be a representation of A.*

(1) *The representation π is nondegenerate if and only if*

$$\bigcap_{a\in A} \ker(\pi(a)) = \{0\}. \tag{10.52}$$

(2) *If A is unital, then π is nondegenerate if and only if $\pi(e) = I_{\mathscr{H}}$.*

Proof. The proof is given below:

(1) Suppose first that π is nondegenerate, and let $x \in \bigcap_{a\in A} \ker(\pi(a))$. For every $a \in A$ and $y \in \mathscr{H}$,

$$\langle x, \pi(a)y\rangle = \langle \pi(a^*)x, y\rangle = 0. \tag{10.53}$$

Thus, $x \in (\pi(A)\mathscr{H})^\perp$. Since π is nondegenerate, $(\pi(A)\mathscr{H})^\perp = \{0\}$, and hence $x = 0$. Conversely, suppose that $\bigcap_{a\in A} \ker(\pi(a)) = \{0\}$. Let $x \in (\pi(A)\mathscr{H})^\perp$. For every $a \in A$ and $y \in \mathscr{H}$,

$$\langle \pi(a)x, y\rangle = \langle x, \pi(a^*)y\rangle = 0, \tag{10.54}$$

because $\pi(a^*)y \in \pi(A)\mathscr{H}$. Therefore, $\pi(a)x = 0$ for every $a \in A$. The assumed kernel condition gives $x = 0$. Hence, $(\pi(A)\mathscr{H})^\perp = \{0\}$, and consequently $\overline{\pi(A)\mathscr{H}} = \mathscr{H}$.

(2) Suppose that A is unital. If $\pi(e) = I_{\mathscr{H}}$, then

$$x = \pi(e)x \in \pi(A)\mathscr{H} \tag{10.55}$$

for every $x \in \mathscr{H}$. Thus, π is nondegenerate. Conversely, suppose that π is nondegenerate. For every $a \in A$ and $x \in \mathscr{H}$,

$$\pi(e)\pi(a)x = \pi(a)x. \tag{10.56}$$

Therefore, $\pi(e)$ agrees with $I_{\mathscr{H}}$ on the dense subspace $\pi(A)\mathscr{H}$. Since both operators are bounded, $\pi(e) = I_{\mathscr{H}}$.

This completes the proof. □

Representations may be combined by taking direct sums, allowing several actions of a C^*-algebra to be viewed simultaneously on a single Hilbert space. We recall this construction in Definition 10.13.

Definition 10.13. Let A be a C^*-algebra, and let $\{(\pi_\iota, \mathscr{H}_\iota) : \iota \in I\}$ be a family of representations of A. Define

$$\bigoplus_{\iota\in I} \mathscr{H}_\iota = \{(x_\iota)_{\iota\in I} : \sum_{\iota\in I} \|x_\iota\|^2 < \infty\}. \tag{10.57}$$

The **direct sum representation**

$$\bigoplus_{\iota\in I} \pi_\iota \colon A \longrightarrow \mathscr{B}(\bigoplus_{\iota\in I} \mathscr{H}_\iota) \tag{10.58}$$

is defined by

$$\left(\bigoplus_{\iota\in I} \pi_\iota\right)(a)(x_\iota)_{\iota\in I} = (\pi_\iota(a)x_\iota)_{\iota\in I}. \tag{10.59}$$

Proposition 10.14 shows that every representation decomposes into a nondegenerate part and a zero part.

Proposition 10.14. *Let $(\pi, \mathscr{H})$ be a representation of a C^*-algebra A. Define $\mathscr{H}_0 := \overline{\pi(A)\mathscr{H}}$ and $\mathscr{H}_1 := \mathscr{H}_0^\perp$. Then $\mathscr{H}_0$ and $\mathscr{H}_1$ are invariant under π, the restriction $\pi_0 := \pi|_{\mathscr{H}_0}$ is nondegenerate, and $\pi|_{\mathscr{H}_1} = 0$. Consequently,*

$$(\pi, \mathscr{H}) \sim_U (\pi_0, \mathscr{H}_0) \oplus (0, \mathscr{H}_1). \tag{10.60}$$

Proof. The subspace $\mathscr{H}_0$ is invariant because

$$\pi(a)\pi(b)x = \pi(ab)x \in \pi(A)\mathscr{H} \tag{10.61}$$

for every $a, b \in A$ and $x \in \mathscr{H}$. Since π preserves the involution, Proposition 10.16 below shows that $\mathscr{H}_1 = \mathscr{H}_0^\perp$ is also invariant. By construction, $\overline{\pi_0(A)\mathscr{H}_0} = \mathscr{H}_0$, so π_0 is nondegenerate. If $x \in \mathscr{H}_1$, then for every $a \in A$ and $y \in \mathscr{H}$,

$$\langle \pi(a)x, y\rangle = \langle x, \pi(a^*)y\rangle = 0, \tag{10.62}$$

because $\pi(a^*)y \in \mathscr{H}_0$. Hence, $\pi(a)x = 0$ for every $a \in A$. Thus, π vanishes on $\mathscr{H}_1$, which proves the decomposition. □

Irreducible representations are the basic building blocks in the representation theory of C^*-algebras, since they admit no nontrivial decomposition into smaller subrepresentations. To formulate this notion, we first recall what it means for a closed subspace to be preserved by a representation. This leads to the definitions of invariant subspaces and irreducible representations in Definition 10.15.

Definition 10.15. Let $(\pi, \mathscr{H})$ be a representation of a C^*-algebra A. A closed subspace $\mathscr{N} \subseteq \mathscr{H}$ is **invariant** under π if $\pi(a)\mathscr{N} \subseteq \mathscr{N}$ for every $a \in A$. The representation $(\pi, \mathscr{H})$ is **irreducible** if it is nonzero and its only closed invariant subspaces are $\{0\}$ and $\mathscr{H}$.

For representations of C^*-algebras, invariant subspaces automatically have invariant orthogonal complements. This is formalized in Proposition 10.16.

Proposition 10.16. *Let $(\pi, \mathscr{H})$ be a representation of a C^*-algebra A, and let $\mathscr{N} \subseteq \mathscr{H}$ be a closed subspace. Then $\mathscr{N}$ is invariant under π if and only if $\mathscr{N}^\perp$ is invariant under π.*

Proof. Suppose that $\mathscr{N}$ is invariant. Let $x \in \mathscr{N}^\perp$, $y \in \mathscr{N}$, and $a \in A$. Then

$$\langle \pi(a)x, y\rangle = \langle x, \pi(a)^*y\rangle = \langle x, \pi(a^*)y\rangle = 0, \tag{10.63}$$

because $\pi(a^*)y \in \mathscr{N}$. Therefore, $\pi(a)x \in \mathscr{N}^\perp$. Thus, $\mathscr{N}^\perp$ is invariant. The converse follows by applying the same argument to $\mathscr{N}^\perp$ and using $(\mathscr{N}^\perp)^\perp = \mathscr{N}$. □

If $\mathscr{N}$ is a closed invariant subspace, then

$$(\pi, \mathscr{H}) \sim_U (\pi|_{\mathscr{N}}, \mathscr{N}) \oplus (\pi|_{\mathscr{N}^\perp}, \mathscr{N}^\perp). \tag{10.64}$$

Thus, an irreducible representation is precisely a representation that cannot be decomposed nontrivially as an orthogonal direct sum of subrepresentations. Definition 10.17 introduces cyclic vectors and cyclic representations.

Definition 10.17. Let $(\pi, \mathscr{H})$ be a representation of a C^*-algebra A. A vector $x \in \mathscr{H}$ is **cyclic** for π if $\overline{\pi(A)x} = \mathscr{H}$, The representation $(\pi, \mathscr{H})$ is **cyclic** if it possesses a cyclic vector.

Irreducibility is equivalent to the stronger condition that every nonzero vector is cyclic. This is formalized in Proposition 10.18.

Proposition 10.18. *Let $(\pi, \mathscr{H})$ be a nondegenerate representation of a C^*-algebra A. Then π is irreducible if and only if every nonzero vector in $\mathscr{H}$ is cyclic.*

Proof. Suppose that π is irreducible, and let $x \in \mathscr{H}$ be nonzero. The subspace $\mathscr{H}_x := \overline{\pi(A)x}$ is closed and invariant under π. By irreducibility, $\mathscr{H}_x = \{0\}$ or $\mathscr{H}_x = \mathscr{H}$. If $\mathscr{H}_x = \{0\}$, then $\pi(a)x = 0$ for every $a \in A$. Since π is nondegenerate, Lemma 10.12 implies that $x = 0$, which is a contradiction. Therefore, $\mathscr{H}_x = \mathscr{H}$, and x is cyclic. Conversely, suppose that every nonzero vector in $\mathscr{H}$ is cyclic. Let $\mathscr{N} \subseteq \mathscr{H}$ be a nonzero closed invariant subspace, and choose a nonzero vector $x \in \mathscr{N}$. Since $\mathscr{N}$ is invariant, $\overline{\pi(A)x} \subseteq \mathscr{N}$. Since x is cyclic, $\overline{\pi(A)x} = \mathscr{H}$. Therefore, $\mathscr{N} = \mathscr{H}$, and hence π is irreducible. □

For unital C^*-algebras, every nondegenerate representation decomposes as a direct sum of cyclic representations. This is formalized in Proposition 10.19.

Proposition 10.19. *Let A be a unital C^*-algebra, and let $(\pi, \mathscr{H})$ be a nondegenerate representation of A. Then π is unitarily equivalent to a direct sum of cyclic representations.*

Proof. By Zorn's lemma, there exists a maximal set $S \subseteq \mathscr{H}$ of unit vectors such that the cyclic subspaces $\mathscr{H}_x := \overline{\pi(A)x}$ for $x \in S$ are pairwise orthogonal. Since π is unital, $x = \pi(e)x \in \mathscr{H}_x$. Each $\mathscr{H}_x$ is invariant under π, and the restriction $\pi_x := \pi|_{\mathscr{H}_x}$ is cyclic with cyclic vector x. We claim that

$$\mathscr{H} = \bigoplus_{x \in S} \mathscr{H}_x. \tag{10.65}$$

Suppose otherwise. Then there exists a unit vector $x_0 \in \left(\bigoplus_{x\in S} \mathscr{H}_x\right)^\perp$. For $a, b \in A$ and $x \in S$,

$$\langle \pi(a)x_0, \pi(b)x \rangle = \langle x_0, \pi(a^*b)x \rangle = 0. \tag{10.66}$$

Thus, $\mathscr{H}_{x_0}$ is orthogonal to $\mathscr{H}_x$ for every $x \in S$. This contradicts the maximality of S. Therefore, $\mathscr{H} = \bigoplus_{x\in S} \mathscr{H}_x$. The decomposition gives

$$\pi \sim_U \bigoplus_{x \in S} \pi_x. \tag{10.67}$$

This completes the proof. □

We conclude the section by introducing intertwiners and proving Schur's lemma for irreducible representations.

Definition 10.20. Let $(\pi_1, \mathscr{H}_1)$ and $(\pi_2, \mathscr{H}_2)$ be representations of a C^*-algebra A. The **space of intertwiners** from π_1 to π_2 is

$$\operatorname{Hom}(\pi_1, \pi_2) := \{T \in \mathscr{B}(\mathscr{H}_1, \mathscr{H}_2) : T\pi_1(a) = \pi_2(a)T \text{ for every } a \in A\}. \tag{10.68}$$

The representations π_1 and π_2 are **disjoint** if $\mathrm{Hom}(\pi_1, \pi_2) = \{0\}$. The **commutant** of $\pi(A)$ is

$$\pi(A)' := \mathrm{Hom}(\pi, \pi) = \{T \in \mathscr{B}(\mathscr{H}) : T\pi(a) = \pi(a)T \text{ for every } a \in A\}. \tag{10.69}$$

Proposition 10.21 (Schur's lemma)**.** *Let $(\pi, \mathscr{H})$ be a nondegenerate representation of a C^*-algebra A. Then π is irreducible if and only if $\pi(A)' = \mathbb{C}I_{\mathscr{H}}$.*

Proof. Suppose first that $\pi(A)' = \mathbb{C}I_{\mathscr{H}}$. Let $\mathscr{N} \subseteq \mathscr{H}$ be a closed invariant subspace. By Proposition 10.16, $\mathscr{N}^{\perp}$ is also invariant. Let $P_{\mathscr{N}}$ denote the orthogonal projection onto $\mathscr{N}$. Since both $\mathscr{N}$ and $\mathscr{N}^{\perp}$ are invariant,

$$P_{\mathscr{N}}\pi(a) = \pi(a)P_{\mathscr{N}} \tag{10.70}$$

for every $a \in A$. Thus, $P_{\mathscr{N}} \in \pi(A)'$. It follows that $P_{\mathscr{N}} = \lambda I_{\mathscr{H}}$ for some $\lambda \in \mathbb{C}$. Since $P_{\mathscr{N}}$ is a projection, $\lambda \in \{0, 1\}$. Therefore, $\mathscr{N} = \{0\}$ or $\mathscr{N} = \mathscr{H}$, and hence π is irreducible. Conversely, suppose that π is irreducible. The commutant $\pi(A)'$ is a unital C^*-subalgebra of $\mathscr{B}(\mathscr{H})$. It suffices to show that every self-adjoint element of $\pi(A)'$ is a scalar multiple of $I_{\mathscr{H}}$. Let $T \in \pi(A)'$ be self-adjoint. Suppose that $\sigma(T)$ contains two distinct points λ and μ. Choose nonzero functions $f, g \in C(\sigma(T))$ with disjoint supports such that $f(\lambda) \neq 0$ and $g(\mu) \neq 0$. Then $fg = 0$. Define $X := f(T)$, and $Y := g(T)$. Since $\pi(A)'$ is a unital C^*-algebra containing T, the continuous functional calculus gives $X, Y \in \pi(A)'$. Moreover, the functional calculus is isometric, so $X, Y \neq 0$. Since $fg = 0$, $XY = YX = 0$. Therefore, $\mathrm{ran}(X) \subseteq \ker(Y)$. Because $Y \in \pi(A)'$, the closed subspace $\ker(Y)$ is invariant under π. Since $X \neq 0$, the kernel $\ker(Y)$ is nonzero. Irreducibility therefore gives $\ker(Y) = \mathscr{H}$. Hence, $Y = 0$, contradicting the choice of g. Thus, $\sigma(T) = \{\lambda\}$ for some $\lambda \in \mathbb{R}$. Since $T - \lambda I_{\mathscr{H}}$ is self-adjoint,

$$\|T - \lambda I_{\mathscr{H}}\| = r(T - \lambda I_{\mathscr{H}}) = 0. \tag{10.71}$$

Therefore, $T = \lambda I_{\mathscr{H}}$. □

Schur's lemma can be used to characterize disjoint irreducible representations. This is established in Corollary 10.22.

Corollary 10.22. *Let $(\pi_1, \mathscr{H}_1)$ and $(\pi_2, \mathscr{H}_2)$ be irreducible representations of a C^*-algebra A. Then π_1 and π_2 are disjoint if and only if they are not unitarily equivalent.*

Proof. Suppose that π_1 and π_2 are not unitarily equivalent. Let $T \in \mathrm{Hom}(\pi_1, \pi_2)$. Then $T^*T \in \pi_1(A)'$ and $TT^* \in \pi_2(A)'$. By Proposition 10.21, there exist $\lambda, \mu \geq 0$ such that $T^*T = \lambda I_{\mathscr{H}_1}$ and $TT^* = \mu I_{\mathscr{H}_2}$. If $T \neq 0$, then $\lambda, \mu > 0$. Moreover,

$$\lambda = \|T^*T\| = \|T\|^2 = \|TT^*\| = \mu. \tag{10.72}$$

Hence, $\lambda = \mu$. Define $U := \frac{1}{\sqrt{\lambda}}T$. Then $U^*U = I_{\mathscr{H}_1}$ and $UU^* = I_{\mathscr{H}_2}$ Thus, U is unitary. Since U is also an intertwiner, we have $U\pi_1(a) = \pi_2(a)U$ for every $a \in A$. This contradicts the assumption that π_1 and π_2 are not unitarily equivalent. Therefore, $T = 0$, and the representations are disjoint. Conversely, if π_1 and π_2 are unitarily equivalent, then the unitary implementing the equivalence is a nonzero element of $\mathrm{Hom}(\pi_1, \pi_2)$. Therefore, unitarily equivalent representations are not disjoint. □

11. Gelfand–Naimark–Segal Theorem

In this section, we construct representations of abstract C^*-algebras on Hilbert spaces. Multiplication allows the elements of a C^*-algebra to act linearly on the algebra itself. However, a C^*-algebra is not generally a Hilbert space. States provide the additional structure required to construct an inner product and thereby produce Hilbert space representations. This is the content of the Gelfand–Naimark–Segal construction. The construction leads to the Gelfand–Naimark representation theorem, which characterizes C^*-algebras as norm-closed $*$-subalgebras of bounded operators on Hilbert spaces. We first observe in Lemma 11.1 that a representation and a unit vector determine a state.

Lemma 11.1. *Let A be a unital C^*-algebra, let $(\pi, \mathscr{H})$ be a nondegenerate representation of A, and let $x \in \mathscr{H}$ satisfy $\|x\| = 1$. Define*

$$\varphi_x \colon A \longrightarrow \mathbb{C}, \tag{11.1}$$

$$a \longmapsto \langle \pi(a)x, x\rangle. \tag{11.2}$$

Then φ_x is a state on A.

Proof. For every $a \in A$, we have

$$\varphi_x(a^*a) = \langle \pi(a^*a)x, x\rangle \tag{11.3}$$

$$= \langle \pi(a)^*\pi(a)x, x\rangle \tag{11.4}$$

$$= \langle \pi(a)x, \pi(a)x\rangle = \|\pi(a)x\|^2 \geq 0. \tag{11.5}$$

Thus, φ_x is positive. Since π is nondegenerate and A is unital, Lemma 10.12 gives $\pi(e) = I_{\mathscr{H}}$. Therefore,

$$\varphi_x(e) = \langle \pi(e)x, x\rangle = \langle x, x\rangle = 1. \tag{11.6}$$

It follows from Corollary 10.10 that φ_x is a state. $\square$

The Gelfand–Naimark–Segal construction establishes a converse to Lemma 11.1. That is, every state arises from a cyclic representation and a cyclic unit vector. The Gelfand–Naimark–Segal construction is detailed in Proposition 11.2.

Proposition 11.2 (Gelfand–Naimark–Segal construction)**.** *Let A be a unital C^*-algebra, and let φ be a state on A. Then there exist a nondegenerate representation*

$$\pi_\varphi \colon A \longrightarrow \mathscr{B}(\mathscr{H}_\varphi) \tag{11.7}$$

and a cyclic unit vector $\xi_\varphi \in \mathscr{H}_\varphi$ such that

$$\varphi(a) = \langle \pi_\varphi(a)\xi_\varphi, \xi_\varphi\rangle_\varphi \tag{11.8}$$

for every $a \in A$. Moreover, this triple is unique up to unitary equivalence. More precisely, suppose that $\pi \colon A \to \mathscr{B}(\mathscr{H})$ is a representation with a cyclic unit vector $x \in \mathscr{H}$ such that $\varphi(a) = \langle \pi(a)x, x\rangle$ for every $a \in A$. Then there exists a unique unitary operator

$$U \colon \mathscr{H}_\varphi \longrightarrow \mathscr{H} \tag{11.9}$$

such that

$$U\xi_\varphi = x, \tag{11.10}$$

$$U\pi_\varphi(a) = \pi(a)U \tag{11.11}$$

for every $a \in A$.

Proof. Define

$$\mathscr{N}_\varphi := \{a \in A : \varphi(a^*a) = 0\}. \tag{11.12}$$

We first prove that $\mathscr{N}_\varphi$ is a closed left ideal of A. Since the map

$$A \longrightarrow \mathbb{C}, \tag{11.13}$$

$$a \longmapsto \varphi(a^*a) \tag{11.14}$$

is continuous, $\mathscr{N}_\varphi$ is closed. Let $a, b \in \mathscr{N}_\varphi$. By Proposition 10.8(2),

$$|\varphi(b^*a)|^2 \leq \varphi(a^*a)\varphi(b^*b) = 0. \tag{11.15}$$

Thus, $\varphi(b^*a) = 0$. It follows that, for every $\lambda, \mu \in \mathbb{C}$,

$$\varphi((\lambda a + \mu b)^*(\lambda a + \mu b)) = |\lambda|^2\varphi(a^*a) + \overline{\lambda}\mu\varphi(a^*b) + \lambda\overline{\mu}\varphi(b^*a) + |\mu|^2\varphi(b^*b) = 0. \tag{11.16}$$

Therefore, $\lambda a + \mu b \in \mathscr{N}_\varphi$, so $\mathscr{N}_\varphi$ is a vector subspace. Now let $a \in \mathscr{N}_\varphi$. Define

$$\psi\colon A \longrightarrow \mathbb{C}, \tag{11.17}$$
$$c \longmapsto \varphi(a^*ca). \tag{11.18}$$

The functional ψ is positive. Indeed, if $c \geq 0$, then by Proposition 10.5(5), there exists $d \in A$ such that $c = d^*d$. Hence,

$$\psi(c) = \varphi(a^*d^*da) = \varphi((da)^*(da)) \geq 0. \tag{11.19}$$

Moreover, $\psi(e) = \varphi(a^*a) = 0$. By Proposition 10.8(3), $\|\psi\| = \psi(e) = 0$. Thus, $\psi = 0$. For every $b \in A$, we therefore have

$$\varphi((ba)^*(ba)) = \varphi(a^*b^*ba) = \psi(b^*b) = 0. \tag{11.20}$$

Hence, $ba \in \mathscr{N}_\varphi$. Thus, $\mathscr{N}_\varphi$ is a closed left ideal. Define a sesquilinear form on A by

$$\langle a, b\rangle_\varphi := \varphi(b^*a). \tag{11.21}$$

This form is linear in the first variable, conjugate-linear in the second variable, and positive semidefinite. We now define an inner product on $V_\varphi := A/\mathscr{N}_\varphi$ by

$$\langle a + \mathscr{N}_\varphi, b + \mathscr{N}_\varphi\rangle_\varphi := \varphi(b^*a). \tag{11.22}$$

To prove that this is well-defined, let $x, y \in \mathscr{N}_\varphi$ and $a, b \in A$. By Proposition 10.8(2),

$$|\varphi(b^*x)|^2 \leq \varphi(x^*x)\varphi(b^*b) = 0, \tag{11.23}$$
$$|\varphi(y^*a)|^2 \leq \varphi(a^*a)\varphi(y^*y) = 0, \tag{11.24}$$
$$|\varphi(y^*x)|^2 \leq \varphi(x^*x)\varphi(y^*y) = 0. \tag{11.25}$$

Therefore,

$$\varphi((b + y)^*(a + x)) = \varphi(b^*a) + \varphi(b^*x) + \varphi(y^*a) + \varphi(y^*x) = \varphi(b^*a). \tag{11.26}$$

Thus, the inner product is well-defined. Moreover,

$$\langle a + \mathscr{N}_\varphi, a + \mathscr{N}_\varphi\rangle_\varphi = \varphi(a^*a) \tag{11.27}$$

vanishes precisely when $a \in \mathscr{N}_\varphi$. Hence, it is an inner product. Let $\mathscr{H}_\varphi$ be the Hilbert space completion of V_φ. For $a, b \in A$, define

$$\pi_\varphi(a)(b + \mathscr{N}_\varphi) := ab + \mathscr{N}_\varphi. \tag{11.28}$$

Since $\mathscr{N}_\varphi$ is a left ideal, this map is well-defined on V_φ. We next prove boundedness. We have $0 \leq \|a\|^2e - a^*a$. Consequently, for every $b \in A$,

$$0 \leq b^*(\|a\|^2e - a^*a)b = \|a\|^2b^*b - b^*a^*ab. \tag{11.29}$$

Applying the positive functional φ gives $\varphi(b^*a^*ab) \leq \|a\|^2\varphi(b^*b)$. Thus,

$$\begin{aligned}
\|\pi_\varphi(a)(b+\mathscr{N}_\varphi)\|_\varphi^2 &= \varphi((ab)^*(ab)) && (11.30)\\
&= \varphi(b^*a^*ab) && (11.31)\\
&\leq \|a\|^2\varphi(b^*b) && (11.32)\\
&= \|a\|^2\|b+\mathscr{N}_\varphi\|_\varphi^2. && (11.33)
\end{aligned}$$

Hence, $\|\pi_\varphi(a)\| \leq \|a\|$. Therefore, $\pi_\varphi(a)$ extends uniquely to a bounded operator on $\mathscr{H}_\varphi$. The map π_φ is linear and multiplicative. For $a, b, c \in A$, we have

$$\begin{aligned}
\langle \pi_\varphi(a)(b+\mathscr{N}_\varphi), c+\mathscr{N}_\varphi\rangle_\varphi &= \langle ab+\mathscr{N}_\varphi, c+\mathscr{N}_\varphi\rangle_\varphi && (11.34)\\
&= \varphi(c^*ab) && (11.35)\\
&= \varphi((a^*c)^*b) && (11.36)\\
&= \langle b+\mathscr{N}_\varphi, a^*c+\mathscr{N}_\varphi\rangle_\varphi && (11.37)\\
&= \langle b+\mathscr{N}_\varphi, \pi_\varphi(a^*)(c+\mathscr{N}_\varphi)\rangle_\varphi. && (11.38)
\end{aligned}$$

Thus, $\pi_\varphi(a)^* = \pi_\varphi(a^*)$. Therefore, $\pi_\varphi\colon A \to \mathscr{B}(\mathscr{H}_\varphi)$ is a representation. Define $\xi_\varphi := e + \mathscr{N}_\varphi$. Since φ is a state,

$$\|\xi_\varphi\|_\varphi^2 = \varphi(e^*e) = \varphi(e) = 1. \qquad (11.39)$$

Thus, ξ_φ is a unit vector. For every $a \in A$,

$$\pi_\varphi(a)\xi_\varphi = a + \mathscr{N}_\varphi. \qquad (11.40)$$

Hence, $\overline{\pi_\varphi(A)\xi_\varphi} = \mathscr{H}_\varphi$, so ξ_φ is cyclic. In particular, π_φ is nondegenerate. Finally,

$$\langle \pi_\varphi(a)\xi_\varphi, \xi_\varphi\rangle_\varphi = \langle a+\mathscr{N}_\varphi, e+\mathscr{N}_\varphi\rangle_\varphi = \varphi(e^*a) = \varphi(a). \qquad (11.41)$$

We now prove uniqueness. Let $\pi\colon A \to \mathscr{B}(\mathscr{H})$ be another representation with a cyclic unit vector $x \in \mathscr{H}$ such that $\varphi(a) = \langle \pi(a)x, x\rangle$ for every $a \in A$. Define

$$\begin{aligned}
U_0\colon V_\varphi &\longrightarrow \pi(A)x, && (11.42)\\
a+\mathscr{N}_\varphi &\longmapsto \pi(a)x. && (11.43)
\end{aligned}$$

For every $a, b \in A$,

$$\begin{aligned}
\langle U_0(a+\mathscr{N}_\varphi), U_0(b+\mathscr{N}_\varphi)\rangle &= \langle \pi(a)x, \pi(b)x\rangle && (11.44)\\
&= \langle \pi(b)^*\pi(a)x, x\rangle && (11.45)\\
&= \langle \pi(b^*a)x, x\rangle && (11.46)\\
&= \varphi(b^*a) && (11.47)\\
&= \langle a+\mathscr{N}_\varphi, b+\mathscr{N}_\varphi\rangle_\varphi. && (11.48)
\end{aligned}$$

Thus, U_0 is well-defined and isometric. Since x is cyclic, $\overline{\pi(A)x} = \mathscr{H}$. Therefore, U_0 extends uniquely to a unitary operator $U\colon \mathscr{H}_\varphi \to \mathscr{H}$. Moreover,

$$U\xi_\varphi = U_0(e+\mathscr{N}_\varphi) = \pi(e)x = x, \qquad (11.49)$$

and, for every $a, b \in A$,

$$\begin{aligned}
U\pi_\varphi(a)(b+\mathscr{N}_\varphi) &= U_0(ab+\mathscr{N}_\varphi) && (11.50)\\
&= \pi(ab)x && (11.51)\\
&= \pi(a)\pi(b)x && (11.52)\\
&= \pi(a)U_0(b+\mathscr{N}_\varphi). && (11.53)
\end{aligned}$$

By density, this gives $U\pi_\varphi(a) = \pi(a)U$ for every $a \in A$. The cyclicity of ξ_φ shows that these conditions uniquely determine U. □

Remark 11.3. A state φ is called pure if it cannot be written as a nontrivial convex combination of two distinct states. One can show that φ is pure if and only if φ *isirreducible*

To construct a faithful representation, we need a sufficiently rich family of states to separate the elements of the C^*-algebra. The existence of such states is established in Lemma 11.4 and Corollary 11.5.

Lemma 11.4. *Let A be a unital C^*-algebra.*

(1) *Let $\varphi \in A^*$ satisfy $\|\varphi\| = \varphi(e) = 1$. Then φ is a state.*
(2) *If $a \in A$ is a nonzero self-adjoint element, then there exists a state ψ on A such that $|\psi(a)| = \|a\|$.*

Proof. The proof is given below:

(1) Skipped.
(2) Since a is self-adjoint, it is normal. The spectrum $\sigma_A(a)$ is compact, and the function
$$\sigma_A(a) \longrightarrow [0, \infty), \tag{11.54}$$
$$\lambda \longmapsto |\lambda| \tag{11.55}$$
attains its maximum. Since $\|a\| = r(a)$, there exists $\lambda_0 \in \sigma_A(a)$ such that $|\lambda_0| = \|a\|$. Let $B := C^*(e, a) \cong C(\sigma_A(a))$. Evaluation at λ_0 defines a state on $C(\sigma_A(a))$ and hence a state $\omega\colon B \to \mathbb{C}$ such that $\omega(a) = \lambda_0$. By the Hahn–Banach theorem, ω extends to a bounded linear functional $\psi \in A^*$ such that $\|\psi\| = \|\omega\| = 1$. Moreover, $\psi(e) = \omega(e) = 1$. (1) implies that ψ is a state. Finally, $|\psi(a)| = |\lambda_0| = \|a\|$.

This completes the proof. □

Corollary 11.5. *Let A be a unital C^*-algebra, and let $a \in A$ be nonzero. Then there exists a state φ_a on A such that $\varphi_a(a^*a) = \|a\|^2$.*

Proof. The element a^*a is nonzero, self-adjoint, and positive. By Lemma 11.4(2), there exists a state φ_a such that
$$|\varphi_a(a^*a)| = \|a^*a\|. \tag{11.56}$$
Since φ_a is positive and $a^*a \geq 0$, we have $\varphi_a(a^*a) \geq 0$. Thus,
$$\varphi_a(a^*a) = \|a^*a\| = \|a\|^2. \tag{11.57}$$
This completes the proof. □

The Gelfand–Naimark–Segal construction from Proposition 11.2 associates a Hilbert space representation to each state on a C^*-algebra. By taking enough of these representations together, one can obtain a representation that detects every nonzero element. This allows us to show in Proposition 11.6 that every abstract C^*-algebra can be realized concretely as a norm-closed $*$-subalgebra of bounded operators on a Hilbert space.

Proposition 11.6 (Gelfand–Naimark–Segal)**.** *Let A be a C^*-algebra. Then there exist a Hilbert space $\mathscr{H}$ and an isometric $*$-isomorphism from A onto a norm-closed $*$-subalgebra of $\mathscr{B}(\mathscr{H})$.*

Proof. Suppose first that A is unital. For every nonzero $a \in A$, choose a state φ_a as in Corollary 11.5. Let
$$\mathscr{H}_u := \bigoplus_{a \in A \setminus \{0\}} \mathscr{H}_{\varphi_a}, \tag{11.58}$$

and define the direct sum representation

$$\pi_u := \bigoplus_{a \in A \setminus \{0\}} \pi_{\varphi_a} \colon A \longrightarrow \mathscr{B}(\mathscr{H}_u). \tag{11.59}$$

We claim that π_u is injective. Let $a \in A$ be nonzero. The component indexed by a satisfies

$$\|\pi_{\varphi_a}(a)\xi_{\varphi_a}\|^2 = \langle \pi_{\varphi_a}(a^*a)\xi_{\varphi_a}, \xi_{\varphi_a} \rangle = \varphi_a(a^*a) = \|a\|^2 > 0. \tag{11.60}$$

Thus, $\pi_{\varphi_a}(a) \neq 0$, and hence $\pi_u(a) \neq 0$. Therefore, π_u is injective. By Corollary 9.7, every injective $*$-homomorphism between C^*-algebras is isometric. Hence, $\|\pi_u(a)\| = \|a\|$ for every $a \in A$. Since A is complete and π_u is an isometry, $\pi_u(A)$ is norm closed in $\mathscr{B}(\mathscr{H}_u)$. Now let A be nonunital, and let A' be its unitization. Applying the unital case to A', we obtain a Hilbert space $\mathscr{H}$ and an injective isometric $*$-homomorphism

$$\rho \colon A' \longrightarrow \mathscr{B}(\mathscr{H}). \tag{11.61}$$

The restriction $rho|_A$ is an injective isometric $*$-homomorphism. Since A is a closed subspace of A', its image $\rho(A)$ is norm closed. This proves the result. $\square$

The representation constructed in Proposition 11.6 is called the universal representation. With some additional work, the states appearing in its construction can be chosen to be pure. Since pure states give rise to irreducible GNS representations, it follows that the universal representation may be realized as a direct sum of irreducible representations.

Part 3. C^*-Algebras II

We now study further examples and structural properties of C^*-algebras. Our focus includes concrete constructions such as finite-dimensional C^*-algebras, universal C^*-algebras, group C^*-algebras, crossed products, and other naturally arising examples.

12. Finite-Dimensional C^*-Algebras

We prove that every finite-dimensional C^*-algebra is isometrically $*$-isomorphic to a finite direct sum of matrix algebras over $\mathbb{C}$. Thus, finite-dimensional C^*-algebras admit a complete and explicit structural description. The proof combines the Gelfand–Naimark–Segal construction with the double commutant theorem[5]. We begin by decomposing finite-dimensional representations into irreducible representations. This is done in Lemma 12.1.

Lemma 12.1. *Let A be a unital C^*-algebra, and let $\pi\colon A \to \mathscr{B}(\mathscr{H})$ be a unital representation on a finite-dimensional Hilbert space $\mathscr{H}$. Then there exist irreducible representations $\pi_1, \dots, \pi_r$ such that*

$$\pi \simeq \pi_1 \oplus \cdots \oplus \pi_r. \tag{12.1}$$

Proof. We proceed by induction on $\dim \mathscr{H}$. If $\dim \mathscr{H} = 1$, then every nonzero unital representation on $\mathscr{H}$ is irreducible. Suppose that the result holds for representations on Hilbert spaces of dimension strictly less than $\dim \mathscr{H}$. If π is irreducible, there is nothing to prove. Otherwise, there exists a nonzero proper closed subspace $\mathscr{N} \subsetneq \mathscr{H}$ that is invariant under π. By Proposition 10.16, the orthogonal complement $\mathscr{N}^\perp$ is also invariant under π. Consequently,

$$\pi \simeq \pi|_{\mathscr{N}} \oplus \pi|_{\mathscr{N}^\perp}. \tag{12.2}$$

Since both $\mathscr{N}$ and $\mathscr{N}^\perp$ have dimension strictly less than $\dim \mathscr{H}$, the induction hypothesis applies to both restricted representations. Therefore, each restriction is a finite direct sum of irreducible representations, and hence so is π. $\square$

To apply Lemma 12.1, we first construct a faithful representation of a finite-dimensional unital C^*-algebra on a finite-dimensional Hilbert space. We do this by first producing a faithful state and then applying the Gelfand–Naimark–Segal construction from Proposition 11.2.

Lemma 12.2. *Let A be a unital finite-dimensional C^*-algebra. Then there exist a finite-dimensional Hilbert space $\mathscr{H}$ and a faithful representation $\pi\colon A \to \mathscr{B}(\mathscr{H})$. Moreover, π may be chosen in the form*

$$\pi = \pi_1 \oplus \cdots \oplus \pi_s, \tag{12.3}$$

where $\pi_1, \dots, \pi_s$ are pairwise disjoint irreducible representations.

Proof. By Proposition 11.6, there exist a Hilbert space $\mathscr{H}$ and a faithful representation $\rho\colon A \to \mathscr{B}(\mathscr{H})$. Since ρ is injective, it is isometric by Corollary 9.7. Let $S_A := \{a \in A : \|a\| = 1\}$ be the unit sphere of A. For each $a \in S_A$, choose a unit vector $\xi_a \in \mathscr{H}$ such that[6] $\|\rho(a)\xi_a\| > 1/2$. For each $a \in S_A$, define

$$U_a := \{b \in S_A : \|\rho(b)\xi_a\| > 1/2\}. \tag{12.4}$$

[5] Let $\mathscr{H}$ be a Hilbert space, and let $B \subseteq \mathscr{B}(\mathscr{H})$ be a unital $*$-subalgebra. The double commutant theorem states that the strong and weak operator closures of B are both equal to B''. Consequently, B is strongly or weakly closed if and only if $B = B''$.

[6] Since ρ is a faithful $*$-homomorphism, it is isometric. Therefore, $\|\rho(a)\| = \|a\| = 1$ for every $a \in S_A$. Hence, there exists a unit vector $\xi_a \in \mathscr{H}$ such that $\|\rho(a)\xi_a\| > 1/2$.

The sets U_a form an open cover of S_A. Since A is finite dimensional, S_A is compact. Therefore, there exist $a_1, \dots, a_m \in S_A$ such that $S_A = U_{a_1} \cup \cdots \cup U_{a_m}$. Set $\xi_j := \xi_{a_j}$ and define

$$\omega(a) := \frac{1}{m}\sum_{j=1}^{m}\langle \rho(a)\xi_j, \xi_j\rangle \tag{12.5}$$

for every $a \in A$. Each summand is positive. Indeed, let $\omega_j(a) := \langle \rho(a)\xi_j, \xi_j\rangle$. for each $j \in \{1, \dots, m\}$. If $a \in A_+$, then by Proposition 10.5(5), there exists $b \in A$ such that $a = b^*b$. Therefore,

$$\omega_j(a) = \langle \rho(b^*b)\xi_j, \xi_j\rangle = \langle \rho(b)^*\rho(b)\xi_j, \xi_j\rangle = \|\rho(b)\xi_j\|^2 \geq 0. \tag{12.6}$$

Thus, each ω_j is positive. Since ω is a positive linear combination of the functionals $\omega_1, \dots, \omega_m$, it is also positive. by Proposition 10.5(3). Moreover, since ρ is unital and each ξ_j is a unit vector,

$$\omega(e) = \frac{1}{m}\sum_{j=1}^{m}\langle \xi_j, \xi_j\rangle = 1. \tag{12.7}$$

Thus, ω is a state by Corollary 10.10 We claim that ω is faithful. That is, if $x \in A_+$ satisfies $\omega(x) = 0$, then $x = 0$. By Proposition 10.5(5), every positive element of A has the form a^*a for some $a \in A$. It therefore suffices to prove that $\omega(a^*a) = 0$ implies $a = 0$. Suppose that $\omega(a^*a) = 0$. Then

$$0 = \omega(a^*a) = \frac{1}{m}\sum_{j=1}^{m}\langle \rho(a^*a)\xi_j, \xi_j\rangle = \frac{1}{m}\sum_{j=1}^{m}\|\rho(a)\xi_j\|^2. \tag{12.8}$$

Hence, $\rho(a)\xi_j = 0$ for every j. If $a \neq 0$, then $a/\|a\| \in S_A$, so there exists j such that $a/\|a\| \in U_{a_j}$. This gives

$$\left\|\rho\left(\frac{a}{\|a\|}\right)\xi_j\right\| > \frac{1}{2}, \tag{12.9}$$

contradicting $\rho(a)\xi_j = 0$. Therefore, $a = 0$, and ω is faithful. Let $(\pi_\omega, \mathscr{H}_\omega, \Omega_\omega)$ be the GNS representation associated with ω. Since $\mathscr{H}_\omega = \overline{\pi_\omega(A)\Omega_\omega}$ and A is finite dimensional, the subspace $\pi_\omega(A)\Omega_\omega$ is finite dimensional and therefore closed. Hence, $\mathscr{H}_\omega$ is finite dimensional. We show that π_ω is faithful. Suppose that $\pi_\omega(a) = 0$. Then

$$\omega(a^*a) = \langle \pi_\omega(a^*a)\Omega_\omega, \Omega_\omega\rangle = \langle \pi_\omega(a)^*\pi_\omega(a)\Omega_\omega, \Omega_\omega\rangle = \|\pi_\omega(a)\Omega_\omega\|^2. \tag{12.10}$$

Since ω is faithful, it follows that $a = 0$. Thus, π_ω is faithful. By Lemma 12.1, there exist irreducible representations $\pi_1, \dots, \pi_r$ such that

$$\pi_\omega \simeq \pi_1 \oplus \cdots \oplus \pi_r. \tag{12.11}$$

Define an equivalence relation on $I := \{1, \dots, r\}$ by declaring $i \sim j$ if and only if π_i and π_j are unitarily equivalent. Choose a subset $R \subseteq I$ containing exactly one representative from each equivalence class, and define

$$\pi := \bigoplus_{i \in R}\pi_i. \tag{12.12}$$

Equivalent representations have the same kernel. Consequently,

$$\ker\pi = \bigcap_{i \in R}\ker\pi_i = \bigcap_{i=1}^{r}\ker\pi_i = \ker\pi_\omega = \{0\}. \tag{12.13}$$

The third equality follows because π_ω is unitarily equivalent to $\bigoplus_{i=1}^{r}\pi_i$, and the kernel of a direct sum of representations is the intersection of the kernels of its summands. Thus, π is faithful. Its irreducible summands are pairwise inequivalent and hence pairwise disjoint by Corollary 10.22. □

By Lemma 12.2, every unital finite-dimensional C^*-algebra admits a faithful representation that decomposes into pairwise disjoint irreducible representations. Computing the commutant and double commutant of this representation identifies the matrix blocks that appear in the algebra. This leads to the classification theorem in Proposition 12.3.

Proposition 12.3. *Let A be a unital finite-dimensional C^*-algebra. Then there exist positive integers $n_1, \ldots, n_s$ such that*

$$A \cong \bigoplus_{i=1}^{s} M_{n_i}(\mathbb{C}) \tag{12.14}$$

isometrically as $*$-algebras.

Proof. By Lemma 12.2, there exist pairwise disjoint finite-dimensional irreducible representations $(\pi_i, \mathscr{H}_i)$, where $1 \le i \le s$, such that the representation

$$\pi := \bigoplus_{i=1}^{s} \pi_i \colon A \longrightarrow \mathscr{B}\left(\bigoplus_{i=1}^{s} \mathscr{H}_i\right) \tag{12.15}$$

is faithful. Write an operator $T \in \mathscr{B}(\bigoplus_{i=1}^{s} \mathscr{H}_i)$ as a block matrix $T = (T_{ij})$, where $T_{ij} \in \mathscr{B}(\mathscr{H}_j, \mathscr{H}_i)$. The condition $T \in \pi(A)'$ is equivalent to

$$T_{ij}\pi_j(a) = \pi_i(a)T_{ij} \tag{12.16}$$

for every $a \in A$ and $1 \le i, j \le s$. Thus, $T_{ij} \in \operatorname{Hom}(\pi_j, \pi_i)$. Since the representations are pairwise disjoint, $T_{ij} = 0$ whenever $i \ne j$. By Schur's lemma (Proposition 10.21), $T_{ii} \in \mathbb{C}I_{\mathscr{H}_i}$. Therefore,

$$\pi(A)' = \{\operatorname{diag}(\lambda_1 I_{\mathscr{H}_1}, \ldots, \lambda_s I_{\mathscr{H}_s}) : \lambda_1, \ldots, \lambda_s \in \mathbb{C}\}. \tag{12.17}$$

An operator commutes with every element of $\pi(A)'$ if and only if it is block diagonal with respect to the decomposition $\bigoplus_{i=1}^{s} \mathscr{H}_i$. Consequently,

$$\pi(A)'' = \{\operatorname{diag}(T_1, \ldots, T_s) : T_i \in \mathscr{B}(\mathscr{H}_i)\} \cong \bigoplus_{i=1}^{s} \mathscr{B}(\mathscr{H}_i). \tag{12.18}$$

Set $n_i := \dim \mathscr{H}_i$. Then $\mathscr{B}(\mathscr{H}_i) \cong M_{n_i}(\mathbb{C})$. Since $\bigoplus_{i=1}^{s} \mathscr{H}_i$ is finite dimensional, the algebra $\pi(A)$ is closed in the weak and strong operator topologies. The double commutant theorem therefore gives $\pi(A) = \pi(A)''$. Since π is faithful, it is an isometric $*$-isomorphism from A onto $\pi(A)$. Hence,

$$A \cong \pi(A) = \pi(A)'' \cong \bigoplus_{i=1}^{s} M_{n_i}(\mathbb{C}). \tag{12.19}$$

This completes the proof. □

The converse of Proposition 12.3 is also true. That is, every finite direct sum of matrix algebras is a unital finite-dimensional C^*-algebra. Indeed, by Example 6.10(7), if $A_1, \ldots, A_s$ are C^*-algebras, then their direct sum is a C^*-algebra under componentwise operations and the norm

$$\|(a_1, \ldots, a_s)\| = \max_{1 \le i \le s} \|a_i\|_{A_i}. \tag{12.20}$$

In particular, the direct sum appearing in Proposition 12.3 carries the norm

$$\|(a_1, \ldots, a_s)\| = \max_{1 \le i \le s} \|a_i\|_{M_{n_i}(\mathbb{C})}. \tag{12.21}$$

Thus, Proposition 12.3 gives a complete classification of unital finite-dimensional C^*-algebras. As a further consequence, the assumption of unitality can in fact be removed.

Corollary 12.4. *Every nonzero finite-dimensional C^*-algebra is unital.*

Proof. Let A be a nonzero finite-dimensional C^*-algebra. If A is already unital, there is nothing to prove. Suppose that A is nonunital, and let A' be its unitization. Since $A' = A \oplus \mathbb{C}$ as a vector space, A' is finite dimensional. By Proposition 12.3, there exist positive integers $n_1, \dots, n_s$ and a $*$-isomorphism

$$\Phi\colon A' \longrightarrow \bigoplus_{i=1}^{s} M_{n_i}(\mathbb{C}). \tag{12.22}$$

Since A is a closed two-sided ideal of A', its image $J := \Phi(A)$ is a two-sided ideal of $\bigoplus_{i=1}^{s} M_{n_i}(\mathbb{C})$. For each i, let p_i denote the central projection whose i-th component is I and whose remaining components are zero. Thus, $p_1 + \cdots + p_s = e$. Since J is a two-sided ideal, $p_i x \in J$ for every $x \in J$. Moreover,

$$x = (p_1 + \cdots + p_s)x = p_1 x + \cdots + p_s x. \tag{12.23}$$

Each element $p_i x$ has only its i-th component possibly nonzero. Therefore, $J = \bigoplus_{i=1}^{s} p_i J$. Indeed, the inclusion from left to right follows from the preceding decomposition, while the reverse inclusion holds because $p_i J \subseteq J$ for every i. The sum is direct because its summands are supported in distinct matrix blocks. Under the natural identification $p_i \left(\bigoplus_{k=1}^{s} M_{n_k}(\mathbb{C})\right) \cong M_{n_i}(\mathbb{C})$, the subspace $J_i := p_i J$ is a two-sided ideal of $M_{n_i}(\mathbb{C})$. Since every matrix algebra $M_{n_i}(\mathbb{C})$ is simple, each J_i is either $\{0\}$ or $M_{n_i}(\mathbb{C})$. Therefore, there exists a nonempty subset $F \subseteq \{1, \dots, s\}$ such that

$$J = \bigoplus_{j \in F} M_{n_j}(\mathbb{C}). \tag{12.24}$$

The algebra on the right is unital, with identity given by the element whose j-th component is I for $j \in F$ and whose remaining components are zero. Hence, J, and therefore A, is unital. $\square$

13. Inductive Limits of C^*-Algebras

Inductive limits allow us to construct a potentially infinite-dimensional C^*-algebra from a sequence of simpler C^*-algebras connected by $*$-homomorphisms. To define inductive limits, we first introduce a bounded product that will serve as the ambient C^*-algebra for the construction. Let $\{A_i\}_{i \in I}$ be a family of C^*-algebras. Define the bounded product

$$\prod_{i \in I}^{\infty} A_i := \left\{ (a_i)_{i \in I} : a_i \in A_i \text{ for every } i \in I \text{ and } \sup_{i \in I} \|a_i\|_{A_i} < \infty \right\}. \tag{13.1}$$

Lemma 13.1. *Let $\{A_i\}_{i \in I}$ be a family of C^*-algebras. Then $\prod_{i \in I}^{\infty} A_i$ is a C^*-algebra with respect to the norm*

$$\|(a_i)_{i \in I}\| := \sup_{i \in I} \|a_i\|_{A_i}, \tag{13.2}$$

and the algebraic operations and involution defined coordinatewise.

Proof. It is straightforward to verify that $\prod_{i \in I}^{\infty} A_i$ is a $*$-algebra and that the given norm is submultiplicative. We verify the C^*-identity. For every $(a_i)_{i \in I} \in \prod_{i \in I}^{\infty} A_i$, we have

$$\|(a_i)_{i \in I}^*(a_i)_{i \in I}\| = \sup_{i \in I} \|a_i^* a_i\|_{A_i} = \sup_{i \in I} \|a_i\|_{A_i}^2 = \left(\sup_{i \in I} \|a_i\|_{A_i}\right)^2 = \|(a_i)_{i \in I}\|^2. \tag{13.3}$$

We now prove completeness. Let $(a^n)_{n\geq 1}$ be a Cauchy sequence in $\prod_{i\in I}^{\infty} A_i$, where $a^n = (a_i^n)_{i\in I}$. For every fixed $i \in I$, the sequence $(a_i^n)_{n\geq 1}$ is Cauchy in A_i. Thus, there exists $a_i \in A_i$ such that $a_i^n \to a_i$. Set $a := (a_i)_{i\in I}$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$\|a^n - a^m\| < \varepsilon \tag{13.4}$$

whenever $n, m \geq N$. Fix $n_0 \geq N$. For every $i \in I$, we have

$$\|a_i\|_{A_i} \leq \|a_i^{n_0}\|_{A_i} + \|a_i^{n_0} - a_i\|_{A_i} \leq \|a^{n_0}\| + \varepsilon. \tag{13.5}$$

Hence,

$$\sup_{i\in I} \|a_i\|_{A_i} \leq \|a^{n_0}\| + \varepsilon < \infty. \tag{13.6}$$

Thus, $a \in \prod_{i\in I}^{\infty} A_i$. For every $n \geq N$ and every $i \in I$, we have

$$\|a_i^n - a_i\|_{A_i} = \lim_{m\to\infty} \|a_i^n - a_i^m\|_{A_i} \leq \limsup_{m\to\infty} \|a^n - a^m\| \leq \varepsilon. \tag{13.7}$$

Therefore, $\|a^n - a\| \leq \varepsilon$ for every $n \geq N$. Hence, $a^n \to a$, and $\prod_{i\in I}^{\infty} A_i$ is complete. □

To construct inductive limits, we require a C^*-algebra that contains finitely supported families of elements from the algebras A_i. Its closure will provide the ambient algebra in which the inductive limit is formed. We define the algebraic direct sum of $\{A_i\}_{i\in I}$ as the subalgebra of $\prod_{i\in I}^{\infty} A_i$ given by

$$\bigoplus_{i\in I}^{\mathrm{alg}} A_i := \left\{ (a_i)_{i\in I} \in \prod_{i\in I}^{\infty} A_i : a_i \neq 0 \text{ for only finitely many } i \in I \right\}. \tag{13.8}$$

The c_0-direct sum is the norm closure of the algebraic direct sum $\bigoplus_{i\in I} A_i := \overline{\bigoplus_{i\in I}^{\mathrm{alg}} A_i}$. Lemma 13.2 shows that $\bigoplus_{i\in I} A_i$ is a C^*-algebra.

Lemma 13.2. *Let $\{A_i\}_{i\in I}$ be a family of C^*-algebras. Then $\bigoplus_{i\in I} A_i$ is a C^*-algebra.*

Proof. The algebraic direct sum $\bigoplus_{i\in I}^{\mathrm{alg}} A_i$ is a two-sided ideal of $\prod_{i\in I}^{\infty} A_i$. Indeed, if $a = (a_i)_{i\in I} \in \bigoplus_{i\in I}^{\mathrm{alg}} A_i$ and $b = (b_i)_{i\in I} \in \prod_{i\in I}^{\infty} A_i$, then both ab and ba are supported on the finite support of a. Hence, $ab, ba \in \bigoplus_{i\in I}^{\mathrm{alg}} A_i$. Therefore, its closure $\bigoplus_{i\in I} A_i$ is a closed two-sided ideal of $\prod_{i\in I}^{\infty} A_i$. In particular, $\bigoplus_{i\in I} A_i$ is a C^*-algebra. □

The quotient by the c_0-direct sum identifies bounded sequences that differ by a sequence converging to zero in norm. Thus, it retains the asymptotic behavior of a bounded sequence while disregarding finitely many terms and, more generally, norm-vanishing errors. Let

$$\pi \colon \prod_{i\in I}^{\infty} A_i \longrightarrow \prod_{i\in I}^{\infty} A_i \Big/ \bigoplus_{i\in I} A_i \tag{13.9}$$

denote the quotient map. Lemma 13.3 is a useful consequence in the case $I = \mathbb{N}$.

Lemma 13.3. *Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of C^*-algebras, and let $a = (a_n)_{n\in\mathbb{N}} \in \prod_{n\in\mathbb{N}}^{\infty} A_n$. Then*

$$\|\pi(a)\| = \limsup_{n\to\infty} \|a_n\|_{A_n}. \tag{13.10}$$

In particular, $a \in \bigoplus_{n\in\mathbb{N}} A_n$ if and only if $\lim_{n\to\infty} \|a_n\|_{A_n} = 0$.

Proof. Since $\bigoplus_{n\in\mathbb{N}}^{\mathrm{alg}} A_n$ is dense in $\bigoplus_{n\in\mathbb{N}} A_n$, the quotient norm gives

$$\|\pi(a)\| = \inf\left\{ \|a-b\| : b \in \bigoplus_{n\in\mathbb{N}}^{\mathrm{alg}} A_n \right\}. \tag{13.11}$$

Every element $b = (b_n)_{n\in\mathbb{N}} \in \bigoplus_{n\in\mathbb{N}}^{\mathrm{alg}} A_n$ satisfies $b_n = 0$ for all sufficiently large n. Therefore,

$$\|a-b\| = \sup_{n\in\mathbb{N}} \|a_n - b_n\|_{A_n} \geq \limsup_{n\to\infty} \|a_n - b_n\|_{A_n} = \limsup_{n\to\infty} \|a_n\|_{A_n}. \tag{13.12}$$

Thus, $\|\pi(a)\| \geq \limsup_{n\to\infty} \|a_n\|_{A_n}$. Conversely, for every $k \in \mathbb{N}$, define $b^k = (b_n^k)_{n\in\mathbb{N}}$ by

$$b_n^k = \begin{cases} a_n, & n \leq k, \\ 0, & n > k. \end{cases} \tag{13.13}$$

Then $b^k \in \bigoplus_{n\in\mathbb{N}}^{\mathrm{alg}} A_n$, and

$$\|\pi(a)\| \leq \inf_{k\in\mathbb{N}} \|a - b^k\| = \inf_{k\in\mathbb{N}} \sup_{n>k} \|a_n\|_{A_n} = \limsup_{n\to\infty} \|a_n\|_{A_n}. \tag{13.14}$$

This proves the first statement. The second statement follows because a belongs to the kernel of π if and only if $\|\pi(a)\| = 0$. □

An inductive sequence is a sequence of objects $(A_n)_{n\geq 1}$ together with morphisms $\varphi_n \colon A_n \to A_{n+1}$, usually written as

$$A_1 \xrightarrow{\varphi_1} A_2 \xrightarrow{\varphi_2} A_3 \xrightarrow{\varphi_3} \cdots. \tag{13.15}$$

For $m > n$, define

$$\varphi_{m,n} := \varphi_{m-1} \circ \varphi_{m-2} \circ \cdots \circ \varphi_n, \tag{13.16}$$

and set $\varphi_{n,n} := \mathrm{Id}_{A_n}$. An inductive limit is a colimit of an inductive sequence, denoted by $\varinjlim(A_n, \varphi_n)$, or more briefly by $\varinjlim A_n$. For the construction below, we extend the notation by setting $\varphi_{m,n} := 0$ whenever $m < n$.

Example 13.4. Let D be a C^*-algebra, and let $A_n \subseteq A_{n+1}$ be an increasing sequence of C^*-subalgebras of D. Let $\varphi_n \colon A_n \hookrightarrow A_{n+1}$ denote the inclusion map. Then

$$A := \overline{\bigcup_{n=1}^{\infty} A_n} \tag{13.17}$$

is the inductive limit of the system (A_n, φ_n). Indeed, let $\mu_n \colon A_n \hookrightarrow A$ denote the inclusion map. Then $\mu_n = \mu_{n+1} \circ \varphi_n$ for every $n \in \mathbb{N}$. Suppose that B is a C^*-algebra and that $\lambda_n \colon A_n \to B$ are $*$-homomorphisms satisfying $\lambda_n = \lambda_{n+1} \circ \varphi_n$. Define

$$\lambda_0 \colon \bigcup_{n=1}^{\infty} A_n \longrightarrow B, \tag{13.18}$$

$$a \longmapsto \lambda_n(a). \tag{13.19}$$

whenever $a \in A_n$. This map is well defined. Indeed, if $a \in A_n \subseteq A_{n+1}$, then

$$\lambda_{n+1}(a) = \lambda_{n+1}(\varphi_n(a)) = \lambda_n(a). \tag{13.20}$$

This map is a $*$-homomorphism because each λ_n is a $*$-homomorphism and the maps agree on the overlaps $A_n \subseteq A_{n+1}$. Moreover, for every $a \in A_n$,

$$\|\lambda(a)\| = \|\lambda_n(a)\| \leq \|a\|, \tag{13.21}$$

since $*$-homomorphisms between C^*-algebras are contractive by Proposition 6.13(2). Thus, λ is contractive on the dense subalgebra $\bigcup_{n\in\mathbb{N}} A_n$. If (a_k) is a sequence in $\bigcup_{n\in\mathbb{N}} A_n$ converging to an element $a \in A$, then

$$\|\lambda(a_k) - \lambda(a_\ell)\| \le \|a_k - a_\ell\|. \tag{13.22}$$

Hence, $(\lambda(a_k))$ is Cauchy in B, and we may define $\lambda(a) := \lim_{k\to\infty} \lambda(a_k)$. The same estimate shows that this definition is independent of the choice of the approximating sequence. Therefore, λ extends uniquely to a contractive $*$-homomorphism $\lambda\colon A \to B$. such that $\lambda \circ \mu_n = \lambda_n$ for every $n \in \mathbb{N}$. The uniqueness of λ follows from the density of $\bigcup_{n=1}^{\infty} A_n$ in A.

We now establish the existence of inductive limits for sequences of C^*-algebras. This result ensures that every inductive sequence admits a C^*-algebra satisfying the universal property. This is proved in Proposition 13.5.

Proposition 13.5. *Let (A_n, φ_n) be an inductive system of C^*-algebras. Then there exists an inductive limit $A = \varinjlim A_n$ with corresponding $*$-homomorphisms $\mu_n\colon A_n \to A$ such that the following statements hold:*

(1) $A = \overline{\bigcup_{n=1}^{\infty} \mu_n(A_n)}$.
(2) *For every $n \in \mathbb{N}$ and every $a \in A_n$,*

$$\|\mu_n(a)\| = \lim_{m\to\infty} \|\varphi_{m,n}(a)\|_{A_m}. \tag{13.23}$$

(3) *For every $n \in \mathbb{N}$,*

$$\ker(\mu_n) = \left\{ a \in A_n : \lim_{m\to\infty} \|\varphi_{m,n}(a)\|_{A_m} = 0 \right\}. \tag{13.24}$$

Moreover, the inductive limit is unique up to a unique $*$-isomorphism.

Proof. Consider the quotient map

$$\pi\colon \prod_{m\in\mathbb{N}}^{\infty} A_m \longrightarrow \prod_{m\in\mathbb{N}}^{\infty} A_m \Big/ \bigoplus_{m\in\mathbb{N}} A_m. \tag{13.25}$$

For each $n \in \mathbb{N}$, define $\nu_n\colon A_n \longrightarrow \prod_{m\in\mathbb{N}}^{\infty} A_m$ by

$$\nu_n(a)_m := \begin{cases} \varphi_{m,n}(a), & m \ge n, \\ 0, & m < n. \end{cases} \tag{13.26}$$

Since every $*$-homomorphism is contractive, we have

$$\|\nu_n(a)_m\|_{A_m} \le \|a\|_{A_n} \tag{13.27}$$

for every $m \in \mathbb{N}$. Thus, $\nu_n(a)$ belongs to the bounded product. It is straightforward to verify that ν_n is a $*$-homomorphism. Define $\mu_n := \pi \circ \nu_n$. For $a \in A_n$, the element

$$c := \nu_n(a) - \nu_{n+1}(\varphi_n(a)) \tag{13.28}$$

has n-th coordinate equal to a and all other coordinates equal to zero. Therefore, $c \in \bigoplus_{m\in\mathbb{N}}^{\mathrm{alg}} A_m$. It follows that

$$\mu_n = \mu_{n+1} \circ \varphi_n. \tag{13.29}$$

Thus, $\mu_n(A_n) \subseteq \mu_{n+1}(A_{n+1})$ for every $n \in \mathbb{N}$. Define

$$A := \overline{\bigcup_{n=1}^{\infty} \mu_n(A_n)}. \tag{13.30}$$

Then A is a C^*-algebra. We now verify the stated properties.

(1) This follows directly from the definition of A.
(2) Let $a \in A_n$. For $m \geq n$, we have

$$\varphi_{m+1,n}(a) = \varphi_m(\varphi_{m,n}(a)). \tag{13.31}$$

Since φ_m is contractive, the sequence $(\|\varphi_{m,n}(a)\|_{A_m})_{m\geq n}$ is decreasing. Therefore, its limit exists. By the quotient-norm formula,

$$\|\mu_n(a)\| = \|\pi(\nu_n(a))\| = \limsup_{m\to\infty} \|\varphi_{m,n}(a)\|_{A_m} = \lim_{m\to\infty} \|\varphi_{m,n}(a)\|_{A_m}. \tag{13.32}$$

(3) This follows immediately from (2).

It remains to verify the universal property. Let B be a C^*-algebra, and let $\lambda_n\colon A_n \to B$ be $*$-homomorphisms satisfying $\lambda_n = \lambda_{n+1} \circ \varphi_n$. For $m \geq n$, we have $\lambda_m \circ \varphi_{m,n} = \lambda_n$. Since λ_m is contractive,

$$\|\lambda_n(a)\| \leq \|\varphi_{m,n}(a)\|_{A_m} \tag{13.33}$$

for every $a \in A_n$ and every $m \geq n$. Taking the limit and applying (2) gives

$$\|\lambda_n(a)\| \leq \|\mu_n(a)\|. \tag{13.34}$$

Hence, $\ker(\mu_n) \subseteq \ker(\lambda_n)$. Therefore, there exists a unique contractive $*$-homomorphism

$$\lambda_n'\colon \mu_n(A_n) \longrightarrow B \tag{13.35}$$

such that $\lambda_n' \circ \mu_n = \lambda_n$. The compatibility of the maps λ_n implies that

$$\lambda_{n+1}'\big|_{\mu_n(A_n)} = \lambda_n'. \tag{13.36}$$

Thus, the maps λ_n' define a contractive $*$-homomorphism on $\bigcup_{n=1}^\infty \mu_n(A_n)$. By continuity, it extends uniquely to a $*$-homomorphism $\lambda\colon A \to B$ such that $\lambda \circ \mu_n = \lambda_n$ for every $n \in \mathbb{N}$. The uniqueness of λ follows from the density of $\bigcup_{n=1}^\infty \mu_n(A_n)$ in A. This proves the universal property. □

Remark 13.6. Let (A_n, φ_n) be an inductive system of C^*-algebras, and let $A = \varinjlim A_n$ have corresponding $*$-homomorphisms $\mu_n\colon A_n \to A$. Suppose that B is a C^*-algebra and that $\lambda_n\colon A_n \to B$ are compatible $*$-homomorphisms. Let $\lambda\colon A \to B$ be the corresponding $*$-homomorphism. Then $\ker(\mu_n) \subseteq \ker(\lambda_n)$ for every $n \in \mathbb{N}$. We state the following results without proof:

(1) The map λ is injective if and only if $\ker(\lambda_n) = \ker(\mu_n)$ for every $n \in \mathbb{N}$.
(2) The map λ is surjective if and only if $B = \overline{\bigcup_{n=1}^\infty \lambda_n(A_n)}$.

Example 13.7. Let $A_n = M_n(\mathbb{C})$ and let $\varphi_n\colon A_n \to A_{n+1}$ be the $*$-homomorphism

$$\varphi_n(a) = \operatorname{diag}(a, 0). \tag{13.37}$$

Let $\mathscr{H} = \ell^2(\mathbb{N})$, and fix an orthonormal basis $(e_i)_{i\geq 1}$ of $\mathscr{H}$. For every $n \in \mathbb{N}$, let $p_n \in \mathscr{K}(\mathscr{H})$ be the projection onto $\operatorname{span}\{e_1, \dots, e_n\}$. Then

$$p_n = \sum_{i=1}^n e_i \otimes e_i. \tag{13.38}$$

Here, if $x, y \in \mathscr{H}$, then $x \otimes y \in \mathscr{B}(\mathscr{H})$ denotes the rank-one operator given by

$$(x \otimes y)(z) = \langle z, x\rangle y. \tag{13.39}$$

Define

$$\mu_n\colon M_n(\mathbb{C}) \longrightarrow \mathscr{K}(\mathscr{H}), \tag{13.40}$$

$$(a_{ij})_{i,j=1}^n \longmapsto \sum_{i,j=1}^n a_{ij}\, e_j \otimes e_i. \tag{13.41}$$

The image of μ_n is $p_n\mathscr{K}(\mathscr{H})p_n$. Indeed, if $u \in p_n\mathscr{K}(\mathscr{H})p_n$, then $u = p_n w p_n$ for some $w \in \mathscr{K}(\mathscr{H})$, and

$$u = \sum_{i,j=1}^n (e_i \otimes e_i)w(e_j \otimes e_j) \tag{13.42}$$

$$= \sum_{i,j=1}^n \langle w(e_j), e_i\rangle\, e_j \otimes e_i \tag{13.43}$$

$$= \mu_n\left((\langle w(e_j), e_i\rangle)_{i,j=1}^n\right). \tag{13.44}$$

It follows that μ_n is a $*$-isomorphism from $M_n(\mathbb{C})$ onto $p_n\mathscr{K}(\mathscr{H})p_n$. Moreover, $\mu_n = \mu_{n+1}\circ\varphi_n$. For every $T \in \mathscr{K}(\mathscr{H})$, we have

$$\|T - p_nTp_n\| \longrightarrow 0. \tag{13.45}$$

Therefore,

$$\mathscr{K}(\mathscr{H}) = \overline{\bigcup_{n=1}^\infty p_n\mathscr{K}(\mathscr{H})p_n} = \overline{\bigcup_{n=1}^\infty \mu_n(M_n(\mathbb{C}))}. \tag{13.46}$$

By Example 13.4, it follows that

$$\mathscr{K}(\ell^2(\mathbb{N})) \cong \varinjlim M_n(\mathbb{C}). \tag{13.47}$$

14. Universal C^*-Algebras

Universal C^*-algebras provide a systematic method for constructing C^*-algebras from generators and relations. We first form an algebra in which the prescribed relations hold formally, and then equip it with the largest C^*-seminorm arising from its $*$-representations.

Definition 14.1. Let $E = \{x_i : i \in I\}$ be a set. A **noncommutative monomial** in E is a word $x_{i_1}\cdots x_{i_m}$, where $m \geq 1$ and $i_1, \dots, i_m \in I$. A **noncommutative polynomial** in E is a finite formal complex linear combination of noncommutative monomials. The **free algebra** on E, denoted by $F(E)$, is the vector space of noncommutative polynomials in E, with multiplication determined by concatenation of words.

The free algebra is free in the sense that its generators satisfy no relations other than those forced by the algebraic operations. In particular, distinct words in the generators represent distinct elements of $F(E)$, so in general $x_ix_j \neq x_jx_i$.

Remark 14.2. The free algebra satisfies the following universal property. If B is an algebra and $\{y_i : i \in I\} \subseteq B$, then there exists a unique algebra homomorphism $\varphi\colon F(E) \to B$ such that $\varphi(x_i) = y_i$ for every $i \in I$.

Let $E^* = \{x_i^* : i \in I\}$ be a set disjoint from E. We define an involution on $F(E \cup E^*)$ by extending the rule

$$(\alpha x_{i_1}^{\epsilon_1}\cdots x_{i_m}^{\epsilon_m})^* = \overline{\alpha}\, x_{i_m}^{\overline{\epsilon_m}}\cdots x_{i_1}^{\overline{\epsilon_1}}, \tag{14.1}$$

where $\alpha \in \mathbb{C}$, $\epsilon_k \in \{1, *\}$, and

$$\overline{\epsilon_k} = \begin{cases} * & \text{if } \epsilon_k = 1, \\ 1 & \text{if } \epsilon_k = *. \end{cases} \tag{14.2}$$

The resulting $*$-algebra is called the **free $*$-algebra** on E and is denoted by $F^*(E)$.

Definition 14.3. Let $E = \{x_i : i \in I\}$ be a set, and let $R \subseteq F^*(E)$ be a set of relations. Let J_R denote the two-sided $*$-ideal of $F^*(E)$ generated by R. The **universal $*$-algebra** generated by E subject to the relations R is

$$A(E \mid R) := F^*(E)/J_R. \tag{14.3}$$

By abuse of notation, we denote the image of each generator $x_i \in E$ in $A(E \mid R)$ again by x_i.

Example 14.4. Let $E = \{x\}$ and let $R = \{x^2, xx^*x - x\} \subseteq F^*(E)$. Then $A(E \mid R)$ is the universal $*$-algebra generated by an element x satisfying $x^2 = 0$ and $xx^*x = x$. Taking adjoints of these relations also gives $(x^*)^2 = 0$ and $x^*xx^* = x^*$. Consequently, every nonzero monomial reduces to one of x, x^*, xx^*, or x^*x. Thus, $A(E \mid R)$ has dimension at most four.

To obtain a C^*-algebra from a universal $*$-algebra, we must first identify a norm compatible with both multiplication and the involution. Lemma 14.5 constructs the universal C^*-seminorm associated with a given set of generators and relations.

Lemma 14.5. *Let E be a set of generators, let $R \subseteq F^*(E)$ be a set of relations, and suppose that*

$$\|x\|_{\mathrm{u}} := \sup\{\|\pi(x)\| : \pi \textit{ is a } *\textit{-representation of } A(E \mid R)\} \tag{14.4}$$

is finite for every $x \in A(E \mid R)$. Then $\|\cdot\|_{\mathrm{u}}$ is a C^*-seminorm on $A(E \mid R)$. Moreover,

$$N := \{x \in A(E \mid R) : \|x\|_{\mathrm{u}} = 0\} \tag{14.5}$$

is a two-sided $*$-ideal of $A(E \mid R)$.

Proof. For $x, y \in A(E \mid R)$ and $\alpha \in \mathbb{C}$, the norm properties of every representation π give

$$\|\alpha x\|_{\mathrm{u}} = |\alpha| \|x\|_{\mathrm{u}}, \tag{14.6}$$

$$\|x + y\|_{\mathrm{u}} \le \|x\|_{\mathrm{u}} + \|y\|_{\mathrm{u}}. \tag{14.7}$$

Moreover,

$$\|xy\|_{\mathrm{u}} = \sup_\pi \|\pi(x)\pi(y)\| \le \sup_\pi \|\pi(x)\| \sup_\pi \|\pi(y)\| = \|x\|_{\mathrm{u}}\|y\|_{\mathrm{u}}. \tag{14.8}$$

Finally,

$$\|x^*x\|_{\mathrm{u}} = \sup_\pi \|\pi(x)^*\pi(x)\| = \sup_\pi \|\pi(x)\|^2 = \|x\|_{\mathrm{u}}^2. \tag{14.9}$$

Thus, $\|\cdot\|_{\mathrm{u}}$ is a C^*-seminorm. If $x \in N$ and $y \in A(E \mid R)$, then

$$\|yx\|_{\mathrm{u}} \le \|y\|_{\mathrm{u}}\|x\|_{\mathrm{u}} = 0, \tag{14.10}$$

$$\|xy\|_{\mathrm{u}} \le \|x\|_{\mathrm{u}}\|y\|_{\mathrm{u}} = 0. \tag{14.11}$$

Also,

$$\|x^*\|_{\mathrm{u}}^2 = \|xx^*\|_{\mathrm{u}} \le \|x\|_{\mathrm{u}}\|x^*\|_{\mathrm{u}} = 0. \tag{14.12}$$

Hence, $yx, xy, x^* \in N$, and N is a two-sided $*$-ideal. ☐

Definition 14.6. Let E be a set of generators and let $R \subseteq F^*(E)$ be a set of relations. Suppose that the seminorm in Lemma 14.5 is finite on $A(E \mid R)$. The **universal C^*-algebra** generated by E subject to R is the completion

$$C^*(E \mid R) := \overline{A(E \mid R)/N}^{\|\cdot\|_{\mathrm{u}}}. \tag{14.13}$$

The construction in Definition 14.6 is characterized by the following universal property.

Proposition 14.7. *Let $E = \{x_i : i \in I\}$ be a set of generators, and let $R \subseteq F^*(E)$ be a set of relations for which $C^*(E \mid R)$ exists. Let B be a C^*-algebra, and let $\{y_i : i \in I\} \subseteq B$ satisfy the relations R. Then there exists a unique $*$-homomorphism*

$$\varphi\colon C^*(E \mid R) \longrightarrow B \tag{14.14}$$

such that $\varphi(x_i) = y_i$ for every $i \in I$.

Proof. The universal property of $F^*(E)$ gives a $*$-homomorphism $F^*(E) \to B$ sending x_i to y_i for every $i \in I$. Since the elements y_i satisfy the relations in R, the ideal J_R belongs to its kernel. Hence, this map induces a $*$-homomorphism

$$\varphi_0\colon A(E \mid R) \longrightarrow B. \tag{14.15}$$

For $x \in A(E \mid R)$, the map $x \mapsto \|\varphi_0(x)\|$ is one of the C^*-seminorms appearing in the supremum defining $\|\cdot\|_{\mathrm{u}}$. Therefore,

$$\|\varphi_0(x)\| \leq \|x\|_{\mathrm{u}}. \tag{14.16}$$

It follows that $N \subseteq \ker \varphi_0$, so φ_0 factors through a contractive $*$-homomorphism $A(E \mid R)/N \to B$. This map extends uniquely by continuity to a $*$-homomorphism

$$\varphi\colon C^*(E \mid R) \longrightarrow B. \tag{14.17}$$

The prescribed values on the generators determine φ uniquely. $\square$

The finiteness condition in Definition 14.6 is essential. A useful sufficient condition is that every generator be uniformly bounded over all $*$-representations satisfying the relations. Indeed, every noncommutative polynomial involves only finitely many generators and finitely many monomials, so submultiplicativity then yields a finite uniform bound for that polynomial.

Example 14.8. The universal C^*-algebra $C^*(x \mid x = x^*)$ does not exist. For every $\lambda > 0$, the element $y_\lambda = \lambda \in \mathbb{C}$ is self-adjoint and satisfies $\|y_\lambda\| = \lambda$. Thus, the $*$-representation determined by $x \mapsto y_\lambda$ gives $\|x\|_{\mathrm{u}} \geq \lambda$. Since λ is arbitrary, $\|x\|_{\mathrm{u}} = \infty$.

Example 14.9. Consider the universal $*$-algebra in Example 14.4. Let π be a $*$-representation. Since $(x^*x)^2 = x^*x$, we have

$$\|\pi(x)\|^2 = \|\pi(x^*x)\| = \|\pi((x^*x)^2)\| = \|\pi(x^*x)\|^2 = \|\pi(x)\|^4. \tag{14.18}$$

Hence, $\|\pi(x)\| \in \{0, 1\}$. It follows that all generators, and hence all noncommutative polynomials in the generators, are uniformly bounded over $*$-representations. Therefore, $C^*(x \mid x^2 = 0,\ xx^*x = x)$ exists. Let

$$E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(\mathbb{C}). \tag{14.19}$$

Since $E_{12}^2 = 0$ and $E_{12}E_{12}^*E_{12} = E_{12}$, Proposition 14.7 gives a $*$-homomorphism

$$\varphi\colon C^*(x \mid x^2 = 0,\ xx^*x = x) \longrightarrow M_2(\mathbb{C}) \tag{14.20}$$

such that $\varphi(x) = E_{12}$. Moreover,

$$\varphi(x^*) = E_{21} \qquad \varphi(x^*x) = E_{22} \qquad \varphi(xx^*) = E_{11}. \tag{14.21}$$

Thus, φ is surjective. Since the domain has dimension at most four and $M_2(\mathbb{C})$ has dimension four, φ is an isomorphism. Therefore,

$$C^*(x \mid x^2 = 0,\ xx^*x = x) \cong M_2(\mathbb{C}). \tag{14.22}$$

Remark 14.10. A universal C^*-algebra may exist but be trivial. For example,

$$C^*(x \mid x^2 = 0,\ x = x^*) \tag{14.23}$$

exists, but it is the zero algebra. Indeed, the relation $x = x^*$ and the relation $x^2 = 0$ give $x^*x = x^2 = 0$. The C^*-identity then implies $x = 0$.

As we have seen in Section 12, matrix algebras provide fundamental finite-dimensional examples of C^*-algebras. The matrix units satisfy simple algebraic relations that completely determine the multiplication and involution in $M_n(\mathbb{C})$. Proposition 14.11 shows that these relations characterize $M_n(\mathbb{C})$ as a universal C^*-algebra.

Proposition 14.11. *Let $n \geq 2$. Then*

$$M_n(\mathbb{C}) \cong C^*(e_{ij} : 1 \leq i,j \leq n \mid e_{ij}^* = e_{ji},\ e_{ij}e_{kl} = \delta_{jk}e_{il} \textit{ for all } i,j,k,l). \tag{14.24}$$

Proof. Let A denote the universal $*$-algebra defined by the displayed generators and relations.For every $*$-representation π and every i, the defining relations give $e_{ii}^* = e_{ii}$ and $e_{ii}^2 = e_{ii}$. Therefore, the C^*-identity gives

$$\|\pi(e_{ii})\|^2 = \|\pi(e_{ii})^*\pi(e_{ii})\| = \|\pi(e_{ii}^2)\| = \|\pi(e_{ii})\|. \tag{14.25}$$

Since $\|\pi(e_{ii})\| \geq 0$, it follows that $\|\pi(e_{ii})\| \in \{0,1\}$. Moreover,

$$\|\pi(e_{ij})\|^2 = \|\pi(e_{ij}^*e_{ij})\| = \|\pi(e_{ji}e_{ij})\| = \|\pi(e_{jj})\| \leq 1. \tag{14.26}$$

Thus, the universal C^*-algebra exists. Every nonzero monomial in the generators reduces, using the defining relations, to one of the elements e_{ij}. Therefore, the universal $*$-algebra, and hence its C^*-completion, has dimension at most n^2. The standard matrix units $E_{ij} \in M_n(\mathbb{C})$ satisfy the defining relations. By Proposition 14.7, there exists a $*$-homomorphism

$$\varphi\colon C^*(e_{ij} : 1 \leq i,j \leq n \mid e_{ij}^* = e_{ji},\ e_{ij}e_{kl} = \delta_{jk}e_{il}) \longrightarrow M_n(\mathbb{C}), \tag{14.27}$$

$$e_{ij} \longmapsto E_{ij}. \tag{14.28}$$

Since the matrix units span $M_n(\mathbb{C})$, the map φ is surjective. As the domain has dimension at most n^2, φ is injective. This proves the result. □

Proposition 14.12 realizes the compact operators as the universal C^*-algebra generated by a countable system of matrix units.

Proposition 14.12. *Let $\mathscr{H}$ be a separable infinite-dimensional Hilbert space. Then*

$$\mathscr{K}(\mathscr{H}) \cong C^*(e_{ij} : i,j \in \mathbb{N} \mid e_{ij}^* = e_{ji},\ e_{ij}e_{kl} = \delta_{jk}e_{il} \textit{ for all } i,j,k,l). \tag{14.29}$$

Proof. Let

$$A := C^*(e_{ij} : i,j \in \mathbb{N} \mid e_{ij}^* = e_{ji},\ e_{ij}e_{kl} = \delta_{jk}e_{il} \text{ for all } i,j,k,l). \tag{14.30}$$

The argument in Proposition 14.11 shows that $\|\pi(e_{ij})\| \leq 1$ for every $*$-representation π of the corresponding universal $*$-algebra and every $i,j \in \mathbb{N}$. Hence, the universal C^*-algebra A exists. Fix an orthonormal basis $(\xi_n)_{n\in\mathbb{N}}$ for $\mathscr{H}$. For $i,j \in \mathbb{N}$, define

$$T_{ij}\colon \mathscr{H} \longrightarrow \mathscr{H}, \tag{14.31}$$

$$T_{ij}\xi_n = \delta_{jn}\xi_i. \tag{14.32}$$

Each T_{ij} has one-dimensional range and is therefore compact. Moreover,

$$T_{ij}^* = T_{ji}, \tag{14.33}$$

$$T_{ij}T_{kl} = \delta_{jk}T_{il}. \tag{14.34}$$

By Proposition 14.7, there exists a $*$-homomorphism

$$\varphi\colon A \longrightarrow \mathscr{K}(\mathscr{H}), \tag{14.35}$$

$$e_{ij} \longmapsto T_{ij}. \tag{14.36}$$

For $n \in \mathbb{N}$, let P_n denote the orthogonal projection onto $\operatorname{span}\{\xi_1, \dots, \xi_n\}$. Then $P_n \to I_{\mathscr{H}}$ strongly. Let $T \in \mathscr{K}(\mathscr{H})$. Since T is compact, the set $T(B_{\mathscr{H}})$ has compact closure. Strong convergence of (P_n), together with $\|I_{\mathscr{H}} - P_n\| \le 1$, is uniform on compact subsets of $\mathscr{H}$. Therefore,

$$\|(I_{\mathscr{H}} - P_n)T\| \longrightarrow 0. \tag{14.37}$$

Applying the same argument to T^* gives

$$\|T(I_{\mathscr{H}} - P_n)\| = \|(I_{\mathscr{H}} - P_n)T^*\| \longrightarrow 0. \tag{14.38}$$

Moreover,

$$T - P_nTP_n = (I_{\mathscr{H}} - P_n)T + P_nT(I_{\mathscr{H}} - P_n). \tag{14.39}$$

Hence,

$$\|T - P_nTP_n\| \le \|(I_{\mathscr{H}} - P_n)T\| + \|T(I_{\mathscr{H}} - P_n)\| \longrightarrow 0. \tag{14.40}$$

For every $n \in \mathbb{N}$, we have

$$P_nTP_n = \sum_{i,j=1}^{n} \langle T\xi_j, \xi_i\rangle T_{ij}. \tag{14.41}$$

Thus, every compact operator is a norm limit of finite linear combinations of the operators T_{ij}. Since the image of φ contains every T_{ij}, it follows that φ is surjective. For $n \in \mathbb{N}$, let

$$A_n := C^*(e_{ij} : 1 \le i, j \le n) \subseteq A. \tag{14.42}$$

The elements $\{e_{ij} : 1 \le i, j \le n\}$ satisfy the matrix-unit relations. Hence, Proposition 14.11 gives a surjective $*$-homomorphism

$$\theta_n\colon M_n(\mathbb{C}) \longrightarrow A_n, \tag{14.43}$$

$$E_{ij} \longmapsto e_{ij}. \tag{14.44}$$

The composition $\varphi \circ \theta_n$ sends E_{ij} to T_{ij}, and is nonzero. Hence, θ_n is nonzero. Since $M_n(\mathbb{C})$ is simple, $\ker\theta_n = \{0\}$. Therefore, θ_n is a $*$-isomorphism. The restriction of φ to A_n is a $*$-isomorphism onto the finite matrix algebra generated by

$$\{T_{ij} : 1 \le i, j \le n\}. \tag{14.45}$$

In particular, φ is isometric on every A_n. The algebraic $*$-subalgebra generated by the elements e_{ij} is contained in $\bigcup_{n\in\mathbb{N}} A_n$. Since this algebraic $*$-subalgebra is dense in A, the union $\bigcup_{n\in\mathbb{N}} A_n$ is dense in A. As φ is isometric on this dense subalgebra, it is isometric on A. Hence, φ is injective and therefore a $*$-isomorphism. $\square$

15. Group C^*-Algebras

16. Crossed Product C^*-Algebras